



ALLAHABAD UNIVERSITY

NOTES

FOR

BSC 3RD YEAR

ZOLOGY

PYQ BASED



PYQ BASED

Topic wise previous
year questions



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MULTIPLE CHOICE QUESTIONS

Question..... Which of the following is a rectal ciliate causing protozoan parasitic disease in man? (a) Euglena (b) Balantidium (c) Volvox (d) Paramecium Answer..... **(b) Balantidium**

Explanation: *Balantidium coli* is a ciliated protozoan parasite that inhabits the large intestine of humans, causing balantidiasis.

Question..... The intermediate host of *Dracunculus medinensis* is an important constituent of: (a) aquatic ecosystem (b) terrestrial ecosystem (c) Forest ecosystem (d) aerial ecosystem Answer..... **(a) aquatic ecosystem** **Explanation:** The intermediate host of the Guinea worm (*Dracunculus medinensis*) is a freshwater crustacean called *Cyclops*, which lives in aquatic ecosystems.

Question..... Name a cestode of economic significance whose life cycle involved 3 hosts: (a) Hymenolepis nana (b) Diphylobothrium latum (c) Trichuris trichiura (d) Proteocephalus Answer..... **(b) Diphylobothrium latum** **Explanation:** *Diphylobothrium latum* (fish tapeworm) requires three hosts: a copepod (*Cyclops*), a freshwater fish, and a human or fish-eating mammal.

Question..... Identify a pest among the following: (a) Rat (b) Vulture (c) Clarias (d) Shark Answer..... **(a) Rat** **Explanation:** Rats (*Rattus rattus*) are major agricultural and domestic pests that damage crops, stored grains, and transmit human pathogens.

Question..... Name shell fish: (a) Anabas testudineus (b) Macrobrachium rosenbergii (c) Tor tor (d) Garra gotyla gotyla Answer..... **(b) Macrobrachium rosenbergii** **Explanation:** *Macrobrachium rosenbergii* (giant river prawn) is a crustacean cultured commercially as shellfish.

Question..... Predominant Non-Methane Volatile organic compounds are primarily emitted from: (a) Human and Animal Activity (b) Oil & gas industry (c) Agricultural activity (d) Forests Answer..... **(d) Forests** **Explanation:** Biogenic emissions from forest trees and plants (such as terpenes and isoprene) are the primary global source of non-methane volatile organic compounds.

Question..... Terracing is employed to control: (a) Air pollution (b) Soil pollution (c) Water pollution (d) Soil erosion Answer..... **(d) Soil erosion** **Explanation:** Terracing constructs step-like levels on steep slopes to slow water runoff and prevent severe soil erosion.

Question..... Biopollution results into: (a) Flourishing of native species (b) Species mixing (c) Control of invasive species (d) Cleaning of river water Answer..... **(b) Species mixing** **Explanation:** Biopollution involves the introduction of invasive non-native species, leading to ecosystem alterations and artificial species mixing.

Question..... Environmental degradation involves: (a) Upgradation of genotype of biota (b) Resource conservation (c) Dwindling of species (d) Abatement of pollution Answer..... **(c) Dwindling of species**

Explanation: Environmental degradation leads to habitat loss, biodiversity decline, and the reduction or dwindling of biological species.

FILL IN THE BLANKS & FULL FORMS

Question..... Malignant tertian malaria is caused by Answer..... Plasmodium falciparum

Question..... Monocystis is found in of earthworm. Answer..... seminal

vesicles

Question..... *Callosobruchus chinensis* is a pest of Answer..... stored pulses / legumes

Question..... Causative organism of tick fever is Answer..... *Babesia bigemina* (or *Rickettsia rickettsii*)

Question..... Which honey bee species produces maximum honey/comb? Answer..... *Apis dorsata* (Rock bee)

Question..... Pebrine is a disease of Answer..... silkworm (*Bombyx mori*)

Question..... Trichogramma is a Answer..... parasitoid (egg parasitoid)

Question..... ICUN (IUCN) stands for Answer..... International Union for Conservation of Nature

Question..... Scabies is caused by Answer..... *Sarcoptes scabiei* (Itch mite)

Question..... Celphos is used for the control of Answer..... stored grain pests / rodents

Question..... *Plasmodium falciparum* causes malaria. Answer..... malignant tertian

Question..... mosquito is the vector of encephalitis. Answer..... *Culex* (or *Aedes*)

Question..... Bioagents used in biological control of pests are, parasitoids and predators. Answer..... pathogens

Question..... The process of soil formation is termed as Answer..... pedogenesis

Question..... The presence of unwanted sound at a place is known as Answer..... noise pollution

Question..... The father of chipko movement is Answer..... Sunderlal Bahuguna (and Chandi Prasad Bhatt)

Question..... A few years back, the most common heavy metal at the places of heavy traffic used be Answer..... Lead (Pb)

Question..... The study of impact of xenobiotics at ecosystem level is known as Answer..... ecotoxicology

Question..... Conservation of wildlife in their natural habitat is conservation. Answer..... in-situ

Question..... Cultivation of aquatic animals for food and products is known as Answer..... aquaculture

Question..... Euryhaline relates to concentration of in water. Answer..... salt / salinity

Question..... Excess fluoride in drinking water causes in teeth. Answer..... dental fluorosis

Question..... Reduction in biodiversity occurs due to species. Answer..... invasive / alien

Question..... Parity in abundance is an attribute of vulnerability of species to Answer..... extinction

Question..... Major natural sources of pollutants responsible for stratospheric ozone depletion include volcanic Answer..... eruptions / emissions

Question..... Combustion of fuels is an important source of heavy metals and trace elements in environment. Answer..... fossil

Question..... Scampi is the common name of the product of aquaculture of genus Answer..... *Macrobrachium*

Question..... Water with nutrient status and productivity is called oligotrophic. Answer..... low

Question..... fish is an important constituent of polyculture. Answer..... Herbivorous / Phytophagous

Question..... Irrational cutting of trees is the major cause of soil Answer..... erosion

Question..... Use of form of insecticide is generally better for the control of stored grain pests. Answer..... fumigants

Question..... is the vector of Dengue fever. Answer..... *Aedes aegypti*

Question..... Use of parasites, predators and other biological agents for the control of pests, is called Answer..... biological control (biocontrol)

Question..... The oyster of the genus produces pearl. Answer..... *Pinctada*

Question..... *Leptocorisa varicornis* (stink bug) is a pest of crop. Answer..... paddy (rice)

Question..... LD_{50} is the unit to measure the of Answer..... lethal dose / toxicity of toxicants / chemicals

Question..... On the banks of fast flowing rivers, type of soil erosion may take place. Answer..... gully / bank erosion

Question..... Sunder Lal Bahuguna is known for Answer..... Chipko Movement

Question..... Lax (Lac) is obtained from the of an insect. Answer..... secretion (lac resin)

Question..... Cyclops is the intermediate host of Answer..... *Dracunculus medinensis* (or *Diphyllbothrium latum*)

Question..... Give the full form of TDS: Answer..... Total Dissolved Solids

Question..... Give the full form of UNEP: Answer..... United Nations Environment Programme

Question..... Give the full form of CITES: Answer..... Convention on International Trade in Endangered Species of Wild Fauna and Flora

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Question..... Give the full form of W.W.F.: Answer..... World Wide Fund for Nature

Question..... The world environment day is celebrated on Answer..... 5th June

Question..... Blue baby syndrome is caused by the presence of in water. Answer..... nitrates

SHORT NOTES

Question..... Describe in brief: Wax moth Answer..... Wax moths (*Galleria mellonella*, Greater Wax Moth; and *Achroia grisella*, Lesser Wax Moth) belong to the family Pyralidae. Their larvae act as major pests in apiculture by invading beehives, constructing silk-lined tunnels through

comb structures, and feeding on wax, pollen, honey, and cast larval skins. Severe infestation leads to comb destruction, weakening of bee colonies, and absconding. Control measures involve strict hive hygiene, mechanical removal, freezing infested frames, and regulated chemical or biological treatments.

Question..... Describe in brief: Pearl oysters Answer..... Pearl oysters belong to the genus *Pinctada* within the family Pteriidae (Class Bivalvia, Phylum Mollusca). Key species include *Pinctada fucata* and *Pinctada margaritifera*. They secrete nacre (mother-of-pearl) as an immune defense mechanism to coat irritating foreign particles lodged between their mantle and shell. In commercial pearl culture, small beads (nuclei) along with mantle tissue grafts are inserted into the host oyster to induce cultured pearl formation over several months.

Question..... Describe in brief: Tassar silk Answer..... Tassar (Tasar) silk is a wild variety of silk produced by the larvae of moths belonging to the genus *Antheraea* (e.g., *Antheraea mylitta* and *Antheraea paphia*). These silkworms feed primarily on host trees like Asan (*Terminalia tomentosa*) and Arjun (*Terminalia arjuna*). The resulting silk fibers possess a characteristic copperish color and coarse texture. Tassar silk culture forms a vital component of vanya (non-mulberry) sericulture, providing sustainable livelihoods in tribal regions.

Question..... Describe in brief: *Bacillus thuringiensis* Answer..... *Bacillus thuringiensis* (Bt) is a Gram-positive, spore-forming soil bacterium widely utilized as a biological pest control agent. During sporulation, it produces crystalline protein inclusions called delta-endotoxins (Cry proteins). When ingested by susceptible insect larvae (such as caterpillars, beetles, and mosquitoes), these toxins solubilize in the alkaline gut region, binding to midgut epithelial cells, causing pore formation, gut paralysis, and larval death.

Question..... Describe in brief: Zoonosis Answer..... Zoonosis (or zoonotic disease) refers to any infectious disease naturally transmissible between vertebrate animals and humans. These pathogens may be viral, bacterial, parasitic, or fungal in nature. Examples include Rabies, Anthrax, Brucellosis, Toxoplasmosis, and Avian Influenza. Transmission pathways involve direct contact, vector-borne routes, inhalation, or ingestion of contaminated animal products, making zoonotic management essential in public health epidemiology.

Question..... Describe in brief: Afforestation Answer..... Afforestation is the process of establishing forests on lands that historically have not been forested or have been devoid of tree cover for a prolonged period. Unlike reforestation (replanting harvested or degraded forests), afforestation transforms bare, degraded, or agricultural landscapes into forest ecosystems. It effectively counters soil erosion, improves regional hydrology, restores lost wildlife habitats, and enhances carbon sequestration to combat global climate change.

Question..... Describe in brief: Green house effect Answer..... The greenhouse effect is a natural thermal warming phenomenon where specific atmospheric trace gases—termed greenhouse gases (GHGs) like CO_2 , CH_4 , N_2O , and water vapor—absorb and re-radiate infrared thermal radiation emitted from the Earth's surface. While the natural process maintains habitability, human activities (fossil fuel burning, deforestation) amplify GHG concentrations, trapping excess heat and driving global warming and climate instability.

Question..... Describe in brief: Radioactive fallout Answer..... Radioactive fallout consists of airborne, radioactive particles and dust that settle on the Earth following nuclear detonations, industrial nuclear plant accidents, or weapon testing. These residues emit harmful ionizing radiation (such as alpha, beta, and gamma rays). Isotopes like Iodine-131, Cesium-137, and Strontium-90 enter food webs through soil and water contamination, inducing severe bioaccumulation, acute radiation syndrome, and long-term genetic mutations in organisms.

Question..... Describe in brief: Ecotoxicology Answer..... Ecotoxicology is an interdisciplinary

branch of environmental science studying the toxic impacts of chemical, physical, and biological agents on living organisms, specifically across populations, communities, and whole ecosystems. Unlike classical toxicology (which focuses primarily on individual organismal effects), ecotoxicology evaluates pollutant migration, bioaccumulation, biomagnification, metabolic fate, and community-level ecosystem disruptions caused by xenobiotics.

Question..... Describe in brief: In vitro conservation of wild life Answer..... In vitro conservation of wildlife involves preserving biological materials (such as gametes, embryos, somatic cells, and genomic DNA) outside of living wild animals under controlled, artificial laboratory conditions. Techniques include cryopreservation in liquid nitrogen (at -196°C), establishing biobanks, assisted reproductive technologies (ART) like artificial insemination, and *in vitro* fertilization (IVF). It serves as a safety net against species extinction and genetic erosion.

MULTIPLE CHOICE QUESTIONS

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Question..... The disease caused due to cadmium pollution of soil is: (a) Organotin (b) Methaemoglobinemia (c) Blue baby syndrome (d) Itai itai Answer..... **(d) Itai itai** **Explanation:** Cadmium accumulation in water and soil causes Itai-itai disease, characterized by severe bone softening, kidney damage, and joint pain.

Question..... Which among the following is one of the Indian major carps? (a) Cyprinus carpio (b) Catla catla (c) Hilsa ilisha (d) Channa punctatus Answer..... **(b) Catla catla**

Explanation: *Catla catla* (Catla) is a primary fast-growing surface-feeding freshwater species classified under Indian Major Carps.

Question..... Which among following is a pest of sugarcane? (a) *Leptocorisa varicornis* (b) *Heliotis armigera* (c) *Pyrilla perpusilla* (d) *Erias vitella* Answer..... **(c) *Pyrilla perpusilla***

Explanation: *Pyrilla perpusilla* (Sugarcane leafhopper) is a major sap-sucking pest damaging sugarcane crops.

Question..... Tassar Silk is derived from: (a) *Bombyx mori* (b) *Antheraea paphia* (c) *Antheraea assama* (d) *Attacus ricinii* Answer..... **(b) *Antheraea paphia*** **Explanation:** Tassar (Tasar) silk is produced by wild silkworm larvae, predominantly *Antheraea paphia* (or *Antheraea mylitta*).

Question..... Filaria is transmitted by: (a) *Ascaris* (b) Tape worm (c) *Culex* (d) *Aedes* Answer..... **(c) *Culex*** **Explanation:** Lymphatic filariasis (caused by *Wuchereria bancrofti*) is transmitted to humans via the bite of infected *Culex* mosquitoes.

Question..... Example of endangered species is: (a) Gangetic Dolphin (b) Blue whale (c) Peacock pheasant (d) Cheetah Answer..... **(a) Gangetic Dolphin** **Explanation:** *Platanista gangetica* (Gangetic River Dolphin) is an endangered aquatic mammal native to the Ganges-Brahmaputra river system.

FILL IN THE BLANKS & FULL FORMS

Question..... Conversion of agricultural lands into built-up areas results in the elimination of species. Answer..... **native / wild**

Question..... Animals provide for one of the most lucrative black Answer..... markets (wildlife trafficking)

Question..... The world environment day is celebrated on Answer..... 5th June

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Question..... Give the full form of I.C.B.P.: Answer..... International Council for Bird Preservation

SHORT ANSWER QUESTIONS (2 MARKS FORMAT)

Question..... Identify four renewable natural resources. Answer..... Renewable natural resources are infinite natural assets capable of self-replenishment through natural biological or

physical cycles within a human timeframe. Four key examples include:

- **Solar Energy:** Continuous electromagnetic radiation harvested for heat and power generation.
- **Freshwater Systems:** Replenished naturally via global hydrological precipitation cycles.
- **Forests and Biomass:** Regenerative biological systems providing timber, oxygen, and habitat.
- **Wind Energy:** Kinetic atmospheric energy produced by uneven solar heating of Earth's surface.

Question..... Enumerate any four measures to check global warming. Answer..... Global warming is caused by elevated greenhouse gas emissions (CO_2 , CH_4). Four mitigation strategies include:

- **Afforestation and Reforestation:** Increasing tree cover enhances natural carbon sequestration from the atmosphere.
- **Transition to Renewable Energy:** Replacing fossil-fuel combustion with solar, hydro, and wind power sources.
- **Energy Efficiency:** Implementing industrial energy-saving practices, smart grids, and fuel-efficient transportation systems.
- **Reducing Industrial Emissions:** Mandating carbon capture technology, lowering CFC usage, and controlling agricultural methane output.

Multiple Choice & Full Forms

Question: Example of endangered species is: (a) Gangetic Dolphin (b) Blue whale (c) Peacock pheasant (d) Cheetah **Answer:** (a) Gangetic Dolphin (*Platanista gangetica*)

Question: Give full forms of I.U.C.N. **Answer:** International Union for Conservation of Nature and Natural Resources

Question: Give full forms of I.B.W.L. **Answer:** Indian Board for Wildlife

Question: Give full forms of R.D.B. **Answer:** Red Data Book

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Question: Give full forms of I.C.B.P. **Answer:** International Council for Bird Preservation (now BirdLife International)

Fill in the Blanks

Question: The World environment day is celebrated on _____. **Answer:** 5th June

Question: Conversion of agricultural lands into built up areas results in elimination of _____ species. **Answer:** Native / Wild (or Endemic)

Question: Excess fluoride in drinking water causes _____ in teeth. **Answer:** Fluorosis (Dental fluorosis)

Question: Plasmodium falciparum causes _____ malaria. **Answer:** Malignant tertian

Question: Monocystis is found in the _____ of earthworm. **Answer:** Seminal vesicles

Question: Blue baby syndrome is caused by the presence of _____ in water. **Answer:** Nitrates (Nitrate contamination)

Question: _____ is the vector of Dengue fever. **Answer:** *Aedes aegypti* (Mosquito)

Question: Cyclops is the Intermediate host of _____. **Answer:** *Dracunculus medinensis* (Guinea worm) / *Diphyllbothrium latum*

Question: The process of soil formation is termed as _____. **Answer:** Pedogenesis

Question: The father of Chipko movement is _____. **Answer:** Sunderlal Bahuguna (along with Chandi Prasad Bhatt)

One-Word Answer Questions

Question: Name first intermediate host in life cycle of *Dracunculus*. **Answer:** *Cyclops* (Water flea)

Question: How many days would first larval stage of *Diphyllbothrium* survive? **Answer:** 1 to 2 days (The free-swimming ciliated *Coracidium* larva must be ingested by a copepod within 24–48 hours to survive)

Question: Name one pollutant in the effluent of dyeing industry. **Answer:** Synthetic Azo Dyes (or Heavy metals such as Chromium/Lead)

Question: What is the normal range of DO of tap water? **Answer:** 6.5 mg/L to 8.0 mg/L (at room temperature)

Question: Name the infective stage of Plasmodium. **Answer:** Sporozoite

Question: The vector of dengue. **Answer:** *Aedes aegypti*

Question: Zoological name of rice weevil. **Answer:** *Sitophilus oryzae*

Question: *Entamoeba histolytica* causes which disease. **Answer:** Amoebiasis (Amoebic)

dysentery)

Differentiations

Question: Differentiate between a pest and an infecting agent.

Answer:

- **Pest:** A pest is an organism (such as an insect, rodent, weed, or nematode) that causes physical damage to crops, stored food, livestock, or human property, primarily leading to economic loss. Example: Locusts damaging crops or rats destroying stored grains.
- **Infecting Agent (Pathogen):** An infecting agent is a disease-causing microorganism (such as a virus, bacterium, fungus, or protozoan) that physically invades a host's body, multiplies, and disrupts normal physiological functions. Example: *Plasmodium* causing malaria or *Mycobacterium* causing tuberculosis.

Question: Differentiate between the parasite that causes dysentery and pyorrhoea.

Answer:

- **Dysentery Parasite:** Amoebic dysentery is caused by the protozoan *Entamoeba histolytica*. It primarily resides in the upper part of the human large intestine (colon) and is transmitted via the ingestion of contaminated food and water.
- **Pyorrhoea Parasite:** Pyorrhoea (periodontal disease) is heavily associated with the protozoan *Entamoeba gingivalis*, along with bacterial complexes. It resides in the tartar and pus pockets of human teeth and gums, and is typically transmitted through direct oral contact.

Question: Differentiate between point and non-point pollution sources.

Answer:

- **Point Source Pollution:** Refers to pollution originating from a single, clearly identifiable, and localized source. Because the source is specific, it is relatively easy to monitor, regulate, and treat. Example: An effluent discharge pipe from a specific chemical factory or a municipal sewage drain.
- **Non-point Source Pollution:** Refers to pollution originating from diffuse, widespread areas rather than a single distinct location. It is heavily influenced by land use and weather conditions, making it difficult to trace and control. Example: Agricultural fertilizer runoff during heavy rain or general urban stormwater drainage.

Question: Differentiate between a pollutant and a toxicant.

Answer:

- **Pollutant:** Any substance (or energy like heat/noise) introduced into the environment by human activity that adversely affects resource quality, environmental health, or ecosystem balance. Not all pollutants are lethal. Example: Plastic waste, domestic sewage, or excess carbon dioxide.
- **Toxicant:** A specific chemical substance (synthetic or natural) that acts as a poison, directly causing injury, adverse physiological effects, or death in living organisms even at low concentrations. Example: Heavy metals like lead and mercury, or synthetic pesticides like DDT. (*All toxicants released into the environment are pollutants, but not all pollutants are toxicants*).

Question: Differentiate between deforestation and desertification.

Answer:

- **Deforestation:** The deliberate, large-scale removal or clearing of forests and tree cover by humans, usually to convert the land for agricultural, industrial, or urban use.
- **Desertification:** A form of land degradation where previously fertile land gradually turns

into an arid, barren, desert-like environment. It is a complex ecological process driven by a combination of factors, including extreme drought, climate change, and human activities like severe deforestation and overgrazing.

Question: Differentiate between parasite and predator. **Answer:**

- **Parasite:** An organism that lives in or on another organism (the host) to derive its nourishment. A parasite typically harms its host but usually does not kill it immediately, as its own survival depends on the host remaining alive. Example: Liver fluke, tapeworm, or head lice.
- **Predator:** An organism that actively hunts, captures, and kills other organisms (prey) for food. The relationship is brief but lethal for the prey. Example: A lion hunting a deer, or an eagle catching a snake.

Question: Differentiate between Wildlife Sanctuary and Park (National Park). **Answer:**

- **Wildlife Sanctuary:** A protected area primarily designated for the conservation of specific animal species or wildlife habitats. Limited human activities, such as controlled grazing, gathering firewood, or private land ownership, may be permitted provided they do not interfere with the well-being of the wildlife.
- **National Park:** A strictly protected area designated for the conservation of the entire ecosystem, including flora, fauna, landscapes, and historical objects. It has a higher degree of protection, and all human exploitation, private land holding, grazing, and hunting are strictly prohibited.

Question: Differentiate between soil erosion and water erosion. **Answer:**

- **Soil Erosion:** The broad, general process involving the detachment, transport, and deposition of the uppermost layer of soil (topsoil) by various natural agents such as wind, water, ice, and gravity, as well as human activities.
- **Water Erosion:** A specific category of soil erosion where the topsoil is washed away exclusively by the physical force of water. It occurs in distinct stages, including splash erosion (from raindrops), sheet erosion, rill erosion, and gully erosion (from surface runoff).

Short Descriptive Answers (Section-A / Short Notes)

Question: Describe the life cycle and zoonotic significance of a tick/mite. **Answer:**

- **Life Cycle:** Ticks and mites (Arachnida: Acari) undergo a 4-stage life cycle: **Egg** → **6-legged Larva** → **8-legged Nymph** → **Adult**. Females feed on blood before laying eggs in soil or debris. Larvae and nymphs must take a blood meal from suitable vertebrate hosts to molt into the subsequent stage.
- **Zoonotic Significance:** Ticks act as vectors for various deadly zoonotic diseases affecting both humans and animals, such as **Kyasanur Forest Disease (KFD)** in India, **Lyme Disease** (*Borrelia*), and **Tick-borne Encephalitis**. Mites cause **Scabies** and transmit **Scrub Typhus** (*Orientia tsutsugamushi*).

Question: Enumerate main beneficial insects of India. Add a note on apiculture or lac-culture practised in India. **Answer:**

- **Beneficial Insects:**
 1. *Apis cerana indica* / *Apis dorsata* (Honeybees) – Honey, wax production, and pollination.
 2. *Bombyx mori* (Silkworm) – Silk production.
 3. *Kerria lacca* (Lac insect) – Resin/lac production.
 4. *Coccinella septempunctata* (Ladybird beetle) – Biological control of crop pests

(aphids).

- **Note on Apiculture:** Apiculture is the commercial rearing of honeybees in modern moveable-frame hives (Langstroth hives). In India, it yields economically valuable products like pure honey, beeswax, royal jelly, and bee venom. Bees also act as crucial pollinators, increasing crop yields by up to 30–40%.

Question: Give an account of economic importance of Protozoa in relation with human.

Answer:

- **Harmful Effects:** Parasitic protozoans cause severe diseases in humans, such as *Plasmodium* (Malaria), *Entamoeba histolytica* (Amoebic Dysentery), *Leishmania donovani* (Kala-azar), and *Trypanosoma* (Sleeping Sickness).
- **Beneficial Effects:** Free-living protozoans serve as a primary food source for aquatic micro-fauna and plankton. Marine protozoans like **Foraminifera and Radiolaria** form thick oceanic sludge/ooze, acting as valuable bio-indicators for petroleum deposits and ocean geological mapping.

Question: Write short note on Pearl culture. **Answer:** Pearl culture is the technique of culturing marine pearl oysters (*Pinctada fucata*) or freshwater mussels to produce natural pearls. An foreign particle/irritant (a shell bead/tissue piece) is surgically inserted into the mantle cavity of the oyster. In response, the mantle secretes continuous concentric layers of **nacre** (calcium carbonate and conchiolin resin) around the irritant over 1–2 years, forming a precious pearl.

Question: Write short note on Economic importance of porifera and mollusca. **Answer:**

- **Porifera (Sponges):** Bath sponges (*Euspongia*) are harvested for commercial bathing, scrubbing, and polishing. Sponges produce novel bioactive compounds (spongothymidine) with antiviral and anticancer properties.
- **Mollusca:** Mollusks provide edible seafood (oysters, squids, clams). Shells of gastropods and bivalves are used in lime industries, ornaments, button manufacturing, and commercial pearl production.

Question: Write short note on Dengue. **Answer:** Dengue is an acute viral infection caused by the **Dengue Virus (DENV 1–4)**, a Flavivirus. It is transmitted to humans via the bite of infected female ***Aedes aegypti*** mosquitoes. Symptoms include high fever, severe headache, retro-orbital pain, joint/muscle pains ("breakbone fever"), and rash. Severe cases can progress to Dengue Hemorrhagic Fever (DHF) marked by a sharp drop in blood platelet count.

One-Word & Objective Questions

Question: Maximum permissible dose of radiation for worker is: (a) 2 rem/yr (b) 3 rem/yr (c) 4 rem/yr (d) 5 rem/yr **Answer:** (d) 5 rem/yr (50 mSv/yr)

Question: Safe level of noise depends on: (a) Area (b) Pitch (c) Exposure to noise (d) Level of noise and exposure to noise **Answer:** (d) Level of noise and exposure to noise

Question: Lyonet's gland found in: (a) *Apis Indica* (b) *Bombyx Mori* (c) *Laccifer Lacca* (d) *Sphenodon* **Answer:** (b) *Bombyx Mori* (Silkworm)

Question: *Trogoderma granarium* is common pest of stored: (a) Maize (b) Bajra (c) Barley (d) Wheat **Answer:** (d) Wheat

Question: Zoological name of mulberry silkworm: **Answer:** *Bombyx mori*

Question: The vector of malaria: **Answer:** Female *Anopheles* mosquito

Question: Zoological name of Guinea worm: **Answer:** *Dracunculus medinensis*

Question: Number of intermediate hosts found in the life cycle of *Diphyllbothrium latum*:

Answer: Two intermediate hosts (1st: *Cyclops/Diaptomus*, 2nd: Fresh water fish)

Question: The zoological name of lac insect: **Answer:** *Laccifer lacca* (or *Kerria lacca*)

Question: The Zoological name of broad fish tape worm: **Answer:** *Diphyllbothrium latum*

Question: The organism which causes amoebic dysentery: **Answer:** *Entamoeba histolytica*

Question: The zoological name of Khapra beetle: **Answer:** *Trogoderma granarium*

Fill-in-the-Blanks

Question: Filariasis is transmitted by _____. **Answer:** Female *Culex* mosquito

Question: _____ mosquito is the vector of encephalitis. **Answer:** *Culex* / *Aedes* mosquito

Question: Cultivation of aquatic animals for food and products is known as _____. **Answer:** Aquaculture

Question: LD₅₀ is the unit to measure the _____ of _____. **Answer:** Toxicity / Lethality of a chemical substance (Lethal Dose required to kill 50% of the test population)

Short Answer Differentiations (2 Marks Format)

Question: Differentiate between Water pollution and soil pollution.

Answer:

- **Water Pollution:** It is the contamination of water bodies (rivers, lakes, oceans, groundwater) by domestic sewage, toxic industrial effluents, and chemical agricultural runoffs. It affects aquatic ecosystems directly, causes waterborne diseases, and spreads rapidly through water currents.
- **Soil Pollution:** It is the degradation of the earth's land surface and topsoil quality due to toxic heavy metals, non-biodegradable plastics, chemical pesticides, and solid waste accumulation. It reduces soil fertility, damages soil microfauna, and enters the human food chain via plant uptake.

Question: Differentiate between Flood and drought.

Answer:

- **Flood:** A natural disaster characterized by an overflow of excessive water submerging normally dry land, usually caused by heavy monsoons, cloudbursts, or river breaches. It leads to rapid destruction of habitat, soil erosion, and immediate life loss.

- **Drought:** A prolonged period of abnormally low or absent rainfall leading to a severe water shortage in an ecosystem. It causes crop failure, falling groundwater levels, land desertification, and famine over an extended duration.

Question: Differentiate between Biodegradable and non-biodegradable pollutants.

Answer:

- **Biodegradable Pollutants:** These are waste materials that can be easily broken down into harmless natural elements by decomposers (bacteria and fungi) within a short period. Example: Sewage, domestic food waste, paper, and animal dung.
- **Non-Biodegradable Pollutants:** These are persistent materials that cannot be broken down or decomposed by biological processes, remaining in the environment for decades or centuries. Example: Plastics, DDT, glass, heavy metals, and electronic waste.

Question: Differentiate between Overgrazing and deforestation.

Answer:

- **Overgrazing:** Occurs when livestock continuously graze on grassland vegetation faster than it can regenerate, removing root cover, destroying soil structure, and triggering severe topsoil erosion.
- **Deforestation:** The intentional, large-scale cutting down or clearing of standing trees and forests by human activity for agriculture, timber, or urban expansion.

Question: Differentiate between Sewage and industrial effluents.

Answer:

- **Sewage:** Domestic liquid waste containing organic matter, human excreta, soaps, detergent residues, and pathogenic microorganisms originating from households. It is largely biodegradable.
- **Industrial Effluents:** Liquid waste discharged from manufacturing industries, chemical processing units, or tanneries. They contain high levels of non-biodegradable chemicals, heavy metals (lead, mercury, chromium), and toxic compounds requiring specialized chemical treatment.

Question: Define Zoonotic diseases.

Answer: Zoonotic diseases are infectious diseases caused by harmful germs (like viruses, bacteria, parasites, or fungi) that are naturally transmitted between vertebrate animals and humans.

- **Examples:** Rabies, Anthrax, and Dracunculiasis.

Question: Define Parasitoid.

Answer: A parasitoid is an organism that lives in or on a host organism, feeding on it continuously and ultimately causing the death of the host as part of its normal development cycle.

- **Example:** *Trichogramma* (parasitic wasp).

Question: Define Pest.

Answer: A pest is any organism—such as an insect, animal, plant, or microbe—that causes economic damage to agricultural crops, stored food products, livestock, or human health.

- **Example:** Rice weevil (*Sitophilus oryzae*), Khapra beetle (*Trogoderma granarium*).

Question: Define AQI (Air Quality Index).

Answer: Air Quality Index (AQI) is a standardized numerical scale used by environmental agencies to measure and communicate daily ambient air pollution levels and their potential human health risks. Key pollutants measured include PM_{2.5}, PM₁₀, SO₂, and NO₂.

Question: Define Biomagnification.

Answer: Biomagnification is the progressive accumulation and increase in concentration of non-biodegradable toxic substances (such as DDT or heavy metals) at successively higher trophic levels of a food chain.

Question: State True or False: All ticks are ectoparasites.

Answer: True — Ticks are obligate external blood-feeding parasites of vertebrate hosts throughout their life stages.

Question: State True or False: Humans are the primary host of the malarial parasite *Plasmodium*.

Answer: False — The female *Anopheles* mosquito is the primary (definitive) host because sexual reproduction occurs inside it; humans are the secondary (intermediate) host.

Question: State True or False: Exposure to calcium cause Itai-itai disease.

Answer: False — Itai-itai disease is caused by chronic exposure to **Cadmium (Cd)** toxicity, not calcium.

Question: State True or False: Honeybee belong to the order Hymenoptera.

Answer: True — Honeybees belong to the insect order Hymenoptera alongside wasps and ants.

Question: State True or False: The reaction of sulfur dioxide (SO₂) and nitrogen oxides (NO_x) with water, oxygen and other chemicals in air cause acid rain.

Answer: True — Atmospheric reaction of these gases forms sulfuric and nitric acids, resulting in acidic precipitation.

Question: Differentiate between Pollutants and toxicants.

Answer:

- **Pollutants:** Chemical, physical, or biological agents introduced into the environment that alter its natural composition and cause harm to ecosystems (e.g., plastic, sewage, smoke).
- **Toxicants:** Specific chemical substances (synthetic or natural) that cause direct biological damage, poisoning, or lethality to living organisms (e.g., heavy metals, pesticides, toxins).

Question: Differentiate between Water pollution and soil pollution.

Answer:

- **Water Pollution:** Contamination of aquatic ecosystems (rivers, groundwater) by sewage, industrial effluents, and chemicals, leading to waterborne diseases and destruction of aquatic life.
- **Soil Pollution:** Degradation of land and soil quality due to heavy metals, non-biodegradable plastics, and agricultural pesticides, leading to lost soil fertility and entry of toxins into the food chain.

Question: Differentiate between Flood and drought.

Answer:

- **Flood:** An overflow of excessive water submerging dry land due to intense rainfall or river overflow, causing swift destruction and soil erosion.
- **Drought:** A prolonged absence or shortage of rainfall leading to severe water scarcity, crop failure, falling groundwater tables, and land degradation.

Question: Differentiate between Deforestation and desertification.

Answer:

- **Deforestation:** Large-scale clearing, logging, or removal of standing forest cover for agriculture or human settlement.
- **Desertification:** Degradation of dryland vegetation and topsoil due to severe climate variations and human activities, causing fertile land to become arid desert.

Question: Differentiate between Parasite and predator.

Answer:

- **Parasite:** An organism that lives on or inside a host body to derive nourishment without killing the host immediately or intentionally.
- **Predator:** An organism that directly hunts, kills, and consumes other animals (prey) for food.

Question: Differentiate between Wildlife Sanctuary and National Park.

Answer:

- **Wildlife Sanctuary:** A protected area dedicated to preserving specific species of wild fauna where limited human activity (like collection of timber or grazing) is permitted.
- **National Park:** A strictly protected natural habitat reserved for overall biodiversity conservation where all private rights and human activities are strictly prohibited.

Question: Differentiate between Soil erosion and water erosion.

Answer:

- **Soil Erosion:** Detachment and removal of topsoil by natural agents such as wind or flowing water.
- **Water Erosion:** Specific detachment and transport of topsoil specifically caused by rain splash, running surface water, and river currents.

Question: Differentiate between Biodegradable and non-biodegradable pollutants.

Answer:

- **Biodegradable Pollutants:** Waste materials that decompose rapidly into harmless natural components through natural biological activity (e.g., sewage, plant waste).
- **Non-Biodegradable Pollutants:** Substances that resist biological decomposition and persist in the environment for decades or centuries (e.g., plastics, DDT, heavy metals).

Question: Differentiate between Overgrazing and deforestation.

Answer:

- **Overgrazing:** Continuous feeding of livestock on grassland vegetation exceeding its natural regrowth capacity, leading to topsoil loss.
- **Deforestation:** Cutting down or clearing of entire trees and forest cover for land

conversion, timber, or urban expansion.

Question: Differentiate between Sewage and industrial effluents.

Answer:

- **Sewage:** Liquid waste containing organic human excreta, household soap residues, and pathogens originating from domestic homes.
- **Industrial Effluents:** Liquid chemical waste containing heavy metals, toxic chemicals, and non-biodegradable compounds discharged from factories and manufacturing units.
- **Space for extra knowledge**

Question/ Answer

Question 1: *Entamoeba histolytica*

1. Systematic Position & Introduction

Entamoeba histolytica is an obligate, pathogenic monogenetic protozoan endoparasite that inhabits the mucosal and submucosal layers of the human large intestine (colon and cecum). It is the causative organism of **amoebic dysentery** (intestinal amoebiasis) and secondary **amoebic liver abscesses** (extra-intestinal amoebiasis).

- **Phylum:** Protozoa
- **Subphylum:** Sarcomastigophora
- **Superclass:** Sarcodina
- **Class:** Rhizopoda
- **Order:** Amoebida
- **Genus:** *Entamoeba*
- **Species:** *histolytica*

2. Morphological Forms

It occurs in three distinct structural stages during its lifecycle:

- **Trophozoite (Magna form):** The active, feeding, growing, and invasive stage measuring 20 to 40 micrometers in diameter. The cytoplasm is differentiated into a clear outer ectoplasm and a granular inner endoplasm containing food vacuoles with ingested human red blood cells (**erythrophagocytosis**). The nucleus exhibits a delicate ring of peripheral chromatin and a distinct, central dot-like karyosome. It moves actively using a single, broad, finger-like pseudopodium (monopodal movement).
- **Precystic Stage (Minuta form):** A small, non-pathogenic, non-feeding, intermediate stage measuring 7 to 10 micrometers. It is round or oval with no ingested blood cells and acts as an immediate precursor to cyst formation.
- **Cystic Stage:** The infective, non-motile, resistant stage measuring 10 to 15 micrometers. A mature infective cyst is spherical, enclosed within a tough, resistant cyst wall, and contains **four nuclei** (quadrinucleate cyst), along with characteristic rod-shaped chromatoid bars with rounded ends and a glycogen vacuole.

3. Life Cycle Dynamics

- **Host:** Monogenetic (completes its entire life cycle in a single host, human).
- **Mode of Infection:** Transmission occurs via the fecal-oral route through ingestion of food, vegetables, or drinking water contaminated with mature quadrinucleate cysts.
- **Excystation:** Upon ingestion, the resistant cyst wall passes through the stomach intact. In the alkaline medium of the small intestine (ileum), the cyst wall is dissolved by trypsin, releasing a single tetranucleate metacyst.
- **Development:** The metacyst divides by binary fission to produce eight small daughter trophozoites, which migrate to the cecum and colon to colonize the intestinal wall.

4. Pathogenicity & Clinical Pathology

- **Intestinal Amoebiasis:** Trophozoites attach to colonic epithelial cells and secrete a powerful proteolytic enzyme called **histolysin**. This enzyme digests and liquefies intestinal mucosal tissue, causing characteristic **flask-shaped ulcers** with narrow necks and broad bases. Symptoms include severe abdominal cramps, flatulence, and stools mixed with blood and mucus (amoebic dysentery).
- **Extra-Intestinal Amoebiasis:** In severe cases, trophozoites erode colonic blood vessels, enter the portal venous circulation, and migrate to distant organs. The primary site is the liver, where they cause **Amoebic Liver Abscess**, characterized by localized tissue necrosis and pus resembling chocolate brown ("anchovy sauce") fluid. Less frequently, metastasis leads to pulmonary or cerebral amoebiasis.

5. Diagnosis, Prevention, and Treatment

- **Diagnosis:** Microscopic examination of fresh stool samples for quadrinucleate cysts and trophozoites showing ingested RBCs.
- **Prevention:** Maintaining personal hygiene, proper sewage disposal, washing raw vegetables, and consuming boiled or filtered water.
- **Treatment:** Administration of luminal antiamoebic drugs like Diloxanide furoate or systemic tissue antiamoebics such as **Metronidazole (Flagyl)** and **Tinidazole**.

Question 2: Beneficial Insects

1. Introduction

Beneficial insects are species that provide direct commercial products, vital ecological services, or biological control mechanisms that support human agriculture, industrial economy, and environmental balance.

2. Commercial / Product-Yielding Insects

- **Honey Bees (*Apis indica*, *Apis mellifera*):** Belonging to the order Hymenoptera, honey bees are social insects that yield honey, a nutrient-rich food containing fructose, glucose, enzymes, and vitamins used in medicine. They also produce natural beeswax, used extensively in cosmetics, candles, cold creams, and pharmaceutical ointments.
- **Silkworms (*Bombyx mori*):** Belonging to the order Lepidoptera, the caterpillars feed exclusively on mulberry leaves and spin protective silken cocoons made of fibroin and sericin proteins. Harvesting these cocoons forms the foundation of the commercial sericulture industry.
- **Lac Insects (*Kerria lacca* / *Laccifer lacca*):** Belonging to the order Hemiptera, the minute resinous secretion produced by female lac insects on host trees (like Palas, Ber, and Kusum) is refined into shellac. Shellac is used in varnishes, electrical insulation, french polishing, sealing wax, and shoe polishes.

3. Insects as Essential Pollinators

Over 80% of wild flowering plants and nearly one-third of human agricultural food crops depend on insect pollination (**Entomophily**). Key pollinator groups include:

- **Hymenopterans:** Honey bees, bumblebees, and solitary bees.
- **Lepidopterans:** Butterflies and moths.
- **Dipterans:** Hoverflies and bombyliid flies. These insects transfer pollen grains from anthers to stigmas while collecting nectar, significantly boosting seed set and fruit yield in mustard, sunflowers, pulses, apples, citrus, and vegetables.

4. Biological Control Agents (Predators & Parasitoids)

- **Predatory Insects:** Active predators like Ladybird beetles (*Coccinella septempunctata*) devour thousands of agricultural pests like aphids, scale insects, and mealybugs. Praying mantises and lacewings (*Chrysoperla*) act as generalist predators in field crops.
- **Parasitoid Insects:** Parasitic wasps such as *Trichogramma* species lay their eggs inside the eggs of damaging lepidopteran crop borers. The emerging wasp larvae feed on internal tissues, killing the pest before it hatches.

5. Soil Fertility & Decomposition

- **Scavengers:** Dung beetles (*Phanaeus*, *Onthophagus*) bury and consume livestock dung, breaking down animal waste, recycling nitrogen, and suppressing pest fly populations.
- **Soil Aerators:** Ground-dwelling ants and termites loosen soil, improving aeration, water infiltration, and organic matter mixing.

Question 3: Dengue

1. Etiology & Virus Characteristics

Dengue is an acute, mosquito-borne febrile viral illness caused by the **Dengue Virus (DENV)**. DENV is a single-stranded positive-sense RNA virus belonging to the genus *Flavivirus* and family *Flaviviridae*. It exists in four antigenically distinct serotypes: DENV-1, DENV-2, DENV-3, and DENV-4. Infection with one serotype provides lifelong immunity against that specific serotype, but secondary infection with a different serotype significantly increases the risk of severe disease due to Antibody-Dependent Enhancement (ADE).

2. Vector Biology & Transmission Dynamics

- **Primary Vector:** The mosquito species *Aedes aegypti* (commonly called the tiger mosquito due to white lyre-shaped markings on its thorax and silvery bands on its legs).
- **Secondary Vector:** *Aedes albopictus*.
- **Biting & Breeding Habits:** *Aedes* mosquitoes are strict day-biters (peak biting activity occurs in early morning and late afternoon). They breed preferentially in artificial clean-water collections around human habitations, such as discarded tires, flower pots, overhead water tanks, cooler trays, and plastic containers.

3. Clinical Manifestations

- **Classic Dengue Fever (Breakbone Fever):** Sudden onset of high fever ($103-105^{\circ}\text{F}$), accompanied by severe frontal headache, retro-orbital pain (pain behind the eyes), intense joint and muscle aches, nausea, vomiting, and a characteristic maculopapular skin rash.
- **Dengue Hemorrhagic Fever (DHF):** A severe form characterized by vascular hyperpermeability, plasma leakage, mucosal bleeding (epistaxis, bleeding gums), and marked **thrombocytopenia** (platelet count dropping below 100,000 per cubic millimeter).
- **Dengue Shock Syndrome (DSS):** The most critical stage resulting from acute plasma leakage, leading to sudden circulatory failure, rapid weak pulse, severe hypotension, hypovolemic shock, and organ failure.

4. Diagnostic Testing

- **NS1 Antigen Test:** Detects circulating viral non-structural protein 1 during the early febrile phase (Days 1 to 5).
- **Serological Testing:** Detection of Dengue-specific IgM antibodies (indicates primary acute infection) and IgG antibodies (indicates past or secondary infection) using ELISA.
- **Molecular Diagnostics:** Reverse Transcription Polymerase Chain Reaction (RT-PCR) to confirm viral RNA and identify serotypes.

5. Prevention & Vector Management

- **Source Reduction:** Eliminating stagnant water containers around homes to destroy mosquito breeding habitats.
- **Chemical Larvicides:** Application of organophosphates like Temephos (Abate) in stored water tanks.
- **Personal Protection:** Use of mosquito repellent creams, screens, and long-sleeved clothing during daytime hours.

Question 4: Zoonoses

1. Definition & Fundamental Concept

Zoonoses (or zoonotic diseases) are infectious diseases caused by pathogenic micro-organisms—including viruses, bacteria, protozoa, fungi, and helminthes—that are naturally transmitted between vertebrate animals and humans.

2. Scientific Classification of Zoonoses

Zoonotic diseases are classified based on the direction of pathogen transmission and primary reservoir host systems:

- **Anthropozoonoses:** Diseases naturally maintained in non-human vertebrate animals that are transmitted from animals to humans. Examples include Rabies (from dogs/bats) and Brucellosis (from cattle).

- **Zooanthroponoses:** Diseases naturally maintained in human populations that are transmitted from humans to non-human vertebrate animals. An example is human Tuberculosis (*Mycobacterium tuberculosis*) transmitted to captive primates or cattle.
- **Amphixenoses:** Diseases maintained in both humans and non-human vertebrates that can be freely transmitted in either direction. An example is *Trypanosoma cruzi* (Chagas disease).

3. Classification Based on Pathogen Type

- **Viral Zoonoses:** Rabies, Ebola virus disease, Avian Influenza, Kyasanur Forest Disease (KFD), and Dengue.
- **Bacterial Zoonoses:** Anthrax (*Bacillus anthracis*), Leptospirosis, Plague (*Yersinia pestis*), and Bovine Tuberculosis.
- **Parasitic Zoonoses:** Toxoplasmosis (*Toxoplasma gondii*), Hydatid disease (*Echinococcus granulosus*), Taeniasis (*Taenia solium*), and Leishmaniasis.
- **Fungal Zoonoses:** Ringworm (*Microsporum canis*) and Dermatophytosis.

4. Transmission Pathways

- **Direct Contact:** Handling infected animal tissues, blood, saliva, or secretions (e.g., Rabies virus transmitted via bite wounds).
- **Indirect Contact:** Exposure to environmental media (soil, water) contaminated with animal urine or feces (e.g., Leptospirosis).
- **Vector-Borne:** Transmission via hematophagous arthropods such as ticks, mosquitoes, fleas, and sandflies (e.g., Plague transmitted by rat fleas).
- **Food-Borne:** Consumption of contaminated meat, unpasteurized milk, or animal products (e.g., *Salmonella*, *Taenia*).

5. Public Health Importance

Zoonoses account for over 60% of known human infectious diseases and nearly 75% of newly emerging human pathogens. Managing zoonotic risks requires a multi-disciplinary **"One Health"** approach integrating human medicine, veterinary science, and environmental ecology.

Question 5: Economic Significance of Nematoda

1. Introduction

Nematodes (roundworms) are unsegmented, pseudocoelomate, bilaterally symmetrical worms inhabiting aquatic, terrestrial, and parasitic environments. They exert substantial positive and negative impacts on global agriculture, veterinary public health, and human medicine.

2. Harmful Aspects: Plant Parasitic Nematodes

Phytophagous nematodes possess a sharp, hollow oral stylet used to pierce plant cell walls and inject digestive enzymes, inflicting severe agricultural losses:

- **Root-Knot Nematodes (*Meloidogyne* species):** Attack the roots of commercial crops

like tomato, eggplant, potato, and cotton. They induce giant cell formation and root galls (knots), disrupting water and nutrient transport, causing wilting, stunting, and reduced yield.

- **Cereal Cyst Nematode (*Heterodera avenae*):** Infests wheat and barley root systems, causing "Molya disease" in cereal crops.
- **Citrus Nematode (*Tylenchulus semipenetrans*):** Causes slow decline of citrus orchards by destroying feeder roots.
- **Lesion Nematodes (*Pratylenchus*):** Cause necrotic root lesions, facilitating secondary fungal and bacterial invasions.

3. Harmful Aspects: Human & Animal Parasites

- **Human Pathogens:**
 - *Ascaris lumbricoides* (Giant intestinal roundworm): Causes ascariasis, intestinal obstruction, and nutrient malabsorption.
 - *Ancylostoma duodenale* (Hookworm): Attaches to intestinal mucosa, consuming blood and causing severe iron-deficiency anemia.
 - *Wuchereria bancrofti* (Filarial worm): Blocks lymphatic vessels, causing permanent swelling of limbs known as **Elephantiasis** (Lymphatic Filariasis).
- **Veterinary Pathogens:** *Haemonchus contortus* (Barber's pole worm) infests the abomasum of sheep and goats, causing blood loss and mortality in livestock herds.

4. Beneficial Economic Aspects of Nematodes

- **Biological Control (Entomopathogenic Nematodes):** Genera like *Steinernema* and *Heterorhabditis* carry symbiotic lethal bacteria (*Xenorhabdus*). When released into soil, they invade agricultural insect pests (cutworms, white grubs), killing them within 48 hours without polluting the environment.
- **Nutrient Cycling & Soil Fertility:** Free-living bacterivorous and fungivorous soil nematodes feed on soil microbes, releasing excess plant-available nitrogen (ammonium) into the soil, thereby increasing soil fertility.
- **Model Organism in Scientific Research:** *Caenorhabditis elegans* is a non-parasitic, transparent nematode widely used globally in genetic research, developmental biology, neurobiology, and drug testing.

Question 6: Constituents of Zoonotic Interrelationship of Ticks

1. Introduction

Ticks (Order Ixodida) are obligate, blood-feeding ectoparasites of wild and domestic vertebrates. They serve as secondary vectors and primary reservoirs for a broad spectrum of zoonotic pathogens including spirochetes, rickettsiae, arboviruses, and protozoa.

2. Core Constituents of the Tick Zoonotic System

The persistence and circulation of a tick-borne zoonotic disease within a natural ecosystem depend on four interconnected constituents:

- **1. The Zoonotic Pathogen:** The infectious agent maintained within the cycle. Examples include *Borrelia burgdorferi* (bacterium causing Lyme disease), *Rickettsia rickettsii* (causes Rocky Mountain spotted fever), Kyasanur Forest Disease (KFD) virus, and *Babesia microti* (protozoan).
- **2. The Arthropod Vector (Hard and Soft Ticks):**
 - **Ixodidae (Hard Ticks):** Possess a scutum (dorsal shield). Genera such as *Ixodes*, *Dermacentor*, *Rhipicephalus*, and *Haemaphysalis* feed for extended periods and serve as principal vector constituents.
 - **Argasidae (Soft Ticks):** Lack a scutum (e.g., *Ornithodoros*), feeding rapidly in animal burrows and transmitting spirochetes.
- **3. Maintenance / Reservoir Hosts (Wild Animals):** Vertebrates that harbor the pathogen in their bloodstream for extended periods without dying, providing blood meals to ticks. Key reservoir constituents include wild rodents (white-footed mice), ground birds, squirrels, and small mammals.
- **4. Accidental / Terminal Host (Humans & Livestock):** Humans enter the zoonotic cycle accidentally when stepping into tick-infested habitats (forests, pastures). Humans are generally dead-end hosts because they do not transmit the pathogen back to uninfected ticks under normal conditions.

3. Internal Transmission & Maintenance Mechanisms

- **Transstadial Transmission:** The pathogen acquired by a larval tick during a blood meal survives through molting into the nymphal stage, and subsequently into the adult stage, remaining infective throughout its lifespan.
- **Transovarial Transmission:** Female ticks pass the pathogen directly through their ovaries into their eggs. The newly hatched larvae are born already infected, allowing the tick population to function as a self-sustaining pathogen reservoir independent of animal hosts.

4. Illustrative Zoonotic Cycles

- **Lyme Disease Cycle:** Wild rodents (reservoir) → *Ixodes scapularis* larva/nymph (vector) → Human (accidental host).
- **Kyasanur Forest Disease (KFD) Cycle:** Monkeys and wild rodents (reservoirs) → *Haemaphysalis spinigera* tick (vector) → Human (accidental host).

Question 7: *Dracunculus medinensis*

1. Systematic Position & Common Names

Dracunculus medinensis is an exceptionally long tissue-inhabiting nematode parasite. Commonly referred to as the **Guinea Worm** or **Fiery Serpent of Medina**, it is the causative agent of **dracunculiasis** (guinea worm disease).

- **Phylum:** Nematoda
- **Class:** Secernentea (Phasmida)

- **Order:** Spirurida
- **Family:** Dracunculidae
- **Genus:** *Dracunculus*
- **Species:** *medinensis*

2. Morphology & Marked Sexual Dimorphism

- **Female:** Exceptionally long and cylindrical, measuring 60 to 120 centimeters in length with a diameter of only 1 to 2 millimeters. The body is smooth and milky-white. The mature female is **viviparous**, her body occupied almost entirely by a gravid uterus packed with up to 3 million coiled, rhabditiform larvae.
- **Male:** Extremely small compared to the female, measuring only 12 to 40 millimeters in length. Males possess a curved tail with spicules and die shortly after copulation in deep connective tissues.

3. Digenetic Life Cycle

- **Definitive Host:** Humans.
- **Intermediate Host:** Freshwater micro-crustaceans belonging to the genus *Cyclops* (water flea).

Path of Transmission and Development:

- **Ingestion:** Infection occurs when a human drinks unfiltered surface water containing *Cyclops* infested with infective third-stage (L₃) larvae.
- **Migration:** Gastric juice in the human stomach digests the *Cyclops*, releasing L₃ larvae. The larvae penetrate the human duodenal wall, cross the retroperitoneal space, enter deep connective tissues, and undergo two molts to mature into adult worms over 10 to 12 months.
- **Mating:** After mating, the male dies. The gravid female migrates slowly through connective tissue pathways to the subcutaneous tissue of parts of the body exposed to water, most commonly the lower legs and feet.

4. Subcutaneous Blister Formation & Larval Release

- The gravid female secretes a toxic substance that causes a painful, burning **subcutaneous blister** on the skin.
- When the patient submerges the affected limb in cold water to relieve the burning sensation, the blister ruptures.
- The anterior end of the female worm emerges through the ulcer, her body contracting to expel thousands of uncoiled first-stage (L₁) larvae into the water.
- Free-swimming L₁ larvae are ingested by fresh *Cyclops*, molting twice inside their body cavity within 2 weeks to become infective L₃ larvae, completing the cycle.

5. Pathogenesis, Prevention, and Eradication

- **Pathology:** Causes localized intense inflammation, painful ulcers, secondary bacterial infections, joint stiffness, and permanent disability if the worm ruptures during manual

extraction.

- **Prevention & Control:** Because there is no vaccine or effective curative drug, eradication relies on public health strategies:
 - Filtering all drinking water through fine mesh cloth or nylon strainers to catch *Cyclops*.
 - Applying larvicides like Temephos (Abate) to drinking water sources.
 - Preventing individuals with active blisters from entering open water sources.

Question 8: Biological Control of Insect Pests

1. Concept & Definition

Biological control is the deliberate, eco-friendly strategy of managing pest populations using their natural living enemies—including predatory insects, parasitoids, pathogenic micro-organisms, and competitors—to suppress pest densities below economically damaging levels.

2. Primary Categories of Biological Agents

A. Predatory Insects

Predators are free-living organisms that search for, attack, and consume multiple prey individuals throughout their development:

- **Ladybird Beetles (*Coccinella septempunctata*):** Both adults and larvae feed heavily on soft-bodied agricultural pests like aphids, scale insects, and mites in mustard and vegetable fields.
- **Green Lacewings (*Chrysoperla carnea*):** Known as "aphid lions," their larvae voraciously consume whiteflies, thrips, and caterpillar eggs.
- **Praying Mantises:** Generalist foliage predators that capture grasshoppers, bugs, and caterpillars.

B. Parasitoid Insects

Parasitoids are specialized insects (primarily Hymenopterans and Dipterans) whose larvae develop on or inside a single host insect, eventually killing it:

- **Egg Parasitoids (*Trichogramma* species):** Tiny wasps that lay eggs inside the eggs of sugarcane stem borers, cotton bollworms, and paddy leaf folders. The emerging wasp larvae devour the egg contents, halting pest development.
- **Larval Parasitoids (*Bracon*, *Cotesia*):** Attack caterpillars of cabbage butterflies and bollworms by ovipositing into their body cavities.

C. Entomopathogenic Micro-organisms (Microbial Control)

- **Bacterial Pathogens (*Bacillus thuringiensis* / Bt):** During sporulation, Bt produces crystalline insecticidal protein delta-endotoxins. When ingested by susceptible lepidopteran caterpillars, alkaline gut fluids dissolve the crystals, releasing toxins that bind to midgut receptors, perforating the gut wall and killing the caterpillar.

- **Fungal Pathogens (*Beauveria bassiana*, *Metarhizium anisopliae*):** Fungal spores attach to the insect cuticle, germinate, penetrate the body wall, fill the hemolymph with hyphae, and kill the insect (muscardine disease).
- **Viral Pathogens (Baculoviruses - NPV and CPV):** Nuclear Polyhedrosis Viruses (NPV) are highly target-specific pathogens used to control *Helicoverpa armigera* (cotton bollworm) and *Spodoptera litura*.

3. Biological Control Techniques

- **Importation (Classical Control):** Introducing a natural enemy from the pest's native geographic region to control an invasive pest.
- **Augmentation:** Periodically releasing mass-reared natural enemies (e.g., releasing factory-reared *Trichogramma* egg cards into crop fields).
- **Conservation:** Modifying pesticide practices and preserving habitat borders to protect native predator and parasitoid populations.

4. Advantages over Chemical Control

- Selective target specificity with no toxic residues on food crops.
- Prevents target pests from rapidly developing pesticide resistance.
- Environmentally safe, preserving beneficial pollinators, wildlife, and water quality.

Question 9: Stored Grain Pests

1. Introduction

Stored grain pests are specialized insects belonging predominantly to the orders Coleoptera (beetles and weevils) and Lepidoptera (moths). They infest, consume, and degrade harvested food grains, cereal pulses, seeds, and processed flour during storage in godowns, silos, and traditional farm bins.

2. Classification Based on Feeding Nature

A. Primary Stored Grain Pests

These pests possess strong mouthparts capable of attacking and boring into whole, intact, undamaged sound food grains:

- **Rice Weevil (*Sitophilus oryzae*):** Order Coleoptera, Family Curculionidae. A small, reddish-brown beetle characterized by a distinct elongated snout (rostrum). The female bores a tiny hole into intact grains of rice, wheat, or maize, lays a single egg inside, and seals it with a gelatinous plug. The emerging legless grubs feed internally, reducing the grain to a hollow shell filled with frass.
- **Lesser Grain Borer (*Rhyzopertha dominica*):** Order Coleoptera, Family Bostrichidae. A cylindrical, dark brown beetle with its head tucked under a prominent turned-down pronotum. Both adults and larvae feed voraciously on whole grains, creating floury frass and large irregular exit holes.
- **Pulse Beetle (*Callosobruchus maculatus*):** Order Coleoptera, Family Bruchidae. The

primary pest of stored pulses (chana, moong, arhar, urad). Oval-bodied beetles lay smooth white eggs glued to the seed coat. Larvae bore directly into the seed, consuming the endosperm and cotyledons, leaving characteristic circular emergence holes ("windows").

- **Angoumois Grain Moth (*Sitotroga cerealella*):** Order Lepidoptera, Family Gelechiidae. Small buff-yellow moths whose caterpillars bore into standing field crops or stored grains, feeding exclusively inside individual kernels.

B. Secondary Stored Grain Pests

These pests cannot bore into intact sound grains; they feed on grains previously damaged by primary pests, broken kernels, or processed grain flour:

- **Red Flour Beetle (*Tribolium castaneum*):** Order Coleoptera, Family Tenebrionidae. Flat, reddish-brown beetles commonly found in wheat flour (atta), suji, and crushed cereal products. Their presence imparts a pinkish color, disagreeable pungent odor, and bitter taste due to quinone secretions.
- **Saw-Toothed Grain Beetle (*Oryzaephilus surinamensis*):** Named for six tooth-like projections on each side of the thorax; feeds on broken cereals, dried fruits, and oilseeds.

3. Nature of Damage & Economic Losses

- **Quantitative Weight Loss:** Direct consumption of grain endosperm and germ reduces total dry weight and germinability of seed stocks.
- **Qualitative Deterioration:** Contamination with insect body fragments, cast larval skins (exuviae), webbing, fecal frass, and uric acid deposits.
- **Moisture Accumulation & Heating:** Insect metabolic activity releases heat and water vapor, creating localized "hot spots" that trigger secondary growth of toxic fungi such as *Aspergillus flavus* (producing carcinogenic **aflatoxins**).

Question 10: Apiculture in India

1. Definition & Scope

Apiculture (or bee-keeping) is the scientific management and commercial rearing of honey bee colonies for the production of honey, beeswax, and associated hive products, as well as for providing cross-pollination services to agricultural crops.

2. Species of Honey Bees Found in India

A. Indigenous Species

- ***Apis dorsata* (Rock Bee):** Large, aggressive wild honey bee. Builds a single huge open-air comb on high tree branches, cliffs, and tall buildings. Yields large quantities of honey (30 to 50 kg per comb annually) but cannot be domesticated in artificial wooden hives due to its migratory nature.
- ***Apis indica* / *Apis cerana indica* (Indian Hive Bee):** Medium-sized, gentle, docile bee. Builds multiple parallel combs in dark cavities. Yields 3 to 5 kg of honey per colony

annually. Widely domesticated across rural India in movable wooden hives.

- ***Apis florea* (Little Bee):** Very small, docile bee that constructs tiny single combs in bushes; yields negligible honey (200 to 500 grams per comb).

B. Introduced Exotic Species

- ***Apis mellifera* (European / Italian Bee):** Introduced into India to boost commercial honey production. Possesses a prolific queen, low swarming tendency, gentle temperament, and high honey-gathering capacity (25 to 40 kg per hive annually). It forms the backbone of commercial apiculture in northern states like Punjab, Haryana, and Uttar Pradesh.

3. Social Organization of a Colony

A honey bee colony is a highly organized polymorphic social unit comprising three distinct castes:

- **The Queen:** Single fertile female per colony with developed ovaries; her sole function is laying eggs (up to 1,500 to 2,000 eggs per day) and secreting "queen substance" (pheromones) to maintain colony cohesion.
- **Drones:** A few hundred stout, stingless, fertile males developed parthenogenetically from unfertilized eggs (haploid); their sole role is mating with a young virgin queen during a nuptial flight.
- **Workers:** Thousands (20,000 to 80,000) of sterile females developed from fertilized eggs. Possessing specialized structural adaptations (pollen baskets/corbiculae on hind legs, wax glands on abdominal sternites, and a venomous sting), workers perform all hive duties including comb building, nursing larvae, cleaning, guarding, and foraging for nectar and pollen.

4. Modern Bee-Keeping Equipment & Methods

Modern apiculture utilizes scientific movable-frame wooden hives (**Langstroth Hive** or **Newton Hive**) operating on the principle of "Bee Space" (6.35 mm):

- **Hive Components:** Bottom board, brood chamber (where queen lays eggs), super chamber (where excess honey is stored), inner cover, and top cover.
- **Frames & Comb Foundation:** Wooden frames fitted with wax comb foundation sheets embossed with hexagonal cell bases.
- **Honey Extractor:** A centrifugal machine that extracts pure honey from honeycombs without damaging or destroying the wax comb structure, allowing empty combs to be reused by bees.

5. Economic Importance of Apiculture

- **Honey Production:** Honey is a carbohydrate-rich energy food containing fructose, glucose, mineral salts, and enzymes, used in Ayurvedic and modern medicine.
- **Beeswax:** Secreted by worker wax glands; used in cosmetics, cold creams, lipsticks, pharmaceutical ointments, shoe polishes, and candle manufacturing.
- **Royal Jelly & Propolis:** High-value bee products harvested for health supplements and anti-bacterial properties.

- **Crop Pollination:** Honey bees are major agricultural pollinators. Placing hives in mustard, sunflower, apple, litchi, pulse, and cucurbit fields enhances crop yields by 20% to 50%.

Question 11: Economic Importance of Birds

1. Introduction

Birds (Class Aves) play essential positive and negative economic roles across wild ecosystems, agricultural production, forestry, industrial applications, and human nutrition.

2. Beneficial Economic Aspects of Birds

A. Biological Control of Agricultural Pests

- **Insectivorous Birds:** Species such as the Black Drongo (*Dicrurus macrocercus*), Cattle Egret, Rosy Pastor, Myna, Swallows, and Swifts consume vast quantities of crop pests including caterpillars, grasshoppers, locusts, grubs, and beetles daily, acting as natural pest control agents.
- **Rodent Control (Raptors):** Nocturnal birds of prey like Owls (*Tyto alba*) and diurnal raptors like Kestrels and Eagles feed primarily on field mice, rats, and voles, preventing severe rodent-induced crop destruction.

B. Pollination of Plants (Ornithophily)

Nectar-feeding birds possess specialized long slender beaks and brush-tipped tongues adapted to feed on flower nectar. While feeding, they transfer pollen grains between flowers. Important pollinating birds include Sunbirds (*Nectarinia*), Hummingbirds, and Flowerpeckers, which pollinate economically valuable tree species such as *Bombax* (Silk cotton), *Erythrina*, and *Butea*.

C. Seed Dispersal & Afforestation

Frugivorous (fruit-eating) birds such as Hornbills, Barbets, Pigeons, Bulbuls, and Toucans consume wild fruits and berries. The indigestible seeds pass through their digestive tracts undamaged and are deposited in droppings across wide geographical distances. This process accelerates natural forest regeneration and tree propagation.

D. Scavenging & Sanitation

Vultures (*Gyps* species), Crows, and Kites act as ecological scavengers. By rapidly consuming livestock carcasses and decaying organic debris, they prevent the spread of infectious bacterial diseases like anthrax and keep natural environments clean.

E. Guano Industry

In dry coastal regions, massive colonies of marine birds (Gannets, Cormorants, Pelicans)

deposit accumulated droppings known as **Guano**. Guano is exceptionally rich in nitrogen, uric acid, and phosphorus, harvested commercially as an organic agricultural fertilizer.

F. Poultry & Feather Industry

- **Food Supply:** Domesticated poultry birds (Chickens, Ducks, Turkeys, Geese, Quails) supply human populations with animal protein through meat and eggs.
- **Down Feathers:** Feathers from geese and ducks are used for insulation in winter clothing, sleeping bags, and pillows.

3. Harmful Economic Aspects of Birds

- **Crop & Orchard Damage:** Frugivorous and granivorous birds such as Parakeets (*Psittacula*), House Sparrows, and Weaver birds feed on standing crops of paddy, wheat, maize, and ripening orchard fruits (mangoes, guava, grapes), causing significant post-harvest and field losses.
- **Vectors of Disease:** Wild birds serve as natural reservoir hosts for zoonotic viral pathogens including Avian Influenza (H5N1 virus), West Nile Virus, and Psittacosis (*Chlamydia psittaci*).
- **Aviation Hazards:** Bird strikes present severe safety risks and cause economic damage to commercial and military aircraft during takeoff and landing.

Question 12: Soil Protozoa

1. Introduction & Micro-Habitat

Soil protozoa are single-celled, microscopic, eukaryotic microorganisms that inhabit the thin water films (hydrophytic micro-environments) surrounding soil particles, organic debris, and plant root zones (rhizosphere). They constitute an abundant component of the living soil micro-fauna, with densities reaching 10^4 to 10^6 individuals per gram of fertile surface soil.

2. Major Taxonomic Groups of Soil Protozoa

Soil protozoa are divided into four main morphological groups based on their structures for locomotion:

- **Flagellates (Mastigophora):** Smallest soil protozoans ($5-20\ \mu\text{m}$ in size), possessing one or more long thread-like flagella (e.g., *Bodo*, *Cercomonas*, *Monas*). They dominate numerically in upper soil horizons and respond quickly to moisture increases.
- **Naked Amoebae (Sarcodina):** Protozoa lacking a rigid cell wall or shell, moving via flexible pseudopodia (e.g., *Acanthamoeba*, *Hartmannella*, *Naegleria*). They are well-adapted to crawling along soil particle surfaces and penetrating narrow soil pores.
- **Testate Amoebae (Arcellinida / Testacea):** Amoeboid protozoa enclosed within a hard protective shell or test constructed of secreted chitinous material or cemented silica mineral grains (e.g., *Arcella*, *Diffugia*, *Euglypha*). Their rigid tests persist in soil after death, serving as bio-indicators in ecological studies.
- **Ciliates (Ciliophora):** Largest soil protozoans ($30-200\ \mu\text{m}$ in size), covered with hair-like cilia used for rapid swimming and filter feeding (e.g., *Colpoda*, *Paramecium*,

Vorticella). They require thicker water films and are active mainly during periods of high soil moisture following rainfall or irrigation.

3. Adaptations to Soil Drying (Encystment)

Soil moisture levels fluctuate rapidly. To survive desiccation, freezing, and food scarcity, soil protozoa undergo **encystment**. The active feeding stage (trophozoite) rounds up, discards locomotory organelles, condenses its cytoplasm, and secretes a thick, resistant, multi-layered cyst wall composed of chitin and proteins. Soil cysts remain dormant for months or years until moisture returns, triggering **excystation** back into active trophozoites.

4. Ecological Functions of Soil Protozoa

- **Regulation of Bacterial Populations:** Protozoa are primary consumers of soil bacteria, consuming thousands of bacterial cells daily and maintaining bacterial communities in active growth phases.
- **Indication of Ecotoxicology:** Because soil protozoa react quickly to heavy metals, pesticides, and soil pollution, changes in their population structure serve as sensitive biological indicators of soil health and environmental degradation.

Question 13: Protozoa and Soil Fertility

1. Overview

Soil protozoa play a central role in maintaining and enhancing agricultural soil fertility. Their primary ecological influence operates through their trophic interactions with bacterial populations, driving nutrient cycling within the root zone (**rhizosphere**).

2. The Micro-Predation Process & The "Microbial Loop"

Protozoa are the principal predators of soil bacteria. Bacteria absorb inorganic mineral nutrients (nitrogen, phosphorus, potassium) from soil pore water and bind them tightly within their cellular biomass—a process known as **nutrient immobilization**. Without protozoan grazing, these essential plant nutrients would remain locked inside bacterial cells, unavailable to crop roots.

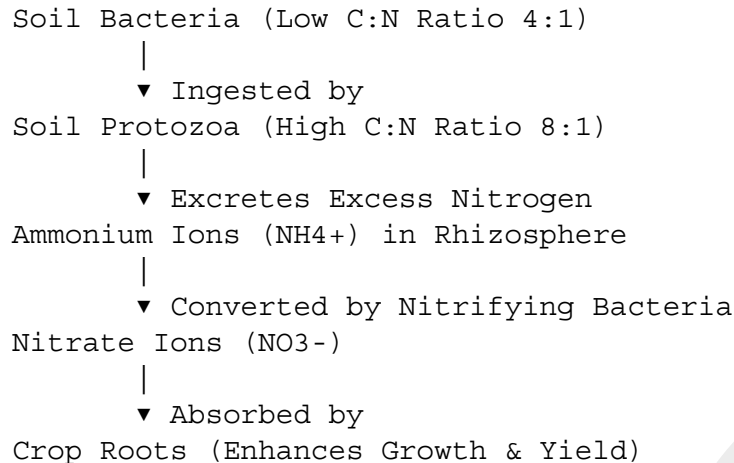
3. Mechanism of Nitrogen Mineralization

The fundamental biochemical mechanism by which protozoa enhance soil fertility is explained by differences in elemental Carbon to Nitrogen (C:N) ratios:

- Soil bacteria possess a low C:N ratio (approximately 3:1 to 5:1), meaning their cytoplasm contains high concentrations of nitrogen relative to carbon.
- Soil protozoa possess a higher C:N ratio (approximately 6:1 to 10:1).
- When a soil protozoan ingests bacteria to meet its carbon energy requirement, it consumes far more nitrogen than its body requires for cellular growth and structural maintenance.
- The protozoan excretes this excess, unneeded nitrogen into the surrounding soil root zone in the form of soluble **ammonium ions** (NH_4^+).

- Soil nitrifying bacteria quickly convert ammonium into **nitrate ions (NO_3^-)**, which are readily absorbed by plant roots to synthesize plant proteins and nucleic acids.

4. Text-Based Flow of Nitrogen Mineralization



5. Stimulation of Bacterial Activity & Organic Matter Decomposition

Un-grazed bacterial colonies quickly enter a stagnant stationary phase due to space and resource competition. By continuously consuming older bacterial cells, protozoan predation keeps the bacterial population in a state of rapid exponential growth. This accelerated metabolic state increases the overall rate of organic matter decomposition, humus formation, and soil mineral recycling.

6. Improvement of Soil Structure

The movement of motile protozoan trophozoites through soil pore spaces, combined with their sticky mucous secretions, binds fine clay particles into stable soil micro-aggregates. This structural aggregation increases soil porosity, water-holding capacity, root penetration depth, and soil aeration.

Question 14: *Plasmodium vivax*

1. Systematic Position & Introduction

Plasmodium vivax is a digenetic sporozoan endoparasite inhabiting human liver cells and red blood cells (RBCs). It is the causative agent of **Benign Tertian Malaria**, characterized by recurrent fever paroxysms occurring every 48 hours.

- **Phylum:** Apicomplexa
- **Class:** Sporozoa
- **Order:** Haemosporida
- **Genus:** *Plasmodium*
- **Species:** *vivax*

2. Life Cycle (Digenetic)

Plasmodium vivax requires two distinct hosts to complete its life cycle:

- **Primary / Definitive Host:** Female *Anopheles* mosquito (where sexual reproduction and sporogony take place).
- **Secondary / Intermediate Host: Human** (where asexual multiplication and schizogony take place).

3. Asexual Phase in Human (Schizogony)

A. Pre-Erythrocytic & Exo-Erythrocytic Schizogony (Liver Phase)

- **Inoculation:** Infection begins when an infected female *Anopheles* bites a human, injecting microscopic, spindle-shaped **sporozoites** present in its salivary glands into the bloodstream.
- **Liver Invasion:** Sporozoites leave the bloodstream within 30 minutes and penetrate liver parenchymal cells (hepatocytes).
- **Schizont Formation:** The sporozoite feeds on liver cytoplasm, grows into a large round **schizont**, and undergoes multiple nuclear divisions (schizogony) to produce thousands of tiny **cryptozoites** (exo-erythrocytic merozoites). The liver cell ruptures, releasing cryptozoites into the bloodstream.
- **Hypnozoites & Malarial Relapse:** Some sporozoites entering liver cells do not divide immediately; instead, they enter a dormant resting stage called **hypnozoites**. Months or years later, hypnozoites reactivate, initiating secondary exo-erythrocytic schizogony and causing **relapse** of malarial fever.

B. Erythrocytic Schizogony (Red Blood Cell Phase)

Cryptozoites penetrate human Red Blood Cells (RBCs) to initiate a repeating 48-hour developmental cycle:

- **Ring Stage:** The merozoite feeds on hemoglobin, forming a central vacuole that pushes the nucleus to one side, resembling a **signet ring**.
- **Amoeboid Stage:** The parasite enlarges, develops pseudopodia, and displays active amoeboid movement. The host RBC expands, pales, and develops diagnostic yellowish-pink stippling known as **Schüffner's dots**. Hemoglobin digestion leaves an insoluble toxic iron-pigment byproduct called **haemozoin**.
- **Erythrocytic Schizont:** The amoeboid stage undergoes multiple nuclear divisions to form an erythrocytic schizont containing 12 to 24 erythrocytic merozoites arranged in a rosette pattern around central haemozoin.
- **Rupture & Paroxysm:** The infected RBC bursts, releasing merozoites along with toxic haemozoin into the blood plasma. Haemozoin acts as a pyrogen, triggering intense **malarial paroxysms** consisting of sequential cold, hot, and sweating stages every 48 hours.

C. Gametogony (Formation of Sexual Stages)

After several schizogonic generations, some merozoites invading RBCs do not form schizonts;

instead, they differentiate into sexual precursor cells called **gametocytes**:

- **Microgametocyte**: Small male gametocyte with a large diffuse nucleus.
- **Macrogametocyte**: Large female gametocyte with a compact dense nucleus. These gametocytes remain inside human RBCs until ingested by a mosquito.

4. Sexual Phase in Mosquito (Gametogony & Sporogony)

- **Exflagellation & Fertilization**: When a female *Anopheles* ingests blood containing gametocytes, the blood meal enters its stomach. The microgametocyte undergoes exflagellation to form 6 to 8 thread-like male gametes. The macrogametocyte matures into a female macrogamete. Fusion of male and female gametes (syngamy) produces a diploid **Zygote**.
- **Ookinete & Oocyst Formation**: The motionless spherical zygote transforms into an elongated, motile, worm-like **Ookinete**. The ookinete pierces the muscular stomach wall of the mosquito, settles beneath the outer epithelial membrane, secretes a protective cyst wall, and transforms into a spherical **Oocyst**.
- **Sporogony**: The oocyst absorbs nutrients and undergoes repeated mitotic divisions (sporogony) to produce over 10,000 sickle-shaped **sporozoites**. The mature oocyst ruptures into the hemolymph, and sporozoites migrate through the body cavity to lodge inside the mosquito's **salivary glands**, ready to infect a new human host during its next blood meal.

5. Pathology, Prevention, and Treatment

- **Pathology**: Severe anemia (due to RBC destruction), enlargement of the spleen (splenomegaly), hepatomegaly, and high fever spikes.
- **Prevention**: Destruction of mosquito vectors using larvicides, biological control (*Gambusia* fish), and indoor residual spraying (IRS); personal protection using Insecticide-Treated Nets (ITNs).
- **Treatment**: Oral **Chloroquine** (to eliminate blood schizont stages) combined with **Primaquine** (to eradicate dormant liver hypnozoites and prevent relapse).

Question 15: Prevention and Control of Encephalitis

1. Introduction & Overview

Encephalitis refers to acute inflammatory damage to the brain parenchyma, primarily caused by neurotropic viral pathogens. In India, **Japanese Encephalitis (JE)**—caused by an arbovirus belonging to the family *Flaviviridae*—represents a major public health challenge, particularly affecting children across agricultural belts.

2. Transmission Biology & Reservoir Hosts

JE virus circulates within a complex zoonotic transmission cycle:

- **Vector**: Mosquitoes belonging to the ***Culex vishnui* group** (primarily *Culex tritaeniorhynchus*), which breed extensively in flooded rice paddies, irrigation ditches, and marshlands.

- **Amplifying Hosts:** Pigs and wading water birds (egrets, herons). Pigs develop high titers of viremia without showing symptoms, providing a rich viral source for feeding *Culex* mosquitoes.
- **Incidental Host:** Humans (dead-end hosts).

3. Integrated Vector Management (IVM)

A. Larval Mosquito Control

- **Chemical Control:** Application of organophosphate larvicides such as Temephos (Abate) or oil films over stagnant water surfaces in paddy fields and ditches.
- **Microbial Control:** Application of *Bacillus thuringiensis israelensis* (Bti) spores into standing water to kill *Culex* mosquito larvae.
- **Biological Control:** Introducing larvicidal surface fishes like **Mosquitofish (*Gambusia affinis*)** and Guppy (*Poecilia reticulata*) into paddy field channels, ponds, and drainage ditches.

B. Adult Mosquito Control

- **Space Spraying:** Thermal fogging or Ultra-Low Volume (ULV) space spraying using Malathion or Pyrethrum insecticides around residential zones during disease outbreaks.
- **Indoor Residual Spraying (IRS):** Spraying synthetic pyrethroids on interior walls of houses and cattle sheds to kill resting adult mosquitoes.

4. Amplifying Host Management & Segregation

- **Pigsty Isolation:** Locating pigpens and livestock shelters at least 4 to 5 kilometers away from human residential habitats to prevent infected mosquitoes from feeding on humans.
- **Physical Barrier Screening:** Installing fine wire-mesh netting around pig housing structures to exclude *Culex* mosquitoes.

5. Active Immunization (Vaccination)

Vaccination is the most effective intervention for long-term population protection in endemic regions:

- **SA 14-14-2 Live Attenuated Vaccine:** Single-dose primary vaccination administered under Universal Immunization Programs for children aged 9 months to 15 years, followed by a booster dose.
- **Inactivated & Recombinant JE Vaccines:** Administered to high-risk laboratory workers and travelers.

6. Personal Protective Measures

- Using Long-Lasting Insecticidal Nets (LLINs) or Insecticide-Treated Nets (ITNs) while sleeping.
- Applying topical chemical mosquito repellents containing DEET (N,N-diethyl-meta-toluamide), Picaridin, or Oil of Lemon Eucalyptus on exposed skin during twilight hours.

- Wearing light-colored, full-sleeved shirts and long trousers to minimize exposed skin surfaces.

Question 16: Management Aspects of Shellfish

1. Definition & Commercial Scope

Shellfish management encompasses the scientific culture, habitat maintenance, water quality control, disease prevention, post-harvest processing, and sustainable harvest of economically valuable aquatic invertebrates. Commercial shellfish fall into two distinct taxonomic groups:

- **Crustacean Shellfish:** Prawns, shrimps, crabs, and lobsters.
- **Molluscan Shellfish:** Oysters, mussels, clams, and scallops.

2. Management of Crustacean Shellfish Culture (Prawn/Shrimp Farming)

A. Site Selection & Pond Construction

- Farm sites should be located in sheltered estuarine or coastal intertidal zones with access to clean brackish water.
- Ponds must feature clay-loam soil bottom textures that resist water seepage and maintain neutral-to-slightly alkaline pH levels (7.5-8.5).

B. Broodstock & Hatchery Management

- Sourcing certified **Specific Pathogen Free (SPF)** post-larvae (PL) of species like the Pacific White Shrimp (*Litopenaeus vannamei*) or Giant Tiger Prawn (*Penaeus monodon*).
- Maintaining strict biosecurity protocols in hatcheries to prevent viral contamination.

C. Water Quality & Feed Management

- **Physico-Chemical Monitoring:** Maintaining salinity (15-25 ppt), water temperature (28-32°C), and high dissolved oxygen levels (>5.0 mg/L) using mechanical paddle-wheel aerators.
- **Nutritional Management:** Feeding formulated nutritionally balanced pelleted feed containing 35% to 40% crude protein, adjusted based on check-tray feeding assessments to avoid overfeeding and water pollution.

3. Management of Molluscan Shellfish Culture

A. Culture Systems

- **Raft & Long-Line Culture:** Used for green mussels (*Perna viridis*) and edible oysters (*Crassostrea madrasensis*). Seed-bearing ropes or mesh socks are suspended from floating wooden rafts into nutrient-rich estuarine waters, maximizing natural plankton feeding.
- **On-Bottom Rack Culture:** Rearing bivalves on raised bamboo racks or bottom tiles in

shallow intertidal zones.

B. Depuration (Sanitary Processing)

- Harvested live bivalve molluscs filter large volumes of water and can concentrate pathogens, heavy metals, and marine toxins.
- **Depuration Process:** Harvested oysters and mussels are placed inside flow-through depuration tanks containing clean, UV-sterilized seawater for 24 to 48 hours. This allows them to purge ingested bacteria, sand, and digestive impurities prior to commercial marketing.

4. Disease Biosecurity & Sustainability

- Monitoring and preventing major viral diseases such as White Spot Syndrome Virus (WSSV) in shrimp aquaculture using probiotics instead of antibiotics.
- Implementing seasonal harvest closures and enforcing minimum mesh size regulations in wild capture fisheries to prevent over-exploitation of natural shellfish stocks.

Question 17: Economic Importance of Porifera and Mollusca

1. Economic Importance of Porifera (Sponges)

A. Commercial Bath Sponge Industry

- The horny fibrous skeleton of certain marine sponges—composed entirely of flexible **spongin fibers** without hard siliceous spicules—is harvested commercially for human use.
- Key commercial species include *Euspongia officinalis* (the common bath sponge) and *Hippospongia* (the wool sponge).
- **Industrial Uses:** Due to their water-holding capacity, durability, and soft texture, cleaned sponge skeletons are used for bathing, surgical swabbing, helmet lining, liquid filtration, and fine polishing of optical lenses and metals.

B. Pharmaceutical & Biomedical Products

- Marine sponges produce unique secondary bioactive metabolites with defensive properties against predators and bacterial colonization.
- **Cytarabine (Ara-C):** An anti-cancer drug used in treating leukemia, derived from nucleosides isolated from the Caribbean sponge *Cryptotethya crypta*.
- **Vidarabine (Ara-A):** An antiviral drug synthesized based on natural sponge-derived nucleoside structures.

C. Ornamental & Scientific Values

- Dried skeletons of *Euplectella aspergillum* ("Venus' Flower Basket") enclose a pair of small commensal deep-sea shrimp (*Spongicola*) within their glass spicule lattice. Dried

specimens are presented as traditional wedding gifts in Japan, symbolizing lifelong companionship.

D. Harmful Aspects of Sponges

- **Boring Sponges (*Cliona* species):** *Cliona celata* bores complex tunnels into calcium carbonate shells of commercial pearl oysters, edible oysters, and coral reefs, causing structural weakening, shell erosion, and mortality in oyster beds.

2. Economic Importance of Mollusca

A. Food Source (Edible Molluscs)

Molluscs represent a high-protein global seafood resource:

- **Bivalves:** Edible oysters (*Crassostrea*), green mussels (*Perna viridis*), clams (*Meretrix*), and scallops.
- **Cephalopods:** Squids (*Loligo*), Octopuses, and Cuttlefishes (*Sepia*), harvested extensively in capture fisheries for human food and frozen exports.
- **Gastropods:** Land snails (*Helix pomatia*) and sea ears (*Abalone*).

B. Pearl Industry

- Pearl oysters—primarily *Pinctada fucata* and *Pinctada maxima*—produce highly valuable natural and cultured pearls when an irritant lodges inside their mantle cavity, stimulating continuous nacre secretion.

C. Industrial Shell Products

- Shells of cowries (*Cypraea*), conchs (*Turbinella pyrum* / Shankha), scallops, and freshwater mussels are processed into buttons, decorative handicrafts, cameos, inlay works, and traditional trumpets.
- Ground mollusc shells yield calcium-rich quicklime used in cement manufacturing, soil acid neutralization, and poultry feed supplements.

D. Harmful Aspects of Molluscs

- **Shipworms (*Teredo navalis*):** Specialized wood-boring bivalves that excavate tunnels into submerged wooden ships, harbor pilings, and wooden docks, causing severe economic damage.
- **Agricultural Pests:** Giant African Land Snails (*Achatina fulica*) and garden slugs devour foliage, vegetable crops, and plantation leaves.
- **Intermediate Vector Hosts:** Freshwater snails like *Lymnaea* and *Planorbis* act as mandatory intermediate hosts for human and veterinary trematode parasites such as *Fasciola hepatica* (liver fluke) and *Schistosoma* (blood fluke).

Question 18: Pearl Culture

1. Introduction & Principle

Pearl culture is the artificial, scientific technique of rearing pearl oysters and surgically inserting an external foreign irritant (nucleus/shell bead) along with a donor mantle tissue graft into the live oyster. This stimulates the oyster's mantle epithelium to secrete consecutive concentric layers of **nacre** (mother-of-pearl) around the irritant, ultimately forming a high-value cultured pearl.

2. Major Pearl Oyster Species

- **Marine Pearl Oysters:** *Pinctada fucata* (the principal Indian pearl oyster), *Pinctada maxima* (gold-lip oyster), and *Pinctada margaritifera* (black-lip oyster).
- **Freshwater Pearl Mussels:** *Lamellidens marginalis* and *Parreysia corrugata*.

3. Mechanism of Pearl Secretion

The body wall of a pearl oyster is lined by the **mantle**. The outer epithelial layer of the mantle secretes nacre, which consists of crystalline **aragonite** (CaCO_3) bound within an organic protein matrix called **conchiolin**. When a foreign particle enters beneath the mantle:

- The mantle epithelium invaginates around the irritant, forming a closed **pearl sac**.
- The cells of the pearl sac continuously secrete thin concentric layers of nacre onto the irritant.
- Over 1 to 3 years, these iridescent nacreous layers harden to produce a cultured pearl.

4. Step-by-Step Procedure of Cultured Pearl Production

A. Collection & Conditioning of Oysters

Healthy pearl oysters are collected from natural sea beds or produced via hatchery technology. They are placed in mesh cages suspended in seawater rafts for 2 to 3 weeks to recover and condition prior to surgery.

B. Graft Preparation (Saibo Preparation)

A donor oyster is sacrificed to obtain a strip of fresh, living mantle tissue. The mantle strip is cut into tiny $2 \times 2 \text{ mm}$ square pieces called **graft tissues** or **saibo**.

C. Surgical Grafting Operation

The recipient oyster is placed in a wooden clamp, and its shell valves are opened gently using a peg:

- A delicate incision is made into the gonad or visceral mass of the recipient oyster.
- Using specialized surgical needles, one piece of donor graft tissue (saibo) is inserted into the incision site.
- A spherical polished shell bead nucleus (usually made from freshwater mussel shell, $3\text{--}6 \text{ mm}$ in diameter) is placed in direct contact with the outer epithelial side of the graft tissue.

D. Post-Operative Care & Rearing

Operated oysters are kept in indoor recovery tanks for 7 to 10 days under antibiotic treatment to heal surgical wounds. Healthy surviving oysters are transferred to raft-suspended mesh cages in calm sea bays. They are reared for 18 to 24 months, during which mantle epithelial cells grow around the shell bead to form a pearl sac that secretes nacre continuously.

E. Harvesting & Processing

Oysters are brought ashore, opened surgically, and the cultured pearls are extracted. Harvested pearls undergo washing in neutral detergent, sorting by shape, size, and color, chemical bleaching, and final polishing.

Question 19: Mites

1. Systematic Position & General Characteristics

Mites are minute arthropods belonging to the Class **Arachnida** and Order **Acarina**. Unlike insects, their body shows no external division into a separate head, thorax, and abdomen; instead, the entire body is fused into a single unsegmented structure called the **idiosoma** (with the anterior mouthpart region termed the **gnathosoma**). Adults possess **four pairs of legs** (larvae have three pairs) and lack antennae and true compound eyes.

2. Medically & Veterinary Important Mites

A. The Human Itch Mite (*Sarcoptes scabiei*)

- **Pathology:** Causes **Scabies**, an intensely itchy skin disease.
- **Mechanism:** Fertilized female mites burrow into the outer epidermal layer (stratum corneum) of human skin, creating tortuous subcutaneous tunnels. The female lays eggs inside these tunnels, excreting fecal pellets.
- **Symptoms:** Severe itching (especially at night) and allergic skin papules around finger webs, wrists, elbows, and waistline. Transmission occurs via direct skin-to-skin contact or contaminated bedding.

B. House Dust Mites (*Dermatophagoides pteronyssinus*)

- **Pathology:** Primary indoor triggers for allergic respiratory illnesses.
- **Mechanism:** Dust mites inhabit household mattresses, carpets, pillows, and upholstered furniture, feeding on shed human skin scales.
- **Symptoms:** Powerful digestive enzymes (Der p 1 allergen) present in their microscopic fecal pellets become airborne, triggering **bronchial asthma**, perennial allergic rhinitis, and eczema in sensitive individuals.

C. Scrub Typhus Vectors (*Chiggers* / *Leptotrombidium*)

- Parasitic larval stages (chiggers) of trombiculid mites bite humans in scrub vegetation zones, transmitting the intracellular bacterium *Orientia tsutsugamushi* (causative agent of **Scrub Typhus**).

3. Agriculturally Harmful Mites (Phytophagous Mites)

- **Spider Mites (*Tetranychus urticae*)**: Infest commercial crops, cotton, vegetables, and fruit orchards. They puncture plant leaf cells to suck cell sap, construct fine silken webbing over leaves, cause leaf bronzing, chlorosis, and severe yield reduction.
- **Eriophyid Mites (*Aceria guerratonis*)**: The coconut mite, which feeds on young coconut floral button tissues, causing fruit scarring, splitting, and premature nut drop.

4. Control Measures

- **Medical Treatment**: Application of topical scabicides such as 5% Permethrin lotion or Gamma Benzene Hexachloride over the body for scabies.
- **Agricultural Control**: Spraying selective **acaricides** (e.g., Dicofol, Fenazaquin, or Wettable Sulphur dusts) to manage plant mite populations.

Question 20: Harmful Insects as Carriers of Disease of Human Beings

1. Introduction

Many insect species act as disease vectors, transferring pathogenic micro-organisms (viruses, bacteria, protozoa, and helminths) to human hosts. Transmission occurs via two primary biological mechanisms:

- **Mechanical Transmission**: The insect acts as a passive carrier, transferring pathogens on its feet, mouthparts, or body hairs without the pathogen undergoing multiplication inside the insect.
- **Biological Transmission**: The pathogen enters the insect body, undergoes mandatory multiplication, developmental changes, or sexual reproduction before becoming infective to humans.

2. Major Insect Carriers & Human Diseases

A. Dipterans: Mosquitoes (Family Culicidae)

- ***Anopheles stephensi* & *Anopheles culicifacies***: Biological vectors of *Plasmodium* protozoa, transmitting **Malaria** via female bites.
- ***Culex quinquefasciatus***: Biological vector of the filarial nematode *Wuchereria bancrofti*, transmitting **Lymphatic Filariasis (Elephantiasis)**.
- ***Aedes aegypti***: Biological vector of arboviruses, transmitting **Dengue**, **Chikungunya**, **Yellow Fever**, and **Zika Virus**.

B. Dipterans: Flies (Flies & Sandflies)

- **Housefly (*Musca domestica*)**: Important mechanical vector. Houseflies feed on human excrement, sewage, and exposed human food. Pathogenic bacteria and cysts stick to their hairy legs, sticky foot pads (pulvilli), and vomit drops, contaminating food and transmitting **Cholera** (*Vibrio cholerae*), **Typhoid** (*Salmonella typhi*), **Amoebic Dysentery**, and **Bacillary Dysentery**.
- **Sandfly (*Phlebotomus argentipes*)**: Biological vector of the protozoan *Leishmania donovani*, transmitting **Kala-azar** (Visceral Leishmaniasis) through blood meals.
- **Tsetse Fly (*Glossina palpalis*)**: Biological vector of *Trypanosoma brucei*, transmitting **African Sleeping Sickness**.

C. Siphonapterans: Fleas

- **Oriental Rat Flea (*Xenopsylla cheopis*)**: Biological vector of the bacterium *Yersinia pestis*, transmitting **Bubonic Plague** from infected rodents to humans when flea gut blockage forces regurgitated bacteria into bite wounds.

D. Anoplurans: Sucking Lice

- **Human Body Louse (*Pediculus humanus humanus*)**: Transmits Rickettsial pathogens causing **Epidemic Typhus** (*Rickettsia prowazekii*) and Trench Fever when louse feces are rubbed into skin abrasion wounds.

3. Integrated Vector Management (IVM)

Managing insect vectors requires combining habitat elimination, chemical larvicides, indoor insecticide sprays, biological control agents (like *Gambusia* fish), and personal protective barriers (mosquito nets and repellents).

Question 21: *Diphyllobothrium latum*

1. Systematic Position & Common Name

Diphyllobothrium latum is an exceptionally long intestinal parasite. Commonly known as the **Fish Tapeworm** or **Broad Tapeworm**, it is the largest tapeworm infecting humans, causing diphyllobothriasis.

- **Phylum**: Platyhelminthes
- **Class**: Cestoda
- **Order**: Pseudophyllidea
- **Genus**: *Diphyllobothrium*
- **Species**: *latum*

2. Morphology

- **Adult Worm**: Measures 10 to 15 meters in length and consists of up to 3,000 to 4,000

proglottids.

- **Scolex (Head):** Small, elongated, almond-shaped organ lacking hooks or circular suckers. Instead, it possesses two deep, longitudinal, suction grooves called **bothria** used to adhere to the human intestinal mucosa.
- **Proglottids:** Mature segments are distinctively **broader than long** (hence "broad tapeworm"). Each mature proglottid contains a central rosette-shaped convoluted uterus with a uterine pore that continuously discharges unembryonated eggs into the host stool.

3. Digenetic Life Cycle (Requiring Three Hosts)

- **Definitive Host:** Humans, dogs, cats, and fish-eating mammals.
- **First Intermediate Host:** Freshwater micro-crustaceans (Copepods: **Cyclops** or *Diaptomus*).
- **Second Intermediate Host:** Freshwater predatory fish (Pike, Perch, Trout, Salmon).

Path of Transmission and Larval Stages:

- **Coracidium Stage:** Eggs passed in human stool hatch in freshwater into spherical, ciliated, free-swimming embryos called **coracidia**.
- **Proceroid Stage:** A *Cyclops* ingests the coracidium. Inside the *Cyclops* body cavity, the parasite develops into an elongated **Proceroid larva**.
- **Plerocercoid Stage (Sparganum):** When a freshwater fish eats the infected *Cyclops*, the proceroid penetrates the fish intestine, migrates into its muscle tissue, and transforms into an infective, unsegmented, solid-bodied **Plerocercoid larva** (Sparganum).
- **Human Infection:** Humans acquire infection by eating raw, inadequately cooked, under-salted, or smoked freshwater fish containing plerocercoid larvae. The larva attaches to the ileal mucosa using its bothria and grows into a mature egg-laying tapeworm within 5 to 6 weeks.

4. Pathogenesis & Pernicious Anemia

- **Intestinal Symptoms:** Most infections cause mild abdominal discomfort, diarrhea, flatulence, and weakness.
- **Vitamin 12 Deficiency:** *Diphyllobothrium latum* absorbs large amounts of host Vitamin 12 across its body tegument, competing directly with host intestinal absorption. Long-standing heavy infections lead to severe Vitamin 12 deficiency, causing **Pernicious Megaloblastic Anemia** characterized by reduced RBC counts and neurological symptoms.

5. Diagnosis & Treatment

- **Diagnosis:** Microscopic identification of operative, unembryonated eggs showing a terminal knob in stool samples.
- **Treatment:** Oral administration of single-dose **Praziquantel** or **Niclosamide** to detach and expel the adult tapeworm.

Question 22: Aquaculture

1. Definition & Basic Concept

Aquaculture is the rational, scientific cultivation, rearing, management, and harvesting of economically valuable aquatic organisms—including finfishes, crustaceans, molluscs, and aquatic plants—under controlled or semi-controlled freshwater, brackishwater, or marine environments.

2. Major Branches of Aquaculture

- **Pisciculture (Fish Farming):** Rearing of edible finfishes (e.g., Major Carps, Catfishes, Tilapia).
- **Crustacean Aquaculture:** Rearing of prawns, marine shrimps (*Penaeus*), and mud crabs.
- **Molluscan Aquaculture:** Rearing of edible oysters, green mussels, clams, and pearl oysters.
- **Mariculture:** Cultivation of marine organisms in open oceans, enclosed sea bays, or coastal waters (e.g., Seaweed farming of *Gracilaria*, sea bass culture).
- **Brackishwater Aquaculture:** Farming in estuarine backwaters and mangroves where seawater mixes with freshwater (e.g., Tiger shrimp farming).

3. Composite Fish Culture System (Polyculture)

Composite fish culture is an advanced pisciculture strategy where six compatible, fast-growing, non-competing edible carp species (three indigenous Indian Major Carps and three exotic carps) are cultured together in a single fish pond. By selecting species that feed at different water depths and consume different natural foods, ecological niches are fully utilized to produce high fish yields per unit area:

- **Surface Feeders:**
 - **Catla (*Catla catla*):** Indian Major Carp; feeds on surface zooplankton.
 - **Silver Carp (*Hypophthalmichthys molitrix*):** Exotic Carp; feeds on surface phytoplankton.
- **Column Feeders:**
 - **Rohu (*Labeo rohita*):** Indian Major Carp; feeds on filamentous algae and decaying plants in middle water layers.
- **Bottom Feeders:**
 - **Mrigal (*Cirrhinus mrigala*):** Indian Major Carp; feeds on bottom detritus, mud, and decaying organic matter.
 - **Common Carp (*Cyprinus carpio*):** Exotic Carp; omnivorous bottom feeder consuming benthic fauna and detritus.
- **Vegetation Feeders:**
 - **Grass Carp (*Ctenopharyngodon idella*):** Exotic Carp; feeds voraciously on aquatic weeds and macro-vegetation, keeping the pond clean.

4. Key Management Phases in Aquaculture

- **Nursery Ponds:** Rearing delicate hatchlings into fry (1-2 cm) over 15 to 20 days.
- **Rearing Ponds:** Growing fry into fingerlings (5-10cm) over 2 to 3 months.
- **Stocking Ponds:** Final grow-out ponds where fingerlings are fed formulated supplementary diets until reaching marketable table size (1-2 kg).
- **Water Quality Maintenance:** Regular monitoring of Dissolved Oxygen (DO), pH, alkalinity, and liming to maintain pond productivity.

5. Economic Importance of Aquaculture

- Enhances food security by providing accessible animal protein rich in essential amino acids and omega-3 fatty acids.
- Generates rural employment opportunities in hatchery operations, feed manufacturing, processing, and seafood export markets.

Question 23: Cattle Tick

1. Systematic Position & Identification

The cattle tick, *Boophilus microplus* (also classified as *Rhipicephalus microplus*), is an economically destructive hard tick belonging to the Class Arachnida, Order Ixodida, and Family Ixodidae. It acts as an obligate, blood-sucking ectoparasite of cattle, buffaloes, and domestic livestock across tropical and subtropical regions.

2. Morphology

- Adult ticks possess a flattened, oval body covered dorsally by a chitinous shield called the **scutum** (which covers the entire dorsal surface in males, but is restricted to the anterior region in females to allow massive abdominal engorgement during blood feeding).
- Possess four pairs of jointed legs ending in claws adapted to grip animal hair.
- Mouthparts (**capitulum**) include a barbed median **hypostome** inserted into host skin to anchor the tick and draw blood.

3. Life Cycle Dynamics (One-Host Tick)

Unlike many ixodid ticks that require two or three host species to complete their development, *Boophilus microplus* is a strict **one-host tick**:

- **Larval Attachment:** Newly hatched six-legged larvae ("seed ticks") climb onto pasture vegetation, transfer onto a passing cattle host, attach to soft skin zones (flanks, udder, dewlap), and begin blood feeding.
- **On-Host Development:** The larva feeds, engorges, and molts directly into an eight-legged nymph on the same individual animal host. The nymph feeds and molts into an adult male or female tick without dropping off the animal.
- **Mating & Engorgement:** Adult males move across the host skin to fertilize feeding females. The fertilized female feeds continuously for 7 to 14 days, engorging heavily with

host blood until her body swells to several times its original size.

- **Egg Laying:** The fully engorged female drops off the cattle host onto surrounding pasture soil, crawls into sheltered soil crevices, lays 2,000 to 4,000 eggs over several days, and dies. Eggs hatch into larvae on the pasture to repeat the cycle.

4. Pathogenicity & Economic Veterinary Losses

- **Direct Damage:** Heavy tick infestations cause progressive **anemia**, weight loss, stunted growth, reduced milk yield (dropping by 20% to 30%), and hide damage that lowers leather quality.
- **Vector Transmission of Cattle Diseases:** *Boophilus microplus* is the primary biological vector for devastating intra-erythrocytic livestock pathogens:
 - ***Babesia bigemina* & *Babesia bovis*:** Transmitted via tick bites, causing **Bovine Babesiosis (Redwater Fever)**, characterized by high fever, severe intravascular hemolysis, hemoglobinuria (red urine), and mortality.
 - ***Anaplasma marginale*:** Causes Bovine Anaplasmosis (Gall Sickness).

5. Control Measures

- **Chemical Control (Acaricides):** Periodically dipping or spraying livestock with synthetic pyrethroids (Cypermethrin), organophosphates, or Amitraz.
- **Pasture Rotation:** Spell pastures for 3 to 4 months to starve unattached larvae present in the field.
- **Biological Control & Breeding:** Utilizing entomopathogenic fungi (*Metarhizium anisopliae*) and breeding tick-resistant cattle varieties (crossbreeding *Bos indicus* breeds like Zebu with European cattle).

Question 24: Diseases of Silkworm

1. Introduction

The mulberry silkworm, *Bombyx mori*, is vulnerable to various infectious diseases caused by microsporidian protozoans, viruses, bacteria, and fungi. Disease outbreaks in rearing houses present serious financial risks to the sericulture industry.

2. Detailed Study of Major Silkworm Diseases

A. Pebrine (Protozoan Disease)

- **Causative Agent:** A microsporidian fungal-protozoan parasite named ***Nosema bombycis***.
- **Symptoms:** Infected caterpillars show sluggish movement, loss of appetite, irregular growth, and fail to molt properly. A diagnostic sign is the appearance of dark, pepper-like black spots across the larval skin.
- **Transmission:**
 - **Horizontal Transmission:** Healthy silkworms ingest mulberry leaves contaminated with *Nosema* spores excreted by infected larvae.

- **Vertical (Transovarial) Transmission:** The parasite invades the ovaries of the female moth, passing inside the egg chorion so that newly hatched larvae are born infected.
- **Control:** Examining mother moths microscopically after egg laying (Fujiwara moth examination method). If spores are detected, the egg batch is discarded and destroyed.

B. Grasserie (Viral Disease / Jaundice)

- **Causative Agent:** *Bombyx mori* Nuclear Polyhedrosis Virus (BmNPV).
- **Symptoms:** The larval body intersegmental membranes swell up, becoming shiny and fragile. The body hemolymph turns turbid and milky white due to polyhedral viral inclusion bodies. Skin ruptures easily upon touch, leaking infectious milky fluid; dying caterpillars crawl to comb tops and hang head-downward.
- **Control:** Maintaining low humidity and optimal temperature in rearing rooms; dusting rearing trays with formal-slaked lime powder.

C. Flacherie (Bacterial & Viral Disease)

- **Causative Agent:** Digestive tract infection caused by viruses (Flacherie virus) combined with opportunistic pathogenic bacteria such as *Streptococcus*, *Staphylococcus*, and *Bacillus thuringiensis*.
- **Symptoms:** Caused by high temperatures, poor airflow, and feeding stale, wet mulberry leaves. The caterpillar's digestive tract becomes paralyzed; larvae become flaccid, lethargic, vomit digestive juice, excrete foul-smelling diarrhea, turn soft and black post-death, and decay rapidly.
- **Control:** Feeding fresh, clean, dry mulberry leaves; ensuring proper ventilation and avoiding overcrowding in rearing trays.

D. Muscardine (Fungal Disease)

- **Causative Agent:** Fungi. **White Muscardine** is caused by *Beauveria bassiana*; **Red Muscardine** is caused by *Metarhizium anisopliae*.
- **Symptoms:** Airborne fungal spores land on larval skin, germinate, penetrate the cuticle, and produce toxins that kill the caterpillar. After death, the body turns stiff and rigid like a piece of chalk, covered entirely by white or red powdery fungal mycelia and conidiophores.
- **Control:** Disinfecting rearing equipment with 2% formalin; dusting anti-fungal powders like Duster Vijetha over silkworms after molting.

Question 25: Stored Grain Pests and Their Control Methods

1. Introduction

Post-harvest food grain losses caused by insect pests during storage range between 10% and 25% globally. Effective protection of stored food stocks requires an **Integrated Pest Management (IPM)** approach combining preventive measures, physical methods, chemical

treatments, and eco-friendly botanical alternatives.

2. Major Stored Grain Pests

Key pest species infesting stored grains include the Rice Weevil (*Sitophilus oryzae*), Lesser Grain Borer (*Rhyzopertha dominica*), Pulse Beetle (*Callosobruchus maculatus*), Red Flour Beetle (*Tribolium castaneum*), and Angoumois Grain Moth (*Sitotroga cerealella*).

3. Comprehensive Control Methods

A. Preventive Hygiene & Storage Sanitation

- Thoroughly cleaning godowns, warehouses, and storage structures prior to loading new grain.
- Removing old grain residues, floor frass, cobwebs, and sealing cracks, wall crevices, and rodent holes.
- Disinfecting empty godown walls by spraying Malathion 50% EC emulsion.

B. Physical & Environmental Control Methods

- **Sun Drying & Moisture Reduction:** Reducing grain moisture content below 10% to 12% prior to storage. Insects cannot multiply in dry grains containing low moisture.
- **Thermal Treatment:** Exposing grain to high temperatures (55-60°C) for several hours to kill all insect life stages without reducing seed germination capacity.
- **Hermetic (Airtight) Storage:** Storing grains in sealed, airtight metallic drums, Pusa bins, or hermetic plastic bags. Insect metabolic respiration consumes oxygen and elevates carbon dioxide levels within the sealed container, causing pest asphyxiation.

C. Chemical Control Methods

Surface Sprays & Contact Insecticide Treatment:

Treating storage bags and wall surfaces with synthetic pyrethroids (Deltamethrin) or organophosphates (Malathion) to prevent pest entry.

Fumigation (Gaseous Chemical Control):

Fumigation involves treating stored grains inside airtight sealed godowns or under gas-proof covers using volatile chemical gases that penetrate deep inside grain sacks to eradicate all life stages (eggs, larvae, pupae, adults):

- **Aluminum Phosphide (Celphos / Quickphos):** The most widely used stored grain fumigant. When solid tablets react with atmospheric moisture, they release **Phosphine gas (PH₃)**, a highly toxic gas that kills stored grain pests. Standard dosage is 1 to 2 tablets (3 grams each) per metric ton of stored grain, maintained under sealed covers for a mandatory 5 to 7 day exposure period.
- **Ethylene Dibromide (EDB):** Liquid fumigant used for grain seed stock treatment.

D. Botanical & Eco-Friendly Biological Control

- **Neem Products (*Azadirachta indica*):** Mixing dried neem leaves, neem seed kernel powder (2% weight ratio), or neem oil with stored grains acts as a natural antifeedant, repellent, and insect growth inhibitor.
- **Inert Dusts:** Mixing activated clay, silica aerogels, or wood ash with grain abrades the protective wax cuticle of insects, leading to desiccation and death.

Question 26: Protozoan Diseases

1. Introduction

Protozoan diseases are human and veterinary infections caused by microscopic, single-celled eukaryotic parasites belonging to various classes of the Phylum Protozoa. These pathogens inhabit host blood, tissues, or the gastrointestinal tract, causing morbidity and mortality across tropical regions.

2. Systematic Survey of Major Human Protozoan Diseases

A. Malaria

- **Causative Organisms:** *Plasmodium vivax*, *Plasmodium falciparum*, *Plasmodium malariae*, and *Plasmodium ovale*.
- **Vector & Transmission:** Transmitted by female *Anopheles* mosquitoes injecting infective sporozoites during blood meals.
- **Clinical Symptoms:** Periodical shivering chills followed by high fever spikes, heavy sweating, severe anemia, jaundice, and splenomegaly. *P. falciparum* causes dangerous **Cerebral Malaria** due to cytoadherence of infected RBCs blocking brain capillaries.
- **Control & Treatment:** Vector management, bed nets, oral Chloroquine, Primaquine, and Artemisinin-based Combination Therapies (ACT).

B. Amoebiasis (Amoebic Dysentery)

- **Causative Organism:** *Entamoeba histolytica*.
- **Mode of Transmission:** Ingestion of food or water contaminated with mature quadrinucleate cysts.
- **Clinical Symptoms:** Erosion of colonic mucosa causing flask-shaped ulcers, severe abdominal cramps, stools with blood and mucus, and potential metastatic Amoebic Liver Abscesses.
- **Control & Treatment:** Improved personal sanitation, clean water supply, Metronidazole, and Tinidazole.

C. Kala-Azar (Visceral Leishmaniasis)

- **Causative Organism:** *Leishmania donovani*.
- **Vector & Transmission:** Transmitted by tiny biting **Sandflies (*Phlebotomus argentipes*)**

injecting flagellated promastigote stages.

- **Clinical Symptoms:** The parasite invades reticuloendothelial cells, causing persistent irregular fever, severe emaciation, skin darkening ("black fever"), anemia, leukopenia, and massive enlargement of the liver (hepatomegaly) and spleen (**splenomegaly**).
- **Control & Treatment:** Sandfly control using residual sprays, treatment with Sodium Stibogluconate, Meglumine antimoniate, or **Amphotericin B**.

D. African Sleeping Sickness (Human African Trypanosomiasis)

- **Causative Organism:** *Trypanosoma brucei gambiense* and *Trypanosoma brucei rhodesiense*.
- **Vector & Transmission:** Transmitted by the bite of the **Tsetse Fly (*Glossina palpalis*)**.
- **Clinical Symptoms:** Parasites multiply in blood and lymph nodes (causing swollen lymph nodes/Winterbottom's sign), then cross the blood-brain barrier into the central nervous system. This causes severe neurological damage, apathy, confusion, sleep cycle reversal (daytime sleepiness, nighttime insomnia), coma, and death.
- **Control & Treatment:** Tsetse fly traps, clearing brushwood, treatment with Suramin, Pentamidine, or Melarsoprol.

E. Giardiasis

- **Causative Organism:** *Giardia lamblia* (commonly called the "Grand Old Man of the Intestine").
- **Mode of Transmission:** Ingestion of cysts in contaminated water supplies.
- **Clinical Symptoms:** Trophozoites use a ventral adhesive disc to attach to the duodenal mucosa, interfering with fat absorption. Causes foul-smelling greasy diarrhea (steatorrhea), flatulence, abdominal distension, and weight loss.
- **Control & Treatment:** Water filtration, boiling drinking water, and treatment with Metronidazole.

F. Trichomoniasis

- **Causative Organism:** *Trichomonas vaginalis*.
- **Transmission:** Sexually transmitted flagellated protozoan (lacks a cyst stage).
- **Clinical Symptoms:** Inhabits the female vagina and male urethra; causes vaginitis characterized by intense burning, itching, and a frothy greenish-yellow vaginal discharge.
- **Control & Treatment:** Safe personal practices and simultaneous treatment of both sexual partners with Metronidazole.

Q27) Economic Importance of Insects

Insects belong to the phylum Arthropoda and represent the most diverse group of animals on Earth. From an economic standpoint, insects exert profound impacts on agriculture, medicine, human welfare, and global ecosystems. Their economic significance is broadly categorized into beneficial and harmful aspects.

Beneficial Insects

Productive Insects

- **Silkworm (*Bombyx mori*):** Mulberry silkworms produce raw silk during pupation. Sericulture supports rural employment and international trade.
- **Honeybee (*Apis indica*, *Apis mellifera*):** Honeybees yield honey (a nutritious food and medicine), beeswax (used in cosmetics and pharmaceuticals), propolis, and royal jelly.
- **Lac Insect (*Laccifer lacca*):** Secretes resinous lac on host trees like *Butea monosperma*. Lac is essential in varnishes, electrical insulation, sealing wax, and shellac products.

Useful Agents in Agriculture and Ecosystems

- **Pollinators:** Over 80% of flowering plants depend on insect pollinators including bees, butterflies, moths, and hoverflies. Major food crops like mustard, sunflower, fruits, and legumes rely entirely on insect pollination for yield maximization.
- **Soil Improvers:** Ants, termites, and burrowing beetles aerate soil, decompose organic debris, and enhance soil fertility by recycling nutrients.
- **Scavengers:** Carrion beetles and dung beetles consume dead animal carcasses and feces, preventing environmental contamination and breeding of pathogens.
- **Biological Control Agents:** Predators like ladybird beetles (*Coccinella*) feed on aphids and scale insects, while parasitic wasps (*Trichogramma*) parasitize pest eggs, reducing chemical pesticide dependency.

Harmful Insects

Agricultural Pests

- **Field Pests:** Insects like the desert locust (*Schistocerca gregaria*), cotton bollworm (*Helicoverpa armigera*), and paddy stem borer (*Scirpophaga incertulas*) destroy standing crops, causing billions of rupees in economic losses annually.
- **Stored Grain Pests:** Rice weevils (*Sitophilus oryzae*) and khapra beetles (*Trogoderma granarium*) destroy stored agricultural produce by feeding and contamination.

Vectors of Human and Animal Diseases

- Mosquitoes (*Anopheles*, *Aedes*, *Culex*) transmit malaria, dengue, chikungunya, and filariasis.
- Sandflies (*Phlebotomus*) transmit Leishmaniasis (Kala-azar).

- Tsetse flies (*Glossina*) spread African sleeping sickness.

Household and Structural Pests

- Termites (*Odontotermes*) damage wooden structures, furniture, and books.
- Cockroaches contaminate food and trigger allergic reactions and gastrointestinal infections.

Q28) Mosquitoes as Vectors of Human Disease: Prevention and Control

Mosquitoes are two-winged dipteran insects that act as obligate vectors for a wide range of protozoan, viral, and helminthic human pathogens. Female mosquitoes require blood meals for egg maturation, during which they transmit pathogenic organisms between human hosts.

Major Mosquito Vectors and Associated Diseases

Anopheles (Female *Anopheles culicifacies*, *A. stephensi*)

- Transmits the protozoan parasite *Plasmodium* (*P. vivax*, *P. falciparum*, *P. malariae*, *P. ovale*), causing malaria.
- Characterized by high fever with chills, anemia, splenomegaly, and potential cerebral complications.

Aedes (*Aedes aegypti*, *Aedes albopictus*)

- Vector for arboviruses responsible for Dengue fever, Chikungunya, Yellow fever, and Zika virus.
- Dengue manifests as high fever, severe joint/muscle pain (breakbone fever), skin rash, and potential dengue hemorrhagic fever.

Culex (*Culex quinquefasciatus*)

- Vector for the filarial nematode *Wuchereria bancrofti*, causing lymphatic filariasis (elephantiasis).
- Transmits Japanese Encephalitis virus (JEV), leading to severe brain inflammation and neurological impairment.

Preventive Measures (Vector Control)

Physical and Environmental Control

- Source reduction by eliminating standing water in coolers, old tires, containers, and open drains to destroy breeding habitats.
- Installation of wire meshes on doors and windows to prevent mosquito entry into households.
- Use of insecticide-treated bed nets (ITNs) and long-lasting insecticidal nets (LLINs) during

sleep.

Biological Control

- Introduction of larvicidal fish such as *Gambusia affinis* (Mosquitofish) and *Poecilia reticulata* (Guppy) into stagnant ponds to feed on larvae.
- Application of bio-insecticides like *Bacillus thuringiensis israelensis* (BTI) which selectively destroys mosquito larval midgut epithelium.

Chemical Control

- Indoor residual spraying (IRS) using synthetic pyrethroids or organophosphates on walls.
- Space spraying and thermal fogging during epidemic outbreaks to knock down adult populations.
- Use of larvicides like Temephos in non-potable water bodies.

Curative Measures

- **Malaria:** Administration of Artemisinin-based Combination Therapy (ACT), Chloroquine, or Quinine depending on the species and resistance profile.
- **Dengue and Chikungunya:** Supportive therapy including intravenous fluids, antipyretics (Paracetamol), rest, and avoidance of NSAIDs like Aspirin to prevent hemorrhage.
- **Filariasis:** Mass drug administration of Diethylcarbamazine (DEC) combined with Albendazole.

Q29) Life Cycle and Economic Significance of *Dracunculus medinensis*

Dracunculus medinensis, commonly known as the Guinea worm or fiery serpent, is a tissue-dwelling nematode parasite responsible for dracunculiasis (Guinea worm disease) in humans.

Systematic Position

- Phylum: Nematoda
- Class: Secernentea
- Order: Spirurida
- Family: Dracunculidae
- Genus: *Dracunculus*
- Species: *medinensis*

Life Cycle of *Dracunculus medinensis*

The life cycle is digenetic, requiring two hosts: Humans (Definitive Host) and freshwater copepods of the genus *Cyclops* (Intermediate Host).

Infection of Human

Infection occurs when humans consume unfiltered water containing *Cyclops* infected with third-stage (L3) infective larvae.

Larval Release and Migration

Inside the human stomach, gastric juices digest the copepod, releasing the L3 larvae. The larvae penetrate the wall of the duodenum and abdominal cavity, migrating to subcutaneous connective tissues.

Maturation and Mating

Larvae mature into adult worms over 10 to 12 months. After fertilization, the male worm dies and degenerates, while the gravid female grows significantly in length (up to 60–100 cm).

Blister Formation

The gravid female migrates to the subcutaneous tissues of the lower extremities (feet or lower legs). She secretes a toxin that forms a painful blister on the host's skin.

Release of Larvae

When the blister ruptures upon contact with cool water, the anterior end of the female worm emerges and releases thousands of active first-stage (L1) larvae into the water.

Infection of Intermediate Host

The free-swimming L1 larvae are ingested by *Cyclops*. Inside the hemocoel of *Cyclops*, the larvae undergo two molts within 2 weeks to become infective L3 larvae, completing the cycle.

Economic Significance and Impact

- **Agricultural Loss:** Symptoms typically peak during planting or harvesting seasons. Disabling ulcerations render farm laborers incapacitated for months, leading to crop yield failure and loss of agricultural productivity.
- **Education and School Absenteeism:** Children infected with Guinea worm are unable to attend school for extended durations due to severe localized pain and secondary bacterial infections.
- **Healthcare Expenditure:** Treating secondary bacterial ulcers, managing chronic disability, and implementing eradication programs pose financial demands on healthcare systems in rural endemic regions.
- **Social and Physical Debility:** While rarely fatal, the disease causes long-term joint stiffness, permanent physical disability, and muscle atrophy, exacerbating poverty in rural communities.

Q30) Role of Predators and Parasites in Insect Pest

Management

Biological control involves utilizing living organisms to maintain pest population densities at lower levels than would occur in their absence. Natural enemies of insect pests are broadly classified into predators and parasites (parasitoids).

Role of Predators in Pest Management

Predators are free-living organisms that consume multiple prey individuals throughout their lifetime. They actively search for, capture, and devour target pests.

Key Characteristics of Predators

- Usually larger than their prey.
- Generalist or specialized feeders with high search efficiency.
- Both immature and adult stages often feed on the target pest.

Examples and Application

- **Ladybird Beetles (*Coccinella septempunctata*):** Both adults and larvae devour soft-bodied insects such as aphids, mealybugs, and scale insects in mustard and fruit crops.
- **Green Lacewings (*Chrysoperla carnea*):** Larvae (known as aphid lions) control whiteflies, thrips, spider mites, and insect eggs in cotton and vegetable crops.
- **Praying Mantis (*Mantis religiosa*):** Polyphagous predators that capture grasshoppers, caterpillars, and flies.
- **Predatory Mites (*Phytoseiulus persimilis*):** Used extensively in greenhouse agriculture to manage phytophagous spider mites.

Role of Parasites and Parasitoids in Pest Management

In entomology, insects that act as parasites are technically called **parasitoids**. Unlike true parasites, parasitoid larvae feed on host tissues and ultimately kill the host organism before reaching maturity.

Key Characteristics of Parasitoids

- Usually smaller than their host.
- Highly host-specific, concentrating on a particular life stage (egg, larva, or pupa) of the host.
- Adult parasitoids are free-living and feed on nectar or honeydew, while immature stages develop internally or externally on a single host.

Classification and Examples

- **Egg Parasitoids:** *Trichogramma* species parasitize eggs of lepidopteran pests like sugar cane borers and cotton bollworms. Eggs laid inside pest eggs hatch and destroy the host

embryo.

- **Larval Parasitoids:** *Bracon hebetor* and *Cotesia flavipes* attack caterpillars of pests by laying eggs inside the larval body cavity.
- **Pupal Parasitoids:** *Xanthopimpla stemmator* attacks pupae of stem borers.

Advantages in Pest Management

- **Sustainability:** Once established, natural enemies reproduce independently, offering long-term pest regulation.
- **Environmental Safety:** Zero chemical residues, preserving soil health, non-target organisms, and human safety.
- **Resistance Avoidance:** Target pests do not easily develop resistance to natural predation or parasitism compared to chemical insecticides.

Q31) Life History and Genetic Significance of Cattle Tick

Cattle ticks, primarily belonging to the family Ixodidae (hard ticks), are obligate ectoparasites of livestock. The most economically damaging cattle tick species is *Rhipicephalus (Boophilus) microplus*.

Life History of Cattle Tick

Rhipicephalus microplus is a one-host tick, meaning all three active lifecycle stages (larva, nymph, adult) develop and feed on a single host animal.

Egg Stage

Female ticks drop off the host animal onto the pasture and lay 2,000 to 4,000 eggs in soil crevices before dying. Eggs hatch into six-legged larvae (seed ticks) in 3 to 4 weeks under favorable warm, moist conditions.

Larval Stage

Larvae climb onto blades of grass (questing behavior) to attach to a passing host. Once attached, larvae feed on host blood, engorge, and undergo their first molt into eight-legged nymphs on the host within 5 to 7 days.

Nymphal Stage

Nymphs attach again, feed on host blood and tissue fluid for 7 to 10 days, and molt into adult male or female ticks without leaving the host.

Adult Stage

Adult ticks mate on the host skin. Fertilized female ticks feed voraciously, engorging with blood over 7 to 14 days. Fully engorged females drop off to the ground to lay eggs, completing the

cycle in 3 to 4 weeks on the host.

Genetic Significance of Cattle Tick

Pesticide and Acaricide Resistance Genetics

- Cattle ticks exhibit remarkable genetic adaptability. Widespread exposure to synthetic acaricides (organophosphates, synthetic pyrethroids, macrocyclic lactones) has selected for resistant tick populations through gene mutations.
- Single nucleotide polymorphisms (SNPs) in the sodium channel gene confer knock-down resistance (kdr) to pyrethroids.
- Amplification of acetylcholinesterase and carboxylesterase genes leads to metabolic detoxification of organophosphates.

Host Specificity and Speciation

Genetic studies on mitochondrial DNA and nuclear ribosomal DNA shed light on the phylogenetic evolution and geographical distribution of tick populations across continents.

Vector-Pathogen Genetic Interactions

Cattle ticks act as vectors for protozoans (*Babesia bigemina*, *Babesia bovis*) and bacteria (*Anaplasma marginale*). Transovarial transmission genetics allows *Babesia* species to pass from female ticks through eggs to the next generation of larval ticks, sustaining disease reservoirs without needing immediate re-infection from an infected host.

Q32) Life Cycle and Zoonotic Significance of *Diphyllobothrium latum*

Diphyllobothrium latum, commonly known as the broad fish tapeworm, is the largest parasite infecting humans, reaching lengths of up to 10 to 15 meters in the human intestine.

Systematic Position

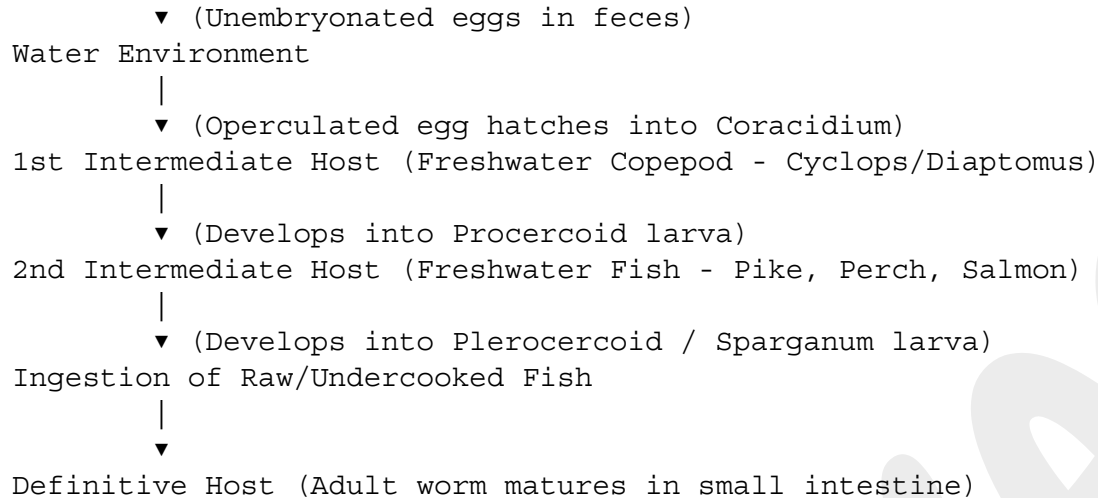
- Phylum: Platyhelminthes
- Class: Cestoda
- Order: Pseudophyllidea
- Family: Diphyllbothriidae
- Genus: *Diphyllobothrium*
- Species: *latum*

Life Cycle of *Diphyllobothrium latum*

The lifecycle is complex and tri-heteroxenous, requiring one definitive host and two intermediate hosts.

Definitive Host (Human / Mammal)

|



Detailed Stages

- **Egg Release:** Adult worms in the small intestine of human/mammalian hosts discharge operculated, unembryonated eggs into the stool.
- **Coracidium Stage:** In freshwater, eggs embryonate and hatch into ciliated, free-swimming larvae called coracidia.
- **Proceroid Stage:** *Cyclops* or *Diaptomus* ingests the coracidium. Inside the copepod's body cavity, it transforms into a proceroid larva.
- **Plerocercoid Stage:** When infected copepods are eaten by small freshwater fish (e.g., minnows), and subsequently by larger predatory fish (pike, perch, salmon), the proceroid larva migrates into the fish muscle tissue and converts into an infective plerocercoid (sparganum) larva.
- **Definitive Infection:** Humans contract infection by ingesting raw, undercooked, or pickled freshwater fish containing plerocercoid larvae. Larvae attach to the mucosal wall of the ileum using two sucking grooves (bothria) on the scolex and mature into adult tapeworms within 5 to 6 weeks.

Zoonotic Significance

- **Bothriocephalus Anemia / Parasitic Vitamin B12 Deficiency:** *D. latum* selectively absorbs vast amounts of dietary Vitamin B12 from the host's intestine. This produces megaloblastic anemia clinically indistinguishable from pernicious anemia, accompanied by neurological symptoms such as subacute combined degeneration of the spinal cord.
- **Zoonotic Reservoir:** The parasite naturally circulates among wild fish-eating mammals such as bears, wolves, dogs, cats, and foxes, which serve as persistent reservoir hosts, making environmental eradication extremely difficult.
- **Public Health and Food Safety Impact:** Increased consumption of raw fish dishes (sushi, sashimi, ceviche) worldwide has expanded the geographic range of human diphyllbothriasis from cold temperature lake regions (Nordic regions, Russia) to global urban centers.

Q33) Pearl Culture in India

Pearl culture is the commercial practice of producing cultured pearls by artificially inserting an irritant (nucleus) into the mantle tissue of pearl oysters. India has a long coast line with rich pearl oyster beds, especially in the Gulf of Mannar, Palk Bay, and the Gulf of Kutch.

Biology of Pearl Oyster

The principal species used for marine pearl culture in India is *Pinctada fucata* (Indian Pearl Oyster). Freshwater pearls are produced using freshwater mussels such as *Lamellidens marginalis* and *Parreysia corrugata*.

Mechanism of Pearl Formation

Pearl formation is a natural defense response to foreign irritants:

1. When a foreign particle enters between the shell and mantle, the outer epithelium of the mantle surrounds the foreign body.
2. The mantle invaginates to form a **pearl sac**.
3. The epithelial cells of the pearl sac secrete concentric layers of **nacre** (mother-of-pearl), composed of aragonite (crystalline calcium carbonate) bound by an organic protein matrix called **conchiolin**.
4. Over time, successive depositions of nacre form a lustrous pearl.

Steps in Pearl Culturing Technique

Collection and Rearing of Oysters

Oysters are collected from wild natural beds (paars) or produced artificially in hatcheries via induced spawning. Young oysters are reared in net cages suspended from floating rafts in sea coastal waters until they reach operational size (about 1.5–2 years old).

Graft Preparation and Nucleus Implantation

- **Donor Oyster Preparation:** A thin strip of outer mantle tissue is excised from a healthy donor oyster and cut into small square pieces (graft tissue or *saibo*).
- **Surgical Implantation:** The recipient oyster is carefully wedged open. A small incision is made in the gonadal tissue or foot region. The prepared graft tissue along with a spherical polished shell bead (nucleus) is surgically inserted into the incision site.

Post-Operative Care and Culturing

Operated oysters are kept in convalescence tanks for observation for a few days. Successfully operated oysters are transferred to sea cages suspended from rafts. They are maintained for 12 to 24 months, during which the mantle graft forms a pearl sac around the bead and secretes layers of nacre.

Harvesting and Processing

Oysters are collected, opened, and pearls are extracted carefully. Harvested pearls undergo washing, sorting, grading based on size, luster, shape, and color, followed by polishing.

Economic Prospects of Pearl Culture in India

- Central Marine Fisheries Research Institute (CMFRI) pioneered indigenous marine pearl culture technology at Tuticorin, Tamil Nadu.
- CIFA (Central Institute of Freshwater Aquaculture), Bhubaneswar, developed accessible freshwater pearl culture methods for rural farmers.
- Provides high economic returns, supports employment in coastal and rural sectors, and yields export value in global gemstone markets.

Q34) Essay on Vitellogenesis

Vitellogenesis is the process of yolk synthesis, transport, and accumulation within the developing oocyte during oogenesis. Yolk serves as the primary nutritional reserve (comprising proteins, lipids, carbohydrates, and minerals) required for embryonic development in oviparous and ovoviviparous animals.

Nature and Composition of Yolk

Yolk is not a single chemical substance, but a heterogeneous mixture deposited in egg cytoplasm. It primarily consists of:

- **Proteins:** Lipovitellin and phosvitin (derived from the precursor protein vitellogenin).
- **Lipids:** Neutral fats, phospholipids, and cholesterol.
- **Carbohydrates:** Glycogen bound to proteins (glycoproteins).
- **Inorganic ions:** Calcium, phosphorus, and iron.

Site of Vitellogenesis

Vitellogenesis occurs via two distinct pathways across the animal kingdom:

Endogenous Vitellogenesis

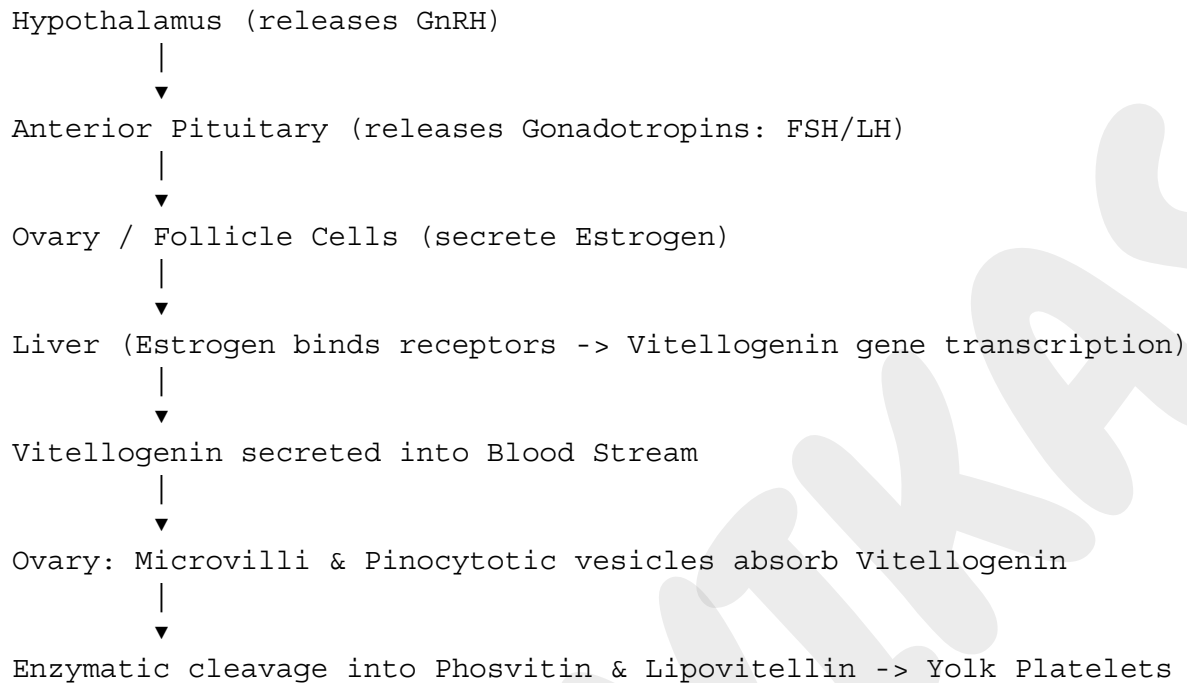
The oocyte synthesizes its own yolk components using its internal organelle machinery (Endoplasmic Reticulum and Golgi Complex). Common in lower invertebrates like cnidarians and primitive flatworms.

Exogenous Vitellogenesis

The predominant mechanism in vertebrates (amphibians, reptiles, birds) and insects. Yolk precursors are synthesized outside the ovary in extra-ovarian organs and transported via the circulatory system to the ovary.

- **In Vertebrates:** The **liver** is the primary site of vitellogenin synthesis.
- **In Insects:** The **fat body** functions as the site of vitellogenin synthesis.

Mechanism of Exogenous Vitellogenesis in Vertebrates



1. **Hormonal Triggering:** Environmental cues trigger the hypothalamus to release Gonadotropin-Releasing Hormone (GnRH), stimulating the pituitary to secrete Follicle-Stimulating Hormone (FSH).
2. **Estrogen Induction:** FSH acts on ovarian follicle cells, stimulating them to secrete **estrogen** into the bloodstream.
3. **Hepatic Synthesis:** Estrogen reaches the liver and binds to specific nuclear estrogen receptors in hepatocytes. This activates the transcription of vitellogenin genes.
4. **Secretion and Transport:** Hepatocytes synthesize the precursor phosphoprotein **vitellogenin**, which is secreted into the blood plasma.
5. **Uptake by Oocyte:** Vitellogenin reaches the ovary through rich capillary networks. It passes through gaps in the follicular cell layer and binds to specific cell-surface receptors on the oolemma.
6. **Endocytosis and Processing:** The oocyte internalizes vitellogenin via receptor-mediated endocytosis through clathrin-coated pinocytotic vesicles. Inside the egg cytoplasm, specific enzymes (e.g., cathepsin D) cleave vitellogenin into its final yolk components: **phosvitin** and **lipovitellin**, which aggregate to form crystalline **yolk platelets**.

Hormonal Regulation in Insects

In insects, juvenile hormone (JH) secreted by the *corpora allata* acts on fat body cells to induce vitellogenin synthesis, while ecdysone plays a synergistic role in stimulating vitellogenin uptake by developing oocytes.

Significance of Vitellogenesis

- Ensures embryonic nutrition and survival in non-placental animals.
- Establishes spatial polarity (animal-vegetal axis) within the egg cytoplasm, which dictates cleavage patterns and embryonic body axis formation during cleavage and gastrulation.

Q35) Biological Control of Insect Pests

Biological control is an eco-friendly pest management method that utilizes natural enemies (predators, parasitoids, and pathogens) to suppress insect pest populations below the Economic Injury Level (EIL).

Categories of Biological Control Agents

Entomophagous Predators

Free-living organisms that feed directly on pest populations:

- **Ladybird Beetle (*Coccinella septempunctata*)**: Controls aphids, scale insects, and mealybugs.
- **Chrysoperla carnea (Green Lacewing)**: Voracious predator of thrips, mites, and caterpillar eggs.
- **Predatory Mites**: Control phytophagous spider mites in orchards and greenhouses.

Parasitoids

Insects whose larvae mature inside or on a host insect, eventually killing it:

- **Trichogramma species**: Egg parasitoids deployed against sugarcane borers and cotton bollworms.
- **Bracon hebetor**: Larval parasitoid targeting storage grain pests and caterpillars.

Entomopathogens (Microbial Control)

- **Bacteria**: *Bacillus thuringiensis* (Bt) produces crystalline endotoxin proteins (Cry toxins) that solubilize in the alkaline gut of caterpillars, perforating midgut membranes and causing lethal septicemia.
- **Fungi**: *Metarhizium anisopliae* (green muscardine fungus) and *Beauveria bassiana* (white muscardine fungus) penetrate pest cuticles and infect beetles, bugs, and locusts.
- **Viruses**: Baculoviruses such as Nucleopolyhedrovirus (NPV) are host-specific viruses that control *Helicoverpa armigera* and *Spodoptera litura*.

Approaches in Biological Control

Importation (Classic Biological Control)

Introduction of exotic natural enemies from a pest's native region to control an invasive pest species. Classic example: Introduction of the Vedalia beetle (*Rodolia cardinalis*) from Australia to California to control the cottony cushion scale (*Icerya purchasi*).

Augmentation

Routine laboratory rearing and release of native natural enemies into fields when natural populations are insufficient.

- **Inundative Release:** Large-scale release of agents for immediate suppression (e.g., mass release of *Trichogramma* cards).
- **Inoculative Release:** Releasing small numbers of agents to establish a reproducing population for seasonal control.

Conservation and Enhancement

Modifying agricultural practices to protect and enhance resident natural enemy populations:

- Minimizing broad-spectrum chemical pesticide use.
- Planting refuge crops, nectar-producing flowers, and cover crops to provide shelter and alternative food sources for adult predators and parasitoids.

Benefits over Chemical Control

- Highly specific without harming non-target organisms, pollinators, or humans.
- Environmentally non-polluting with zero toxic residues in food crops.
- Prevents secondary pest outbreaks and delays pesticide resistance development.

Q36) Essay on Protozoa and Human Diseases

Protozoa are single-celled, eukaryotic microorganisms belonging to the kingdom Protista. While many protozoans are free-living, several parasitic species cause debilitating human diseases globally.

Major Protozoan Human Diseases

Amoebiasis

- **Causative Agent:** *Entamoeba histolytica* (Amoebic protozoan).
- **Transmission:** Ingestion of food or water contaminated with quadrinucleate cysts via the fecal-oral route.
- **Pathogenesis and Symptoms:** Cysts excyst in the small intestine, releasing trophozoites that invade the large intestinal mucosa. Trophozoites secrete proteolytic enzymes (histolysin) causing "flask-shaped" mucosal ulcers.
- **Clinical Manifestations:** Amoebic dysentery characterized by stool with blood and mucus, abdominal cramps, and potential liver abscesses (extra-intestinal amoebiasis) if trophozoites migrate via the portal vein.

Malaria

- **Causative Agent:** *Plasmodium* species (*P. falciparum*, *P. vivax*, *P. malariae*, *P. ovale*).
- **Vector:** Female *Anopheles* mosquito.
- **Pathogenesis and Symptoms:** Sporozoites injected by mosquito bite infect liver

parenchymal cells (hepatic schizogony) and subsequently invade red blood cells (erythrocytic schizogony). Merozoite rupture releases pyrogens (hemozoin toxin), causing periodic febrile paroxysms (chills, high fever, profuse sweating). *P. falciparum* causes malignant tertian malaria, which can plug brain capillaries leading to fatal cerebral malaria.

Leishmaniasis (Kala-azar)

- **Causative Agent:** *Leishmania donovani* (Flagellated protozoan).
- **Vector:** Phlebotomine sandfly (*Phlebotomus argentipes*).
- **Pathogenesis and Symptoms:** Promastigotes injected by sandflies are phagocytosed by macrophages, transforming into non-flagellated amastigotes (LD bodies). Amastigotes multiply intracellularly, lysing macrophages and infecting the reticuloendothelial system (spleen, liver, bone marrow).
- **Clinical Manifestations:** Hepatosplenomegaly, prolonged irregular fever, severe anemia, emaciation, and skin hyperpigmentation ("black fever").

Giardiasis

- **Causative Agent:** *Giardia lamblia* (Flagellated protozoan with characteristic binucleated teardrop shape).
- **Transmission:** Fecal-oral route via cyst-contaminated drinking water.
- **Pathogenesis and Symptoms:** Trophozoites attach to the duodenal and jejunal mucosa via a ventral sucking disc, causing mechanical blockage of nutrient absorption.
- **Clinical Manifestations:** Malabsorption syndrome, chronic foul-smelling fatty diarrhea (steatorrhea), flatulence, and abdominal distension.

Human African Trypanosomiasis (Sleeping Sickness)

- **Causative Agent:** *Trypanosoma brucei gambiense* / *rhodesiense*.
- **Vector:** Tsetse fly (*Glossina* species).
- **Pathogenesis and Symptoms:** Flagellated trypomastigotes multiply in blood and lymph nodes, eventually crossing the blood-brain barrier into the central nervous system.
- **Clinical Manifestations:** Daytime somnolence, nocturnal insomnia, progressive neurological impairment, coma, and death.

Q37) Zoonotic Significance and Transmission of Tapeworm (*Taenia solium*)

Definition of Zoonotic Significance

Zoonotic significance refers to the medical and ecological impact of **zoonoses**—diseases and infections naturally transmitted between vertebrate animals and humans. A parasite has high zoonotic significance when animal reservoirs sustain its transmission cycle, presenting risks to human populations through dietary, occupational, or environmental contact.

Transmission and Zoonotic Characteristics of *Taenia solium*

Taenia solium (Pork Tapeworm) is a classic zoonotic helminth causing two distinct clinical conditions in humans: **Taeniasis** and **Cysticercosis**.

Infected Human (Definitive Host)

| (Discharges Gravid Proglottids / Eggs in Feces)



Environment / Soil Contamination



| Ingestion by Pigs (Intermediate Host)
by Humans (Accidental)



Onchosphere penetrates pig intestine
penetrates human intestine



Develops into *Cysticercus cellulosae* in pig muscle
Cysticercus cellulosae in human brain/eyes



▼ Ingestion of undercooked "measly" pork
Human develops Adult Tapeworm (TAENIASIS)
develops CYSTICERCOSIS / NEUROCYSTICERCOSIS

| Ingestion



Onchosphere



Develops into



Human

Role of the Intermediate Host (Pig)

1. **Environmental Contamination:** Humans infected with adult *T. solium* pass gravid proglottids or eggs in their feces. Pigs, acting as coprophagic intermediate hosts, ingest food or soil contaminated with human feces containing these eggs.
2. **Onchosphere Migration:** Inside the pig's stomach, digestive enzymes dissolve the egg shell, releasing six-hooked embryos called **oncospheres**.
3. **Encystment in Muscle:** Oncospheres penetrate the intestinal mucosa, enter the systemic bloodstream, and migrate to striated muscle tissues (tongue, neck, heart, thigh). Here, they develop into fluid-filled larval cysts known as ***Cysticercus cellulosae*** (producing "measly pork").
4. **Transmission to Humans:** When humans consume undercooked pork infected with viable cysticerci, gastric juices dissolve the cyst wall, the scolex evaginates and attaches to the jejunal mucosa, and matures into an adult tapeworm over 2–3 months, causing **Taeniasis**.

Accidental Autoinfection and Neurocysticercosis in Humans

The zoonotic complexity of *T. solium* is uniquely dangerous because humans can also accidentally serve as intermediate hosts:

- **Mechanism:** Humans ingest infective *T. solium* eggs via contaminated water/food or through autoinfection (reverse peristalsis or poor hand hygiene).

- **Pathology:** Oncospheres hatch in the human intestine, cross the gut wall into the circulation, and encyst in human tissue—predominantly the Central Nervous System (CNS), eyes, and subcutaneous tissue.
- **Neurocysticercosis (NCC):** Cysts developing in the brain parenchyma or ventricles cause severe neurological illness, serving as a primary cause of acquired adult epilepsy and hydrocephalus in developing nations.

Q38) Integrated Freshwater Prawn and Fish Aquaculture

Integrated aquaculture combines the farming of complementary aquatic species to optimize nutrient utilization, reduce economic risks, maximize land yield, and ensure long-term environmental and financial sustainability.

Concept of Integrated Freshwater Prawn and Fish Culture

Polyculture of the **Giant Freshwater Prawn (*Macrobrachium rosenbergii*)** alongside compatible **Carps** (Indian Major Carps and Exotic Carps) utilizes distinct ecological niches within a single pond ecosystem without interspecies competition for food and space.

Niche Complementarity in the Pond Ecosystem

- **Surface Feeder:** Catla (*Catla catla*) feeds on surface zooplankton.
- **Column Feeder:** Rohu (*Labeo rohita*) feeds on column phytoplankton and organic detritus.
- **Bottom Feeder / Benthic Niche:** Freshwater Prawn (*Macrobrachium rosenbergii*) operates exclusively at the pond bottom, feeding on benthic detritus, leftover fish feed, aquatic insect larvae, and natural benthic flora.

Detailed Combination Example: Freshwater Prawn with Indian Major Carps

Recommended Species Mix and Stocking Ratios

For a 1-hectare pond:

- ***Macrobrachium rosenbergii* (Prawn):** 10,000 – 15,000 post-larvae (PL)/ha (stocked at bottom).
- ***Catla catla*:** 1,500/ha (surface niche).
- ***Labeo rohita*:** 2,000/ha (column niche).
- ***Cirrhinus mrigala* (Mrigal):** Omitted or restricted to very low stocking density (500/ha) to prevent direct bottom-space competition with prawns.

Economic and Environmental Sustainability Mechanics

1. **Waste Nutrient Recycling:** Uneaten supplementary feed and nutrient-rich fecal matter from carps sink to the bottom, stimulating benthic bio-film development. *Macrobrachium*

- feeds on this detrital matter, converting waste into high-value crustacean protein.
2. **Ecological Health and Water Quality Maintenance:** Prawns scavenge decaying organic debris off the pond floor, preventing anaerobic bottom mud formation, toxic gas build-up (ammonia, hydrogen sulfide), and micro-habitat degradation.
 3. **Dual Income Streams and Regular Cash Flow:**
 - **Carps** produce high bulk biomass sold at regular market intervals, covering operational expenses (feed, energy, labor).
 - **Giant Freshwater Prawns** command premium market prices per kilogram (2 to 3 times higher price realization than carps), yielding profit margins.
 4. **Risk Diversification:** If fish prices fluctuate or a fish disease outbreak occurs, prawn production buffers the farmer against total economic loss.

Q39) Mosquito Vectors, Pathogens, and Vector-Pathogen Interactions

Zoological Names of Important Vector Mosquitoes

- **Malaria:** *Anopheles culicifacies*, *Anopheles stephensi*, *Anopheles fluviatilis*
- **Dengue:** *Aedes aegypti*, *Aedes albopictus*
- **Encephalitis (Japanese Encephalitis):** *Culex tritaeniorhynchus*, *Culex vishnui*

Interrelationship of Mosquitoes with the Causative Agent of Dengue Virus

Dengue is caused by the Dengue Virus (DENV 1-4), an enveloped single-stranded RNA virus belonging to the family Flaviviridae. The transmission relies on an intrinsic biological relationship with its primary vector, female *Aedes aegypti*.

Extrinsic Incubation Period (EIP) and Viral Transmission Steps (In Words)

1. **Ingestion of Infected Blood Meal:** A non-infected female *Aedes aegypti* feeds on an infected human during the viremic phase (when virus concentration in human blood is high).
2. **Infection of Mosquito Midgut Epithelium:** The ingested virus enters the mosquito's midgut lumen, attaches to specific surface receptors on midgut epithelial cells, and undergoes endocytosis.
3. **Intracellular Viral Replication:** Inside midgut epithelial cells, the viral positive-sense RNA replicates, translates viral proteins, and synthesizes new viral particles.
4. **Midgut Escape Barrier:** Replicated virions exit midgut epithelial cells into the surrounding hemolymph (mosquito body fluid).
5. **Dissemination Through Hemocoel:** The virus circulates within the hemolymph, infecting target tissues including neural tissue, fat bodies, and hemocytes.
6. **Infection of Salivary Glands:** The virus crosses the salivary gland barrier and selectively infects acinar epithelial cells of the mosquito's salivary glands, where intensive viral replication occurs.
7. **Transmission to Secondary Host:** After an **Extrinsic Incubation Period** of 8 to 12

days, the mosquito becomes permanently infectious. When probing a healthy human skin for a blood meal, the mosquito injects saliva containing virions into dermal capillaries, initiating a new human infection.

Q40) Life History, Damage, and Prevention of Cattle Tick (*Boophilus microplus*)

Boophilus microplus (synonym *Rhipicephalus microplus*), the tropical cattle tick, is an ectoparasite affecting livestock production across tropical and subtropical regions.

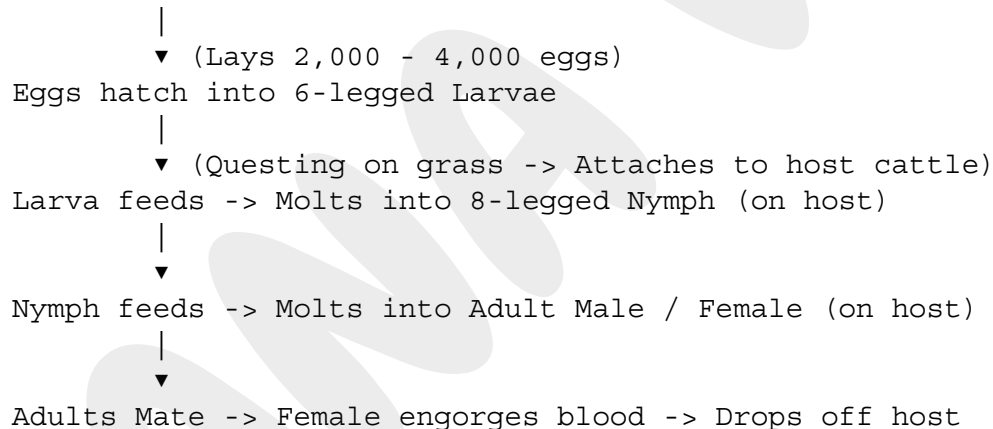
Systematic Position

- Phylum: Arthropoda
- Subphylum: Chelicerata
- Class: Arachnida
- Order: Ixodida
- Family: Ixodidae
- Genus: *Boophilus* / *Rhipicephalus*
- Species: *microplus*

Life History

Boophilus microplus is a **one-host tick** completing all development on a single animal host.

Engorged Female drops to soil



1. **Egg Deposition:** Engorged females drop off the host into pastures and lay up to 4,000 eggs in soil crevices before dying.
2. **Larval Stage:** Eggs hatch into 6-legged larvae (seed ticks) which climb vegetation to attach to passing cattle. They attach, suck blood, and molt into 8-legged nymphs in 5–7 days.
3. **Nymphal Stage:** Nymphs feed on blood/tissue fluid and molt into adults within 7–10 days on the host.
4. **Adult Stage:** Adult ticks mate on the host skin. Fertilized females engorge with blood over 7–14 days, increasing significantly in weight before dropping to the ground to lay eggs.

Nature of Damage Caused

- **Direct Blood Loss and Anemia:** Heavy tick infestations cause severe blood loss (each female tick consumes up to 0.5–3 mL blood), resulting in progressive normocytic normochromic anemia.
- **Tick Worry and Anorexia:** Continuous irritation, painful bites, and skin inflammation cause severe distress ("tick worry"), reduced grazing time, diminished feed intake, and drop in milk production and meat weight gain.
- **Hide Damage:** Tick bite wounds create skin lesions, reducing the commercial market value of leather hides and providing entry points for secondary bacterial infections and blowfly strike (myiasis).
- **Disease Transmission (Vector Role):** Transmits major blood pathogens:
 - *Babesia bigemina* and *Babesia bovis* (causing Bovine Babesiosis / Texas Redwater fever).
 - *Anaplasma marginale* (causing Bovine Anaplasmosis).

Preventive and Control Measures

- **Chemical Control (Acaricides):** Strategic spraying, dipping, or pour-on applications of synthetic pyrethroids (Cypermethrin), organophosphates (Amitraz), or macrocyclic lactones (Ivermectin).
- **Pasture Rotation (Spelling):** Vacating infested pastures for 3 to 4 months starves larval seed ticks, breaking the transmission cycle.
- **Breeding Resistance:** Cross-breeding susceptible European cattle (*Bos taurus*) with tick-resistant Zebu cattle (*Bos indicus*), which display grooming habits and innate skin immune responses against tick attachment.
- **Anti-Tick Vaccination:** Application of recombinant vaccines targeting tick midgut membrane antigens (e.g., Bm86 antigen), which disrupt tick digestion and reduce female egg production.

Q41) Life Cycle and Zoonotic Significance of Protozoan Parasites (*Leishmania donovani*)

Leishmania donovani is a dimorphic, obligate intracellular protozoan parasite responsible for **Visceral Leishmaniasis (Kala-azar)** in humans.

Life Cycle of *Leishmania donovani*

The parasite completes its digenetic lifecycle in two hosts: Humans/Mammals (Definitive Host) and female Sandflies (*Phlebotomus argentipes*) (Vector Host).

Female Sandfly takes blood meal (Injects Promastigotes)

↓
Promastigotes phagocytosed by Human Macrophages

↓
▼

Transform into Amastigotes (LD Bodies) inside phagolysosome



Amastigotes multiply -> Lyse Macrophages -> Infect new RES cells
(Spleen, Liver, Bone Marrow)



Sandfly takes blood meal (Ingests Macrophages with Amastigotes)



Amastigotes transform into Promastigotes in Sandfly Midgut



Promastigotes multiply and migrate to Sandfly Proboscis

In the Human/Mammalian Host

1. **Inoculation:** When an infected sandfly bites a human, it injects flagellated, motile **promastigotes** into the skin.
2. **Intracellular Invasion:** Macrophages phagocytose the promastigotes. Inside the macrophage's phagolysosome, promastigotes lose their flagella and transform into oval, non-motile **amastigotes** (Leishman-Donovan or LD bodies).
3. **Multiplication and Tissue Destruction:** Amastigotes multiply by binary fission within macrophages until the host cell ruptures. Released amastigotes invade neighboring reticuloendothelial cells in the spleen, liver, bone marrow, and lymph nodes.

In the Sandfly Host

1. **Ingestion:** A female sandfly ingests blood containing amastigote-infected macrophages from a host.
2. **Transformation:** In the midgut of the sandfly, amastigotes transform back into motile, elongated **promastigotes**.
3. **Development:** Promastigotes multiply, migrate to the sandfly's foregut and proboscis, ready for injection during the next blood meal.

Zoonotic Significance

- **Mammalian Reservoirs:** In several geographic regions (particularly the Mediterranean basin, Central Asia, and South America), *Leishmania* species are maintained in wild and domestic animal reservoirs including **dogs (canine leishmaniasis)**, rodents, jackals, and foxes.
- **Transmission Dynamics:** Domestic dogs act as urban reservoirs, where sandflies acquire parasites from infected canine skin lesions and transmit them to humans.
- **Public Health Control Challenges:** Eradicating visceral leishmaniasis is difficult because animal reservoirs maintain the cycle independently, leading to human re-infections even after successful vector reduction campaigns.

Q42) Sericulture in India and Its Role in the Rural Economy

Sericulture is an agro-based industry involving the commercial cultivation of host plants, rearing of silkworms (*Bombyx mori*), extraction of raw silk reeled from silkworm cocoons, and subsequent weaving into silk fabrics.

Silkworm Species and Host Plants in India

India holds the distinction of producing all four commercial varieties of natural silk:

- **Mulberry Silk:** Produced by *Bombyx mori* feeding on mulberry leaves (*Morus alba*). Accountable for over 70% of Indian silk production.
- **Tasar Silk:** Produced by *Antheraea mylitta* feeding on *Terminalia arjuna* and *Terminalia tomentosa* trees in forest belts.
- **Eri Silk:** Produced by *Samia cynthia ricini* feeding on castor leaves (*Ricinus communis*).
- **Muga Silk:** Exclusive to Assam, produced by *Antheraea assamensis* feeding on Som (*Machilus bombycina*) and Soalu plants.

Role of Sericulture in the Indian Rural Economy

High Employment Generation

- Sericulture is labor-intensive across its value chain: host plant cultivation, silkworm egg hatching (chawki rearing), cocoon harvesting, silk reeling, and weaving.
- Provides gainful employment to over 8.5 million rural people in India, predominantly targeting marginal farmers and landless agricultural laborers.

Empowerment of Women

- Women constitute more than 50–60% of the sericulture workforce, participating in indoor silkworm rearing, leaf harvesting, and cocoon sorting. This fosters financial independence and rural socio-economic upliftment.

High Economic Yield with Low Gestation Period

- Mulberry cultivation allows multiple crops per year (4 to 5 harvests annually).
- Provides small-scale farmers with regular returns at short intervals compared to traditional seasonal agricultural crops.

Eco-Friendly and Forest-Based Sustainable Livelihoods

- Non-mulberry silks (Tasar, Eri, Muga) are cultivated in tribal and forest regions (e.g., Jharkhand, Chhattisgarh, North-Eastern states), preventing deforestation by providing indigenous communities with forest-based income.

Foreign Exchange Earnings

- Export of Indian silk goods (sarees, garments, carpets) generates significant national foreign exchange revenue.

Q43) Life Cycle of a Platyhelminth Worm Involving Two Intermediate Hosts

The broad fish tapeworm, *Diphyllbothrium latum* (or alternatively, the Chinese Liver Fluke, *Clonorchis sinensis*), is a platyhelminth parasite whose lifecycle involves **two distinct intermediate hosts**. Below is the detailed account for *Clonorchis sinensis* (Chinese Liver Fluke).

Systematic Position

- Phylum: Platyhelminthes
- Class: Trematoda
- Order: Opisthorchiida
- Family: Opisthorchiidae
- Genus: *Clonorchis*
- Species: *sinensis*

Lifecycle of *Clonorchis sinensis*

The lifecycle is tri-heteroxenous, requiring one definitive host (Human/Mammal) and **two sequential intermediate hosts**:

1. **First Intermediate Host:** Freshwater Snail (*Parafossarulus* / *Bithynia* species).
2. **Second Intermediate Host:** Freshwater Cyprinid Fish (e.g., Carp).

Definitive Host (Human / Cat / Dog)

| (Discharges embryonated eggs in feces)



Water Body



▼ Ingested by Snail

1st Intermediate Host (Freshwater Snail - Bithynia)

| (Miracidium -> Sporocyst -> Redia -> Cercaria)



Cercaria exits snail & swims free



▼ Penetrates skin/scales of Fish

2nd Intermediate Host (Freshwater Cyprinid Fish)

| (Encysts as Metacercaria in fish muscle)



Ingestion of Raw / Undercooked Fish



Definitive Host (Human - matures in Bile Ducts of Liver)

Stage 1: Egg Release and Ingestion by Snail

Adult flukes inhabiting the bile ducts of humans lay fully embryonated eggs containing a **miracidium larva**. Eggs are excreted in feces into freshwater bodies. The first intermediate host, a freshwater snail (*Parafossarulus*), ingests the eggs.

Stage 2: Intramolluscan Larval Multiplication

Inside the snail's intestine, the egg hatches into a miracidium, which penetrates the snail gut tissue and undergoes asexual multiplication stages:

- **Miracidium** → **Sporocyst** → **Redia** → **Cercaria**.
- Free-swimming, lophocercous **cercariae** emerge from the snail into the water.

Stage 3: Infection of Second Intermediate Host (Fish)

The free-swimming cercaria comes into contact with a freshwater cyprinid fish. It penetrates the scales and skin, sheds its tail, and encysts in the flesh and muscle tissue of the fish as a **metacercaria larva**.

Stage 4: Infection of Definitive Host

Humans or fish-eating mammals (cats, dogs) become infected by consuming raw, undercooked, or salted freshwater fish containing viable metacercariae.

- In the human duodenum, the metacercaria excysts.
- The juvenile fluke migrates up the biliary tree into the **bile ducts of the liver**, where it matures into an adult hermaphroditic fluke within 1 month, causing **clonorchiasis** (cholecystitis, bile duct obstruction, and cholangiocarcinoma).

Q44) Basic Concepts of Integrated Aquaculture and Shellfish Management

Basic Concepts of Integrated Aquaculture

Integrated aquaculture involves combining aquatic farming systems with agriculture, livestock, or secondary aquatic species to achieve recycling of organic waste, maximum energy transfer, environmental sustainability, and economic optimization.

Core Principles

- **Waste-to-Resource Conversion:** Effluents or solid organic waste from one farming component (e.g., livestock manure or feed residues) act as direct feed or nutrient fertilizer for another component (e.g., fish ponds or aquatic plants).
- **Polyculture and Ecological Niche Optimization:** Rearing species occupying different trophic levels (surface, column, bottom feeders) prevents resource competition and

optimizes ecosystem services.

- **Environmental Balance:** Minimizes nutrient accumulation and eutrophication in nearby natural water systems.

Management Aspects of Shellfish

Shellfish aquaculture encompasses commercially important **Crustaceans** (Prawns, Shrimps, Crabs) and **Molluscs** (Oysters, Mussels, Clams). Effective management requires meticulous oversight of water quality, stocking, feeding, and disease containment.

Site Selection and Pond Construction

- Non-polluted water supply with optimal salinity (15–30 ppt for marine shrimps; freshwater for *Macrobrachium*).
- Soil with adequate clay content (>20%) to ensure minimal seepage.

Seed Selection and Nursery Management

- Stocking healthy, disease-free, PCR-tested post-larvae (PL) from certified hatcheries.
- Acclimatization of PL to pond salinity and temperature prior to release to prevent osmotic shock and mortality.

Water Quality Management

Continuous monitoring of critical physico-chemical water parameters:

- **Dissolved Oxygen (DO):** Maintained above 5 mg/L using mechanical paddle-wheel aerators.
- **pH:** Maintained in the range of 7.5 to 8.5 with minimal diurnal fluctuation.
- **Un-ionized Ammonia (NH₃):** Kept strictly below 0.01 mg/L to prevent crustacean toxicity and shell soft-shell syndrome.

Feed and Biofloc Management

- Utilizing high-protein (35–40%) balanced artificial diets delivered via feed trays to evaluate consumption rates and minimize feed wastage.
- Implementation of **Biofloc Technology (BFT)**: Adding carbon sources (molasses) to promote beneficial heterotrophic bacterial growth, which converts toxic nitrogenous wastes into bacterial protein consumed by shellfish.

Health and Disease Control

- Biosecurity measures including bird netting, crab fencing, and water filtration reservoirs.
- Strict monitoring for viral pathogens such as **White Spot Syndrome Virus (WSSV)** and **Acute Hepatopancreatic Necrosis Disease (AHPND)**.

Q45) Polyphagous Pests: Life Cycle and Damage of

Helicoverpa armigera

Definition of Polyphagous Pests

Polyphagous pests are destructive insect species that do not restrict their feeding to a single host plant or botanical family, but instead feed on a wide range of taxonomically unrelated cultivated crops and wild plants. Their wide host range allows them to migrate seamlessly between seasonal crops, making management challenging.

Detailed Study of a Polyphagous Pest: *Helicoverpa armigera* (Gram Pod Borer / Cotton Bollworm)

- **Host Plants:** Over 180 plant species, including chickpea (gram), cotton, tomato, pigeon pea, maize, sunflower, and okra.

Life Cycle of *Helicoverpa armigera*

The life cycle is complete (Holometabolous): Egg → Larva → Pupa → Adult.

Adult Female lays eggs on leaves / buds

↓

▼ (Hatch in 3 - 5 days)

Larval Stage (6 Instars, 14 - 20 days)

- * Voracious feeder on reproductive plant organs

↓

▼ (Drops to soil)

Pupal Stage (In soil, 10 - 14 days)

↓

▼

Adult Moth emerges → Mating & Egg Laying

Egg Stage

Female moths lay 500 to 1,500 spherical, yellowish-white eggs singly on tender leaves, flower buds, and young shoots during night hours. Eggs hatch in 3 to 5 days.

Larval Stage

- The larval phase lasts 14 to 20 days, progressing through 6 distinct instars.
- Young caterpillars feed initially on tender leaves, while older larvae feed directly on flowers, pods, and fruits.
- Larvae exhibit variable coloration (green, yellowish-brown, dark brown with lateral stripes).

Pupal Stage

Full-grown 6th instar larvae drop off the plant into the soil, constructing an earthen chamber to pupate. The pupal stage lasts 10 to 14 days (or undergoes diapause during harsh winters).

Adult Stage

The adult moth is stout, yellowish-brown with a dark, kidney-shaped mark on the forewings and a distinct border on the hindwings. Adult longevity is 7 to 10 days.

Nature of Damage Caused

- **Targeting Reproductive Organs:** The larvae prefer feeding on high-protein plant structures—flower buds, green pods, and developing fruits.
- **Gram/Chickpea:** Caterpillar bores into young pods, thrusting its head inside to devour developing seeds while keeping the rest of its body hanging outside the pod hole.
- **Cotton:** Known as the American bollworm, it bores into cotton bolls, causing premature shedding and staining of lint.
- **Tomato:** Bores into tomato fruits (Tomato fruit borer), rendering harvested produce unmarketable due to internal rot and fungal contamination.

Q46) Role of Mosquitoes in Transmission of Malaria and Environmental Factors

Role of Mosquitoes in Malaria Transmission

Malaria is caused by the intracellular protozoan *Plasmodium*, which requires two hosts to complete its life cycle: humans (intermediate host undergoing asexual schizogony) and female *Anopheles* mosquitoes (definitive host undergoing sexual sporogony).

Transmission Process

1. **Ingestion of Gametocytes:** A female *Anopheles* mosquito feeds on a malarious human, ingesting microgametocytes (male) and macrogametocytes (female).
2. **Exflagellation and Fertilization:** Inside the mosquito midgut lumen, microgametocytes undergo exflagellation to form male gametes, which fertilize macrogametes to produce a motile zygote called an **ookinete**.
3. **Oocyst Formation:** The ookinete penetrates the midgut epithelial wall and encysts as an **oocyst** on the outer stomach wall.
4. **Sporogony:** The oocyst undergoes nuclear division, forming thousands of active **sporozoites**. The mature oocyst ruptures, releasing sporozoites into the hemolymph.
5. **Salivary Gland Invasion:** Sporozoites migrate through the hemolymph and penetrate the acinar cells of the mosquito's **salivary glands**.
6. **Inoculation into Human:** When the mosquito feeds on another human, it injects saliva containing sporozoites into skin capillaries, initiating the liver stage (exo-erythrocytic schizogony).

Influence of Environmental Factors on Malaria Transmission

Temperature

- **Extrinsic Incubation Period (EIP):** The rate of *Plasmodium* sporogony inside the mosquito is temperature-dependent. At optimal temperatures (25–30°C), sporogony completes in 8 to 10 days. Below 16°C, *P. vivax* development ceases, and below 19°C, *P. falciparum* development halts.
- **Mosquito Survival and Activity:** High ambient temperatures speed up digestion and shorten the gonotrophic cycle, increasing feeding frequency. Extremely high temperatures (>35°C) reduce adult mosquito lifespan.

Humidity and Rainfall

- **Relative Humidity:** Mosquitoes require a relative humidity above 60% for survival. Low humidity causes rapid desiccation and adult mortality before the EIP completes.
- **Rainfall:** Creates stagnant water pools, puddles, choked drains, and river bed pools, which serve as larval breeding grounds. Heavy torrential rainfall can wash away mosquito larvae, temporarily reducing populations.

Land Use and Vegetation

Deforestation, irrigation projects, dams, and urban construction create artificial water collections, fostering localized vector breeding (*Anopheles stephensi* in urban settings, *A. culicifacies* in rural settings).

Q47) Economic Importance of Protozoa in Relation to Humans

Protozoa exert major positive and negative impacts on human welfare, agriculture, sanitation, and global ecological cycles.

Beneficial Protozoa

Role in Water Purification and Sewage Treatment

Free-living ciliated protozoa (*Vorticella*, *Paramecium*, *Opercularia*) play an essential role in biological wastewater treatment plants. They consume bacteria, clarifying sewage effluent and improving water quality before environmental discharge.

Soil Fertility and Nutrient Cycling

Soil-dwelling flagellates, amoebae, and ciliates feed on soil bacteria. This predation regulates bacterial population densities and releases excess bio-available nitrogen and phosphorus back into the rhizosphere, enhancing crop growth.

Rumen Protozoa in Livestock

Symbiotic ciliates (*Entodinium*, *Epidinium*) inhabit the rumen of ruminant livestock (cows,

sheep). They assist in breaking down complex cellulose and plant polymers into short-chain fatty acids, providing primary energy sources for milk and meat production.

Marine Protozoa and Geological Deposits

Planktonic Foraminifera (*Globigerina*) and Radiolaria possess calcareous and siliceous tests (shells). Over geological epochs, settled tests formed deep ocean sediments, creating **chalk, limestone, and petroleum deposits**. These fossil deposits serve as index fossils for oil exploration.

Harmful Protozoa

Human Diseases

Pathogenic protozoans cause severe human morbidity and mortality:

- *Plasmodium falciparum*: Causes cerebral malaria.
- *Entamoeba histolytica*: Causes amoebic dysentery and liver abscesses.
- *Leishmania donovani*: Causes Kala-azar (Visceral Leishmaniasis).
- *Trypanosoma brucei*: Causes African Sleeping Sickness.
- *Giardia lamblia*: Causes chronic malabsorptive diarrhea.

Veterinary Pathogens

- *Babesia bigemina*: Causes Texas redwater fever in cattle.
- *Eimeria tenella*: Causes severe coccidiosis in poultry farms, leading to intestinal hemorrhages and significant economic losses in poultry production.

Q48) Concept of Pond Management and Importance of Aquaculture in India

Concept of Pond Management in Aquaculture

Pond management involves ecological, chemical, and biological practices applied to a water body to maximize fish yield, prevent disease, and maintain environmental stability.

Pre-Stocking Management

- **Drying and Desilting**: Draining and sun-drying the pond bottom to oxidize organic sludge, eliminate pathogenic bacteria, and eradicate unwanted pests.
- **Liming**: Applying agricultural lime (CaCO_3 or Ca(OH)_2) to neutralize soil acidity, stabilize pH, and improve water clarity.
- **Eradication of Weed Fish and Predators**: Using biodegradable plant-derived fish poisons (e.g., Mahua oil cake containing saponin) to remove predatory fish prior to stocking.
- **Manuring and Fertilization**: Applying organic manures (cow dung) and inorganic fertilizers (urea, single superphosphate) to trigger natural plankton blooms (phyto- and zooplankton).

Stocking Management

- Selecting compatible species combinations (Polyculture of Indian Major Carps: Catla, Rohu, Mrigal).
- Stocking healthy fingerlings (10–15 cm) at optimal stocking densities (usually 5,000–10,000 fingerlings/ha) during cooler morning hours.

Post-Stocking Management

- **Supplementary Feeding:** Providing nutritionally balanced feed pellets (groundnut oil cake, rice bran) at 2–3% of total body weight daily.
- **Water Quality Monitoring:** Maintaining dissolved oxygen through aeration, preventing algal blooms, and managing nitrogenous wastes.
- **Disease Surveillance:** Regular netting sampling to monitor fish growth, check for skin lesions, fungal infections, or parasitic ectoparasites (*Argulus*).

Importance of Aquaculture in India

- **Nutritional Security:** Delivers affordable animal protein, essential amino acids, polyunsaturated fatty acids (Omega-3 PUFA), and micronutrients to battle malnutrition.
- **Economic Contribution:** Contributes roughly 1% to India's total GDP and over 6.7% to agricultural GDP.
- **Livelihood and Employment:** Supports over 28 million direct and indirect fishers, fish farmers, and allied vendors in rural areas.
- **Foreign Exchange Earnings:** Marine products and farmed shrimp (*Litopenaeus vannamei*) constitute one of India's largest agricultural export commodities.
- **Blue Revolution Initiative:** Strategic government schemes (PMMSY - Pradhan Mantri Matsya Sampada Yojana) drive modern technology integration, such as Recirculating Aquaculture Systems (RAS) and Biofloc technology.

Q49) Erythrocytic Cycle of *Plasmodium* in Humans and Disease Transmission

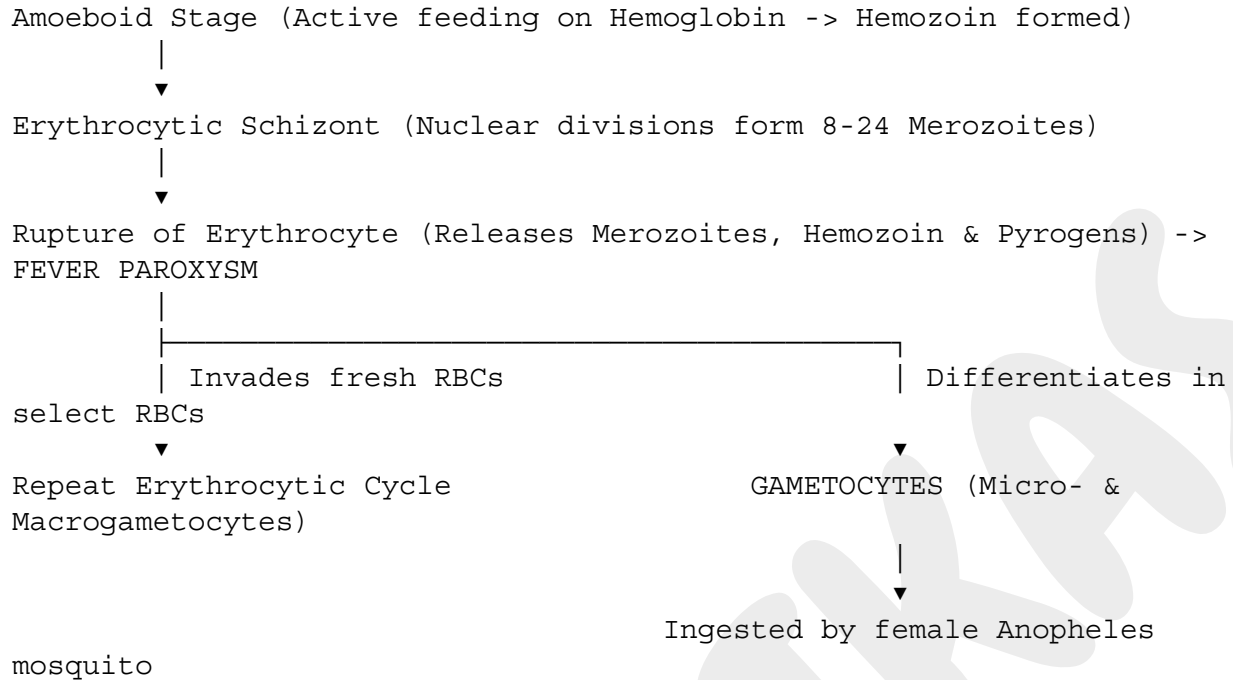
The erythrocytic cycle (erythrocytic schizogony) represents the intra-erythrocytic developmental phase of *Plasmodium* parasites inside human red blood cells (RBCs). This cycle is directly responsible for clinical malaria symptoms and the generation of sexual stages essential for mosquito transmission.

Step-by-Step Erythrocytic Cycle in Human RBCs

Merozoites released from Liver (Exo-erythrocytic schizogony)

▼ Invades Red Blood Cell (RBC)
Ring Stage (Early Trophozoite)

▼



Invasion of Red Blood Cell

Exo-erythrocytic merozoites released from ruptured hepatocytes enter the systemic bloodstream. Merozoite surface proteins (MSP-1) bind to specific receptors on the surface of erythrocytes. The merozoite reorients its apical complex, releases rhoptry and microneme secretions, and invades the RBC.

Trophozoite Stage

- **Ring Stage:** Inside the RBC, the young parasite appears as a translucent vesicle with a peripheral ring of cytoplasm and a dark chromatin dot (Ring Trophozoite).
- **Amoeboid Stage:** The parasite grows, developing an active amoeboid cytoplasm. It ingests host erythrocyte cytoplasm via a cytostome.

Hemoglobin Digestion and Hemozoin Formation

The parasite degrades host hemoglobin inside its digestive vacuole to obtain amino acids. Toxic free heme is converted into an insoluble, dark-brown crystalline pigment called **hemozoin** (malarial pigment).

Erythrocytic Schizogony

The mature trophozoite undergoes multiple nuclear divisions (schizogony) without immediate cytoplasmic division, forming an **erythrocytic schizont**. The cytoplasm then divides around each nucleus, forming 8 to 24 sub-microscopic **merozoites**.

Rupture of RBC and Clinical Paroxysm

The infected erythrocyte becomes fragile, swells, and ruptures, releasing:

- Free merozoites into blood plasma (which invade fresh RBCs).
- Parasitic waste products, cellular debris, and toxic **hemozoin**.

The release of these pyrogenic substances triggers host macrophages to release TNF- α and Interleukin-1, causing classical malarial paroxysms: cold stage (chills), hot stage (high fever up to 104–105°F), and sweating stage.

Progression to Mosquito Transmission

After several erythrocytic cycles, a subpopulation of merozoites invading RBCs stops undergoing asexual division and instead differentiates into sexual precursor cells called **gametocytes**:

- **Microgametocytes (Male)**: Small, with diffuse nuclear chromatin.
- **Macrogametocytes (Female)**: Larger, with dense, compact nuclear chromatin.

Gametocytes circulate passively in peripheral blood capillaries. When a female *Anopheles* mosquito feeds on the human host, it ingests these gametocytes, initiating the sexual cycle (sporogony) inside the mosquito gut and continuing disease transmission.

Q50) Biological Control of Insect Pests

Biological control is an eco-friendly component of Integrated Pest Management (IPM) that uses living organisms—predators, parasitoids, and entomopathogens—to keep pest populations below the Economic Injury Level (EIL).

Major Categories of Biological Control Agents

Entomophagous Predators

Predators kill and feed on multiple prey items during their lifecycle.

- **Ladybird Beetles (*Coccinella septempunctata*)**: Adults and larvae feed on aphids, jassids, and scale insects.
- **Green Lacewing Larvae (*Chrysoperla carnea*)**: Known as "aphid lions", these consume whiteflies, spider mites, and soft-bodied caterpillars.
- **Predatory Mites (*Phytoseiulus persimilis*)**: Used to control tetranychid spider mites in greenhouse crops.

Parasitoids

Parasitoids require a single host individual to complete larval development, eventually killing the host before adult emergence.

- **Egg Parasitoids (*Trichogramma* species)**: Female wasps lay eggs inside the eggs of lepidopteran crop pests (*Helicoverpa*, *Chilo*). The parasitoid larva devours the host egg embryo, emerging as an adult wasp.
- **Larval Parasitoids (*Bracon hebetor*, *Cotesia flavipes*)**: Parasitize caterpillars of stem borers and stored product pests.

Microbial Pathogens (Entomopathogens)

- **Bacterial Pathogens (*Bacillus thuringiensis* - Bt):** Bt spores contain crystalline protein inclusions (Cry toxins). Upon ingestion by caterpillars in alkaline gut conditions, toxins dissolve, bind to midgut epithelial receptors, form pores, cause osmotic cell lysis, and induce lethal septicemia.
- **Fungal Pathogens:** *Beauveria bassiana* and *Metarhizium anisopliae* adhere to insect cuticles, germinate fungal hyphae, penetrate directly through the integument via enzymatic action (chitinases), and cause muscardine disease in beetles and bugs.
- **Viral Pathogens (Baculoviruses):** Nucleopolyhedrovirus (NPV) and Granulovirus (GV) are highly host-specific viruses deployed against *Helicoverpa armigera* and *Spodoptera litura*.

Methodological Strategies in Biological Control

- **Importation (Classic Biological Control):** Introduction of exotic natural enemies to control introduced invasive pests (e.g., *Rodolia cardinalis* introduced against *Icerya purchasi*).
- **Augmentation:** Mass rearing and periodic field release of native natural enemies.
 - *Inundative Release:* Releasing mass quantities for immediate knock-down of severe pest surges.
 - *Inoculative Release:* Releasing smaller numbers to establish natural reproduction for seasonal suppression.
- **Conservation:** Modifying pesticide regimes (switching to selective bio-pesticides) and maintaining floral biodiversity (nectar plants) to preserve resident natural enemy communities.

Key Advantages over Synthetic Chemicals

- Highly targeted mode of action with no risk to pollinators or wildlife.
- Leaves zero chemical residues on harvested agricultural food products.
- Pests do not easily develop resistance to biological predation or parasitism.
- Long-term sustainable regulation of agro-ecosystem balance.

Q51) Greenhouse Effect

The **Greenhouse Effect** is a natural process that warms the Earth's surface. Atmospheric gases trap solar radiation, keeping the planet warm enough to support life (average global temperature around 15°C instead of -18°C). However, excessive emissions from human activities have enhanced this effect, leading to global warming.

Key Greenhouse Gases (GHGs)

- **Carbon Dioxide (CO_2)**: Primary contributor (~60%), emitted via fossil fuel combustion and deforestation.
- **Methane (CH_4)**: Emitted from paddy fields, enteric fermentation in cattle, and landfills (25x more potent than CO_2).
- **Nitrous Oxide (N_2O)**: Originates from chemical fertilizers and agricultural practices.
- **Chlorofluorocarbons (CFCs)**: Synthetic industrial compounds used in refrigeration.

Mechanism

1. Short-wavelength solar radiation passes through the atmosphere and warms the Earth's surface.
2. The Earth re-radiates this energy as long-wavelength infrared (thermal) radiation.
3. Greenhouse gases absorb and re-emit this infrared radiation back toward Earth, trapping heat within the troposphere.

Consequences

- **Global Warming & Climate Change**: Shifts in weather patterns and extreme climate events.
- **Melting of Ice Caps**: Causes polar ice retreat and thermal expansion of oceans, leading to rising sea levels.
- **Disruption of Ecosystems**: Coral bleaching, loss of biodiversity, and altered migration routes for animals.

Q52) Chipko Movement

The **Chipko Movement** was a non-violent socio-ecological movement that started in 1973 in the Chamoli district of Uttarakhand (then part of Uttar Pradesh). The word "*Chipko*" means "*to hug*", referring to villagers clinging to trees to prevent them from being logged.

Historical Context & Leadership

- Rooted in historical precedents like the 18th-century **Bishnoi movement** led by Amrita Devi in Rajasthan.

- **Prominent Leaders:** Sundarlal Bahuguna, Chandi Prasad Bhatt, and Gaura Devi.
- **Motto:** *"What do the forests bear? Soil, water, and pure air."*

Key Objectives & Mechanism

- To stop large-scale commercial deforestation in fragile Himalayan slopes.
- To secure traditional forest rights for indigenous communities whose livelihoods depended on forest produce.
- Villagers formed human chains and hugged trees when commercial contractors arrived with axes.

Significance & Impact

- Led to a **15-year ban on green felling** in the Himalayan region ordered by Prime Minister Indira Gandhi in 1980.
- Inspired eco-movements across India, such as the **Appiko Movement** in Karnataka.
- Transformed environmental conservation into a grassroots social movement in India.

Q53) Environmental Hazards of Radiation

Radiation hazards stem from the emission of ionizing energy (Alpha, Beta, Gamma rays, X-rays) or non-ionizing energy (UV, Microwaves). Ionizing radiation possesses sufficient energy to remove tightly bound electrons from atoms, causing molecular damage in living organisms.

Sources of Exposure

- **Natural:** Cosmic rays, radon gas from soil and granite rocks.
- **Anthropogenic:** Nuclear power plant accidents (e.g., Chernobyl, Fukushima), nuclear weapons testing, medical X-rays, and improper disposal of radioactive waste.

Biological Effects on Living Organisms

1. **Somatic Damage:** Affects body cells directly, causing tissue necrosis, skin burns, radiation sickness, hair loss, and acute leukemia.
2. **Genetic Damage:** Causes gene mutations and chromosomal aberrations in gametes, leading to congenital defects in future generations.
3. **Ecosystem Biomagnification:** Radioactive isotopes like Strontium-90 (Sr^{90}) substitute for calcium in bones, while Iodine-131 (I^{131}) accumulates in thyroid tissue.

Preventive Measures

- Shielding with dense lead barriers during exposure.
- Safe containment and deep geological disposal of radioactive waste.
- Monitoring radiation worker exposure using personal dosimeters.

Q54) Difference Between a National Park and a Wildlife

Sanctuary

Comparison Table

Attribute	National Park	Wildlife Sanctuary
Primary Focus	Protection of entire ecosystem (flora, fauna, landscape, historical objects).	Protection of specific species or group of species (fauna focus).
Protection Status	Highly protected; stricter legal status under Wildlife Protection Act (1972).	Moderate protection compared to a National Park.
Human Activity	Strictly prohibited (no grazing, forestry, or private ownership allowed).	Limited activity allowed (grazing, timber harvesting with permission).
Boundaries	Fixed and clearly demarcated by legislation.	Boundaries are often not sharply demarcated.
Upgradation	Cannot be downgraded to a sanctuary.	Can be upgraded to a National Park.
Examples	Jim Corbett National Park, Kaziranga National Park.	Periyar Wildlife Sanctuary, Bharatpur Bird Sanctuary.

Q55) Noise Pollution

Noise pollution is defined as any unwanted, unpleasant, or excessive sound that interferes with human comfort, physiological well-being, or ecological communication. Sound intensity is measured in **Decibels (dB)**. Exposure above **85 dB** over long periods can cause permanent hearing impairment.

Major Sources

- **Industrial:** Heavy machinery, textile mills, air compressors.
- **Transport:** Jet aircraft take-offs (130–140 dB), automotive traffic honking, railway engines.
- **Domestic/Community:** Loudspeakers, construction site drills, firecrackers.

Physiological & Psychological Impacts

- **Auditory Damage:** Noise-induced hearing loss (NIHL), acoustic trauma, and persistent tinnitus.
- **Non-Auditory Damage:**
 - Elevated blood pressure, hypertension, and increased cardiovascular stress.
 - Insomnia, fatigue, and psychological irritability.
 - Disruption of animal echolocation and acoustic communication in birds and marine mammals.

Control Measures

- Installation of silencers and acoustic baffles in heavy machinery.
- Creation of **Green Belts** (planting dense trees along highways to absorb noise).
- Setting legal decibel limits and strict zoning laws for Quiet Zones (near hospitals/schools).

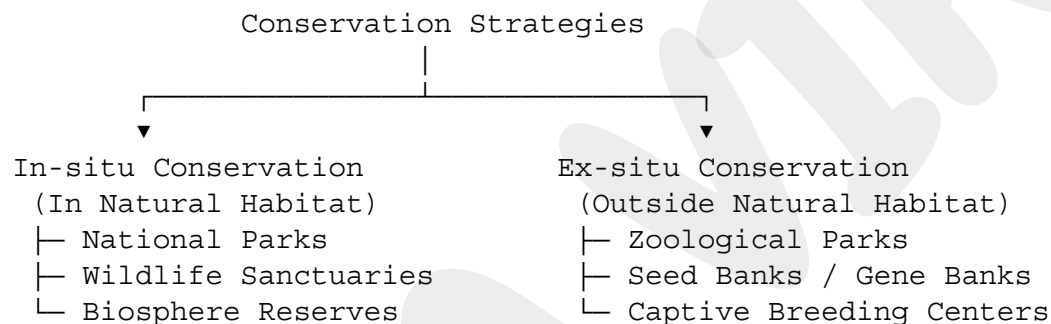
Q56) Conservation of Endangered Species

An **endangered species** is a species categorized by the IUCN (International Union for Conservation of Nature) as facing a very high risk of extinction in the wild. Protecting these species requires multi-tiered conservation strategies.

Causes of Species Decline

- Habitat loss, fragmentation, and degradation.
- Over-exploitation, poaching, and illegal wildlife trade.
- Invasive alien species competition and climate change.

Conservation Strategies



1. In-situ Conservation (On-Site):

- Protecting the species within its natural habitat through National Parks, Sanctuaries, and Biosphere Reserves.
- *Example:* Project Tiger, Project Elephant.

2. Ex-situ Conservation (Off-Site):

- Preserving threatened species outside their natural habitats.
- *Examples:* Zoological gardens, botanical gardens, gene/seed banks, and cryopreservation of gametes.

Legal Measures

- Enforcement of the **Wildlife Protection Act, 1972** in India.
- Implementation of **CITES** (Convention on International Trade in Endangered Species) protocols globally.

Q57) Volcanic Eruptions

A **volcanic eruption** is a geological process where molten rock (magma), ash, steam, and

volatile gases escape to the Earth's surface through a rupture or vent in the crust.

Environmental Effects

- **Atmospheric Impact:** Ash clouds contain Sulfur Dioxide (SO_2), Carbon Dioxide (CO_2), and Hydrogen Fluoride (HF). SO_2 reacts with moisture to form volcanic aerosols that reflect sunlight, leading to **volcanic winter** (short-term global cooling).
- **Acid Rain:** Volcanic gases form atmospheric acids, resulting in localized acid precipitation.
- **Habitat Destruction:** Pyroclastic flows, lahars (mudflows), and basalt lava burn and bury ecosystems instantaneously.

Ecological Restoration & Succession

- **Primary Succession:** Volcanic lava forms bare rock surfaces. Over time, pioneer organisms like lichens and mosses break down rock into soil, initiating ecological succession that eventually restores flora and fauna.

Q58) Urbanization Stress and Health

Rapid **urbanization** leads to population concentration in cities, often outstripping infrastructure. This transformation exerts structural stress on natural systems and creates health hazards for urban populations.

Stressors Associated with Urbanization

1. **Environmental Degradation:** Concentrated air pollution ($\text{PM}_{2.5}$, PM_{10}), industrial effluents, and urban heat island (UHI) effects.
2. **High Population Density:** Leads to slum development, inadequate sanitation, and water contamination.
3. **Loss of Green Spaces:** Deprives urban populations of natural ecosystem services and air filtration.

Impact on Human & Animal Health

- **Respiratory Diseases:** Higher rates of asthma, bronchitis, and lung diseases due to urban smog.
- **Vector-borne Diseases:** Poor drainage creates breeding sites for *Aedes* and *Anopheles* mosquitoes, increasing dengue and malaria transmission.
- **Psychological Stress:** Overcrowding, light pollution, and noise pollution elevate cortisol levels, contributing to anxiety and cardiovascular disorders.
- **Urban Wildlife Alterations:** Species adapt behaviorally or suffer local extinctions; synanthropic animals (e.g., pigeons, rats) proliferate while native species decline.

Q59) Environmental Monitoring

Environmental Monitoring is the systematic collection of physical, chemical, and biological data using standardized quantitative techniques to assess environmental quality, detect degradation, and manage pollution.

Types of Monitoring

1. **Air Quality Monitoring:** Measures levels of criteria pollutants like SO_x , NO_x , ground-level O_3 , carbon monoxide, and particulate matter ($\text{PM}_{2.5}$, PM_{10}).
2. **Water Quality Monitoring:** Evaluates parameters such as dissolved oxygen (DO), biological oxygen demand (BOD), chemical oxygen demand (COD), pH, heavy metal concentration, and coliform counts.
3. **Biological Monitoring (Biomonitoring):** Uses living organisms to assess ecosystem health.
 - *Example:* Lichens act as bioindicators for SO_2 air pollution; benthic macroinvertebrates indicate stream water quality.

Applications & Importance

- Establishes baseline data for Environmental Impact Assessments (EIA).
- Provides early warning systems for ecological disasters and toxic leaks.
- Helps regulatory agencies (e.g., CPCB) enforce environmental standards and policies.
-

Q60) Nesting Behaviour

Nesting behaviour encompasses the sequence of activities performed by animals—particularly birds, insects, and fishes—to construct, maintain, and protect a structure (nest) used for egg laying, incubation, and rearing offspring.

Types & Examples

- **Avian Nests:** Range from simple scrapes on the ground (e.g., plovers) to elaborate woven structures (e.g., Weaverbird *Ploceus philippinus* using intricate knots) and mud nests (e.g., swallows).
- **Fish Nests:** The male Three-spined Stickleback (*Gasterosteus aculeatus*) uses kidney secretions (spiggin) to bind plant materials into a tubular nest.
- **Insect Nests:** Social insects like termites (*Macrotermes*) build complex mud mounds with internal ventilation channels, while honeybees construct hexagonal wax cells.

Hormonal & Evolutionary Control

- Triggered by environmental cues (photoperiod, temperature) that induce hormonal secretion, primarily **prolactin** and **estrogen**.
- Serves crucial evolutionary purposes: protection from predators, microclimate regulation for embryo development, and reproductive success.

Q61) Social Hierarchies

A **social hierarchy** (or dominance hierarchy) is an organizational structure within an animal group where individuals are ranked relative to one another. This arrangement reduces energy expenditure and injury from physical combat over resources.

Types of Hierarchies

1. **Linear Hierarchy (Pecking Order):** A clear rank order ($A > B > C > D$) where individual A dominates all, B dominates all except A, and so on. Classic example: chickens established by Schjelderup-Ebbe.
2. **Complex / Despotic Hierarchy:** One dominant individual (Alpha) rules over all other group members, with little clear structure among subordinates (e.g., wolf packs).

[Alpha (α)] → Dominates all members

|
[Beta (β)] → Subordinate to Alpha, dominates Gamma

|
[Gamma (γ)] → Subordinate to Alpha & Beta

Mechanisms & Maintenance

- Established initially through physical contests, threats, or displays.
- Maintained via submissive postures, vocalizations, and behavioral rituals without ongoing actual fights.
- Dominant individuals gain priority access to food, mates, and prime shelter locations.

Q62) Territory Marking

A **territory** is a defended space that an animal or group uses exclusively for feeding, mating, or rearing young. **Territory marking** involves signaling mechanisms used to convey ownership to competitors without engaging in direct combat.

Modes of Territory Marking

1. **Chemical Marking (Scent Marking):**
 - Mammals use specialized glands or bodily fluids (urine, feces).
 - *Examples:* Canids (dogs/wolves) use urine scent posts; felids (tigers) spray urine mixed with glandular secretions; deer use preorbital glands on branches.
2. **Visual Marking:**
 - Visible changes made to the environment.
 - *Examples:* Bears clawing bark off trees; antelopes creating dung heaps; visual display of bright coloration in territorial reef fish.
3. **Acoustic Marking:**
 - Long-distance vocalizations declaring space occupied.
 - *Examples:* Bird song displays at dawn, howler monkey choruses in rainforest

canopy, lion roars.

Q63) Fixed Action Patterns (FAP)

A **Fixed Action Pattern (FAP)** is an unlearned, innate, highly stereotyped sequence of behavioral acts defined by ethologists Konrad Lorenz and Niko Tinbergen. Once triggered, it runs to completion even if the triggering stimulus is removed.

Key Characteristics

1. **Innate:** Genetically hardwired, requiring no prior experience or learning.
2. **Stereotyped:** Performed in a uniform manner across all members of a species.
3. **Triggered by Sign Stimulus:** Initiated by a specific external cue known as a **Sign Stimulus** or **Releaser**.
4. **Independent of Feedback:** Once initiated, the action sequence completes regardless of environmental changes.

External Cue (Sign Stimulus) → Innate Releasing Mechanism (IRM) → Fixed Action Pattern (FAP)

Classic Examples

- **Egg-Rolling in Greylag Goose:** If an egg rolls out of the nest, the goose uses its beak in a specific motion to roll it back. If the egg is removed mid-action, the goose completes the rolling motion toward the empty nest anyway.
- **Stickleback Fish Attack:** Male sticklebacks attack any object with a red underside (red acts as the sign stimulus triggering territorial defense).

Q64) Chemical Defence

Chemical defence is an adaptation where organisms produce or accumulate volatile, distasteful, or toxic chemical compounds to deter predators and herbivores.

Mechanisms & Classification

1. **Endogenous Synthesis (Autogenous):**
 - Organisms synthesize toxins internally using metabolic pathways.
 - **Examples:**
 - Bombardier Beetle (*Brachinus*) mixes hydroquinones and hydrogen peroxide inside a reaction chamber, blasting boiling noxious gas at enemies.
 - Poison dart frogs (*Dendrobatidae*) secrete batrachotoxins through cutaneous glands.
2. **Exogenous Acquisition (Sequestration):**
 - Organisms ingest toxins from plant diets and store them safely within their tissues.
 - **Example:** Monarch butterfly caterpillars (*Danaus plexippus*) feed on milkweed (*Asclepias*) to sequester cardiac glycosides, rendering both caterpillar and adult unpalatable to birds.

Warning Mechanisms

- Chemical defence is frequently coupled with **Aposematism** (bright warning coloration) to signal unpalatability to visual predators.

Q65) Pheromones

Pheromones are ectohormones—chemical substances secreted by an individual into the external environment that elicit a specific behavioral or physiological response in other individuals of the **same species**.

Classification of Pheromones

1. **Releaser Pheromones:** Cause an immediate, direct behavioral response in the recipient.
 - *Sex Pheromones:* **Bombykol**, released by female silk moths (*Bombyx mori*), attracts male moths over long distances.
 - *Alarm Pheromones:* Released by damaged worker honeybees (isopentyl acetate) to alert the hive to defend against threats.
 - *Trail Pheromones:* Used by ants (e.g., formic acid derivatives) to mark routes to food sources.
2. **Primer Pheromones:** Initiate long-term physiological and developmental changes in the recipient via neuroendocrine systems.
 - *Queen Substance (9-ODA):* Secreted by queen honeybees to suppress ovarian development in worker bees.
 - *Bruce Effect in Mice:* Exposure to a novel male's urine causes pregnancy block in recently inseminated female mice.

Q66) Soil Erosion

Soil erosion is the detachment, transport, and deposition of topsoil—the fertile layer rich in organic matter—by physical agents like water and wind. Accelerating soil loss reduces agricultural yields and alters ecological balances.

Agents and Types of Soil Erosion

1. **Water Erosion:**
 - *Splash Erosion:* Raindrops dislodge soil particles.
 - *Sheet Erosion:* Uniform removal of thin topsoil layers over large areas.
 - *Rill & Gully Erosion:* Runoff carves narrow channels (rills) that expand into deep ravines (gullies).
2. **Wind Erosion:** Prevalent in arid zones; lifts fine particles into the air (saltation, suspension).

Causes

- Deforestation, overgrazing by livestock, intensive agricultural practices, and construction activity without soil cover.

Control Measures

- **Terrace farming** and **contour plowing** on hilly terrains.
- Planting **shelterbelts** (windbreaks) and growing ground cover crops to bind soil structure.

Q67) Water Pollution

Water pollution is the addition of organic, inorganic, biological, or thermal substances into water bodies, altering their physical, chemical, or biological properties and making them unsafe for consumption or aquatic life.

Major Types of Pollutants

- **Organic Wastes:** Domestic sewage, agricultural runoff containing animal wastes.
- **Inorganic Chemicals:** Toxic heavy metals (Pb , Hg , Cd , As), nitrates, and phosphates.
- **Pathogens:** Pathogenic bacteria (*Vibrio cholerae*), viruses, and protozoans.
- **Thermal Effluents:** Heated discharge from industrial cooling systems that lowers dissolved oxygen levels.

Excess Nutrients (N & P) → Algal Bloom → High Decomposition →
Low Dissolved Oxygen → Fish Kills

Critical Effects

- **Eutrophication:** Nutrient enrichment triggers rapid algal growth (algal blooms), causing oxygen depletion and aquatic dead zones.
- **High BOD (Biological Oxygen Demand):** High organic pollution leads to rapid oxygen consumption by microbial decomposers.

Q68) Deforestation and Desertification

Deforestation and **Desertification** are linked ecological degradation processes that destroy global vegetation and transform productive lands into barren desert environments.

Deforestation

- **Definition:** Large-scale clearance of forest cover for agriculture, timber extraction, urbanization, or mining.
- **Ecological Consequences:**
 - Loss of biodiversity and wild habitats.
 - Disruption of local hydrological cycles and reduced atmospheric transpiration.
 - Increased carbon emissions, accelerating the global greenhouse effect.

Desertification

- **Definition:** Land degradation in arid, semi-arid, and dry sub-humid areas resulting from climatic variations and human activities.
- **The Link:** Deforestation removes vegetation cover, exposing soil to wind and water erosion. Loss of organic topsoil reduces water retention, leading to land degradation and dry, desert-like conditions.

Q69) Biological Factors Affecting Toxicity

The physiological impact and severity of a toxicant depend on environmental factors and the intrinsic **biological attributes** of the organism exposed to the toxic compound.

Key Biological Factors

1. **Species Differences:** Variations in metabolic pathways affect toxicity. For example, organophosphate insecticides are more toxic to insects than to mammals because mammals possess higher concentrations of detoxifying esterase enzymes.
2. **Age & Stage of Development:** Very young (neonates) and elderly organisms are generally more susceptible due to underdeveloped or declining hepatic enzyme detoxifying systems (e.g., cytochrome P450) and immature blood-brain barriers.
3. **Sex & Hormonal Status:** Differences in body fat distribution, liver metabolism, and sex hormones influence toxicant storage and elimination rates.
4. **Health & Nutritional Status:** Malnutrition or pre-existing liver/kidney disease lowers the body's capacity to detoxify and excrete xenobiotics.
5. **Genetic Factors / Polymorphisms:** Genetic variations in metabolic enzymes cause differing responses to identical chemical exposures across populations.

Q70) Social Interference and Environmental Degradation

Social interference refers to human activities—such as industrial expansion, unstructured urbanization, resource over-exploitation, and unsustainable tourism—that disrupt natural ecosystem balances, accelerating environmental degradation.

Key Aspects of Social Interference

- **Habitat Fragmentation:** Building linear infrastructure (roads, railways, power lines) splits large wildlife ranges into small, isolated patches, causing genetic isolation and higher roadkill mortality.
- **Resource Over-exploitation:** Overfishing, groundwater extraction beyond recharge rates, and rapid logging exceed ecosystem carrying capacities.
- **Pollution Load:** Expanding consumerism generates non-biodegradable waste (plastics, e-waste) that overwhelms natural waste processing capacities.

Societal Impacts

- Leads to loss of ecosystem services (clean air, natural water filtration).

- Triggers human-wildlife conflicts as animals enter urban areas in search of food.
- Causes socio-economic displacement, creating environmental refugees.

Q71) Volcanic Eruptions and Fluid Ion Influence on Natural Resources

Volcanic eruptions release immense thermal energy along with various gases and **fluid ions** (e.g., F^- , Cl^- , SO_4^{2-} , Na^+ , K^+ , Ca^{2+} , heavy metal cations). These emissions interact directly with surrounding natural resources.

Impact on Natural Water Resources

- **Acidification:** Effluent volcanic gases containing HCl and SO_2 dissolve into surface waters, forming acid solutions that lower pH levels and harm aquatic life.
- **Fluoride Contamination:** High releases of fluoride ions (F^-) contaminate nearby groundwater and surface streams, causing fluorosis in livestock and humans.

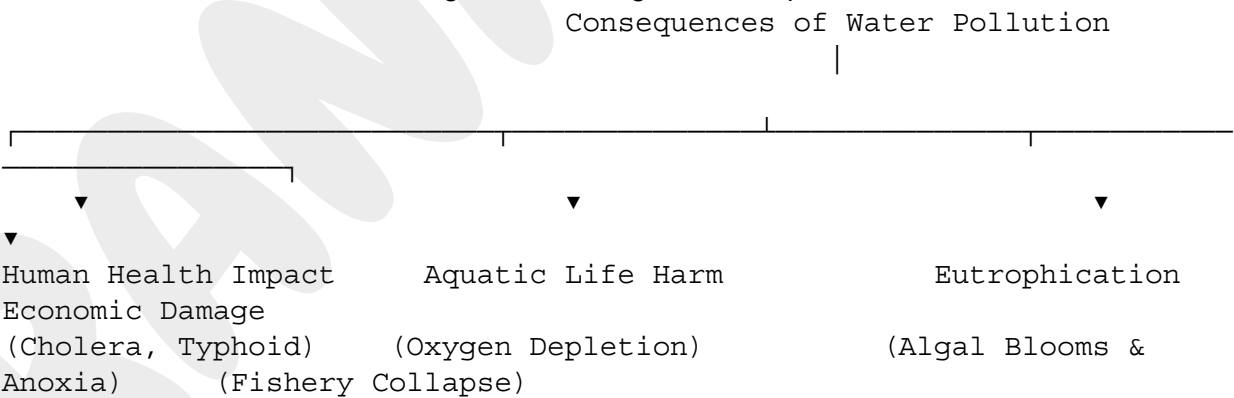
Impact on Soil & Mineral Resources

- **Short-Term Contamination:** Deposition of ash rich in heavy metal ions can temporarily poison soil and make crop yields unsafe.
- **Long-Term Soil Fertility:** Over time, volcanic ash breaks down to enrich soils with mineral ions (K^+ , Ca^{2+} , Mg^{2+}), producing rich, fertile soils ideal for agriculture.

Q72) Consequences of Water Pollution

Introduction

Discharging untreated municipal sewage, industrial effluents, and agricultural runoff into natural water bodies causes severe ecological and biological consequences.



Major Consequences

1. **Human Health Diseases:** Waterborne pathogens cause outbreaks of cholera, dysentery,

typhoid, and hepatitis. Heavy metal poisoning leads to targeted diseases:

- **Minamata Disease:** Caused by Methylmercury (Hg) consumption.
 - **Itai-Itai Disease:** Caused by Cadmium (Cd) toxicity.
2. **Eutrophication & Anoxia:** Excess nitrogen and phosphorus trigger rapid algal blooms. When algae die, microbial decomposition consumes dissolved oxygen, creating hypoxic "dead zones" where fish cannot survive.
 3. **Bioaccumulation & Biomagnification:** Persistent toxic substances (e.g., organochlorine pesticides) accumulate in lower trophic levels and concentrate as they move up the food chain.
 4. **Economic & Fishery Losses:** Degradation of aquatic ecosystems destroys local fisheries and increases water treatment costs for domestic use.

Q73) Pesticide

A **pesticide** is a chemical or biological substance used to prevent, destroy, repel, or mitigate pests such as insects, weeds, fungi, or rodents. While useful in agricultural production, excessive pesticide application causes environmental contamination.

Major Classes of Chemical Pesticides

- **Organochlorines (e.g., DDT, Endosulfan):** Highly persistent in the environment, lipid-soluble, and prone to bioaccumulation up the food chain.
- **Organophosphates (e.g., Malathion, Parathion):** Inhibit the **Acetylcholinesterase (AChE)** enzyme in synapses, causing continuous nerve stimulation.
- **Carbamates (e.g., Carbaryl):** Act similarly to organophosphates by inhibiting AChE, but their binding is reversible.
- **Synthetic Pyrethroids (e.g., Cypermethrin):** Synthetic derivatives of natural pyrethrins; lower mammalian toxicity but toxic to aquatic life.

Environmental & Biological Risks

- **Target Resistance:** Pests develop genetic resistance, requiring higher doses or stronger chemicals (the "pesticide treadmill").
- **Non-Target Toxicity:** Kills beneficial pollinators (e.g., honeybees), soil microbes, and natural predators.
- **Biomagnification:** Organochlorine residues weaken bird eggshells by interfering with calcium metabolism.

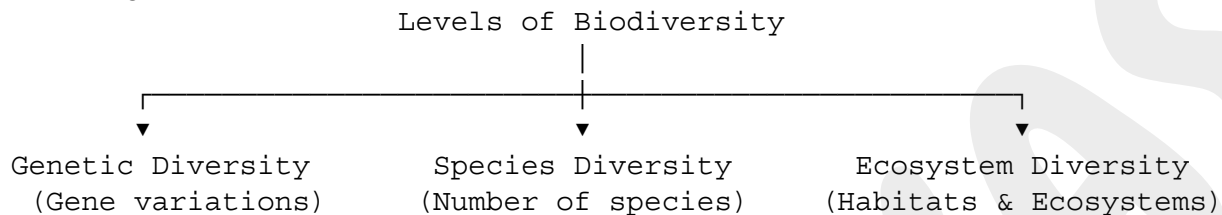
Q74) Biodiversity

Introduction

Biodiversity (biological diversity) refers to the variety and variability of life on Earth. It encompasses all biological levels, from genetic variations within species to entire functioning ecosystems.

Levels of Biodiversity

1. **Genetic Diversity:** Variation of genes within a single species (e.g., thousands of distinct rice varieties in India).
2. **Species Diversity:** Variety of different species within a region, measured by species richness and species evenness.
3. **Ecosystem Diversity:** Diversity of habitats and ecological processes across biomes (e.g., estuaries, rainforests, coral reefs, deserts).



Importance of Biodiversity

- **Ecosystem Services:** Nutrient cycling, soil formation, crop pollination, climate regulation.
- **Economic Values:** Source of food, pharmaceuticals, timber, and industrial raw materials.
- **Ecological Stability:** High species richness enhances ecosystem resilience against natural environmental disturbances.

Q75) Ozonosphere

The **Ozonosphere** (or Ozone Layer) is a region within the Earth's **stratosphere** located approximately 15 to 30 kilometers above Earth's surface. It contains a higher concentration of ozone gas (O₃) compared to other atmospheric layers.

Mechanism of Protection

The ozonosphere absorbs hazardous **Ultraviolet-B (UV-B)** and **UV-C** radiation emitted by the Sun via the **Chapman Cycle**

This conversion converts solar UV radiation into harmless heat energy.

Ozone Layer Depletion

- **Depleting Agents:** **Chlorofluorocarbons (CFCs)**, Halons, and Nitrogen Oxides (NO_x).
- **Mechanism:** Free chlorine radical released from CFCs by UV light acts as a catalyst, breaking down thousands of O₃ molecules.

Consequences of Depletion

- **Human Health:** Increased rates of skin cancers (melanoma), cataract development, and immune suppression.
- **Ecological Damage:** Inhibits photosynthesis in phytoplankton, disrupting marine food webs.

Q76) National Park and Sanctuary

(Refer to Q54 for full comparative text and details).

Key Highlights

- **National Park:** Created under strict legal protections where **all exploitation and human activities** (grazing, wood collection, private landholding) are strictly banned. Focuses on overall ecosystem conservation.
- **Wildlife Sanctuary:** Individual species-oriented preservation where **limited human activities** (traditional grazing rights, controlled timber harvesting) may be permitted by the Chief Wildlife Warden as long as they do not harm the target species.

Q77) Extinct and Endangered Species

Definitions and Comparison

[Extinct Species] → No living individual exists worldwide (e.g., Dodo)

[Endangered Species] → High risk of extinction in the near future (e.g., Asian Elephant)

1. Extinct Species

- **Definition:** A species is declared extinct when there is no reasonable doubt that its last surviving individual has died worldwide.
- **Examples:** Dodo (*Raphus cucullatus*), Passenger Pigeon (*Ectopistes migratorius*), Tasmanian Tiger (*Thylacinus cynocephalus*).
- **Causes:** Historic over-hunting, extreme habitat destruction, introduction of invasive predators.

2. Endangered Species

- **Definition:** A species categorization by the IUCN indicating a very high risk of extinction in the wild in the near future if habitat degradation or poaching pressures continue.
- **Examples:** Bengal Tiger (*Panthera tigris*), Asian Elephant (*Elephas maximus*), Great Indian Bustard (*Ardeotis nigriceps*).
- **Conservation Needs:** Immediate in-situ and ex-situ habitat protection, captive breeding, and anti-poaching enforcement.

Q78) Renewable and Non-Renewable Resources

Comparison Table

Attribute	Renewable Resources	Non-Renewable Resources
Definition	Resources that regenerate naturally over short time periods.	Resources that exist in fixed amounts and take millions of years to form.
Exhaustibility	Inexhaustible if managed sustainably.	Depleted over time through continuous extraction.
Environmental Impact	Lower carbon footprint; minimal environmental pollution.	High carbon footprint; releases greenhouse gases and pollutants.
Rate of Generation	Rate of replenishment \ge Rate of consumption.	Rate of formation \ll Rate of consumption.
Examples	Solar energy, wind power, biomass, hydro energy.	Fossil fuels (coal, crude oil, natural gas), uranium.

Q79) CNG and SMG

1. CNG (Compressed Natural Gas)

- **Composition:** Composed mainly of **Methane** (CH_4 , ~85–90%) compressed to high pressures (200–250 bar).
- **Environmental Advantages:**
 - Burns much cleaner than petrol or diesel, producing significantly lower amounts of carbon monoxide (CO), hydrocarbons, and particulate matter.
 - Leaves virtually no soot or lead emissions, helping reduce urban smog.
- **Application:** Used extensively as a alternative transport fuel in cities.

2. SMG (Synthetic Methane Gas / Substitute Natural Gas)

- **Composition:** Synthetic natural gas composed primarily of **Methane** (CH_4), produced synthetically from carbon sources.
- **Production Process:** Manufactured via the gasification of solid fossil fuels (like coal) or biomass, followed by a catalytic **methanation reaction**:

$$\text{CO} + 3\text{H}_2 \rightarrow \text{CH}_4 + \text{H}_2\text{O}$$
- **Application:** Serves as a direct pipeline substitute for natural gas in industrial fuel systems and power plants.

Q80) Biomagnification

Biomagnification (also known as bioamplification) is the biological process where persistent, non-biodegradable, lipophilic toxic substances accumulate in increasing concentrations at higher trophic levels of a food chain.

Conditions Required for Biomagnification

1. The toxicant must be persistent and long-lived in the environment.
2. The toxicant must be fat-soluble (lipophilic) and poorly metabolized or excreted by organisms.

3. The chemical must be biologically active.

Water (0.000003 ppm DDT) → Zooplankton (0.04 ppm) → Small Fish (0.5 ppm) → Large Fish (2.0 ppm) → Fish-Eating Bird (25.0 ppm)

Classic Example: DDT Biomagnification

- **DDT in Aquatic Food Chains:** Low concentrations of DDT in water absorb into zooplankton. Small fish consume large quantities of zooplankton, concentrating the DDT.
- **Top Predators:** Fish-eating birds (e.g., Ospreys, Bald Eagles) end up with high concentrations of DDT in their tissues (~25 ppm).
- **Biological Effect:** High levels of DDT inhibit the enzyme **calcium adenosine triphosphatase (Ca-ATPase)**, resulting in thin-shelled eggs that crack prematurely during incubation, causing severe population declines.

Q81) Toxicants Containing Food

Foodborne toxicants are natural or synthetic chemical substances present in food items that can cause adverse physiological reactions or chronic health damage upon ingestion.

Categories of Toxicants in Food

1. **Natural Endogenous Toxins:**
 - **Solanine:** A glycoalkaloid found in green or sprouting potatoes.
 - **Cyanogenic Glycosides:** Present in cassava roots and bitter almonds; releases toxic hydrogen cyanide (HCN) upon hydrolysis.
 - **Tetrodotoxin:** A neurotoxin present in the liver and ovaries of pufferfish (*Fugu*).
2. **Mycotoxins (Fungal Toxins):**
 - **Aflatoxins:** Produced by *Aspergillus flavus* in improperly stored peanuts, maize, and tree nuts; acts as a potent hepatotoxin and liver carcinogen.
3. **Anthropogenic / Contaminant Toxins:**
 - Heavy metals (Pb , Cd , As), pesticide residues on fruits, and adulterants added during processing.

Q82) Photochemical Smog

Photochemical Smog (also known as Los Angeles Smog) is a yellowish-brown atmospheric haze produced by the interaction of solar ultraviolet light with primary air pollutants—mainly **Nitrogen Oxides (NO_x)** and **Volatile Organic Compounds (VOCs)**—released from automobile exhaust.

Chemical Formation Process

1. Internal combustion engines release Nitric Oxide (NO), which oxidizes in air to form Nitrogen Dioxide (NO_2).
2. Sunlight splits NO_2 to release free oxygen atoms, which combine with molecular oxygen to form ground-level **Ozone O₃**.

1. Ozone reacts with unburnt hydrocarbon radicals (VOCs) to generate secondary air pollutants, including **Peroxyacetyl Nitrate (PAN)** and aldehydes.

$\text{NO}_x + \text{VOCs} + \text{Sunlight (UV)} \longrightarrow \text{Ozone (O}_3\text{)} + \text{PAN} + \text{Formaldehyde}$
(Photochemical Smog)

Health & Ecological Effects

- **Human Health:** Causes severe eye irritation, coughing, chest tightness, and respiratory damage.
- **Botanical Damage:** PAN and ground-level ozone damage plant leaf tissues (causing chlorosis and necrosis) and depress photosynthetic output.

Q83) Socio-Economic Aspects of Environmental Degradation

Environmental degradation is closely tied to socio-economic conditions. Environmental damage often stems from human economic activities, and its impacts unevenly affect different social classes.

The Vicious Cycle of Poverty and Degradation

- **Poverty as a Cause:** Impoverished communities often rely directly on local natural resources for daily survival, leading to overgrazing, deforestation for fuelwood, and soil depletion.
- **Poverty as an Effect:** Degraded soils, contaminated water, and diminished fish stocks reduce agricultural yields and income, driving affected populations deeper into poverty.

Poverty $\longrightarrow$ Dependence on Natural Resources $\longrightarrow$ Environmental
Over-exploitation $\longrightarrow$ Environmental Degradation $\longrightarrow$ Reduced Resource
Availability $\longrightarrow$ Deeper Poverty

Socio-Economic Consequences

- **Health Costs:** Low-income communities bear a disproportionate burden of exposure to industrial waste, air pollution, and unsafe drinking water, leading to higher medical expenses and lost productivity.
- **Environmental Refugees:** Land degradation and water scarcity force rural populations to migrate into urban slums, straining city infrastructure.
- **Economic Losses:** Ecosystem degradation reduces agricultural productivity, damages fisheries, and requires costly environmental remediation.

Q84) Effects of Deforestation and Overgrazing

Deforestation Effects

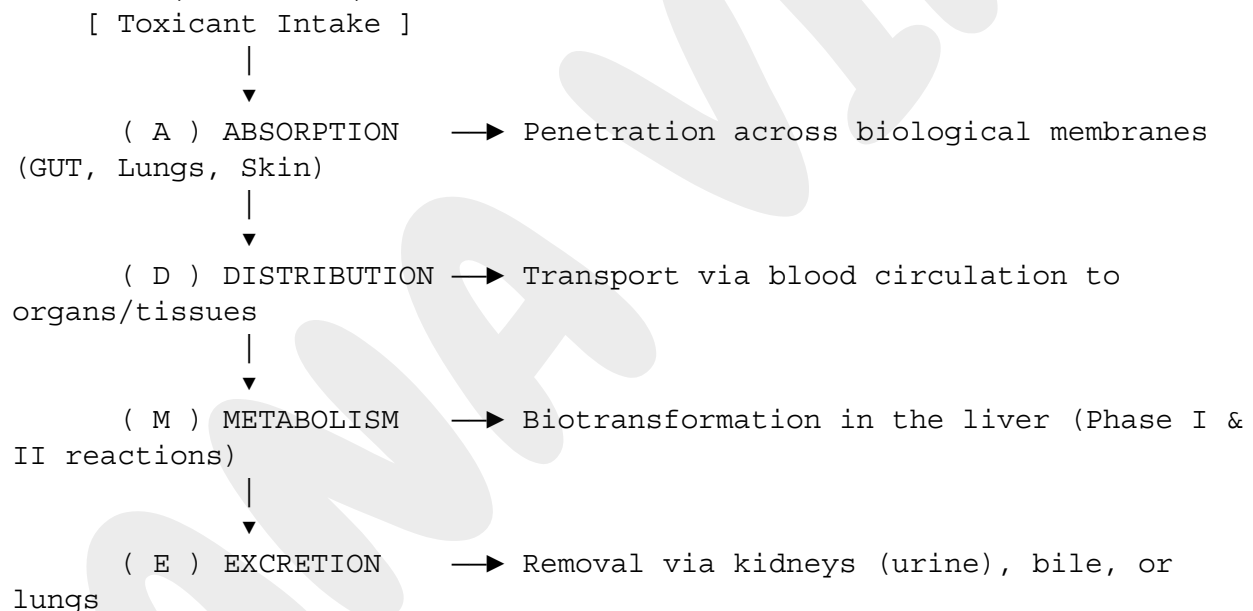
1. **Soil Erosion:** Removing tree roots exposes topsoil to rainfall and wind, washing away fertile layers.
2. **Disruption of Hydrological Cycles:** Decreased tree transpiration lowers local atmospheric humidity and rainfall, altering regional climate patterns.
3. **Biodiversity Loss:** Destroys habitat for forest species, accelerating local extinctions.

Overgrazing Effects

1. **Compaction of Soil:** Livestock hooves compact the soil surface, reducing air porosity and water infiltration rates while increasing surface runoff.
2. **Destruction of Vegetation Cover:** Persistent consumption of young shoots prevents natural plant regeneration, allowing unpalatable weeds to take over.
3. **Desertification:** The combined impact of soil exposure and compaction transforms semi-arid pastures into unproductive, desert-like lands.

Q85) Toxicokinetics

Toxicokinetics is the quantitative study of how a toxic substance enters, moves through, and leaves an organism over time. It is summarized by the **ADME** process: **A**bsorption, **D**istribution, **M**etabolism, and **E**xcretion.



The ADME Process

1. **Absorption:** The movement of a toxicant from its site of administration (gastrointestinal tract, lungs, skin) into systemic blood circulation.
2. **Distribution:** The movement of the absorbed toxicant throughout body fluids and tissues, influenced by blood flow rates and tissue binding affinities.
3. **Metabolism (Biotransformation):** Enzymatic alteration of the toxicant, primarily in the liver.

- **Phase I Reactions:** Oxidation, reduction, or hydrolysis reactions (often mediated by Cytochrome P450 enzymes) that expose polar functional groups.
 - **Phase II Reactions:** Conjugation with endogenous compounds (e.g., glutathione, glucuronic acid) to render the toxicant water-soluble.
4. **Excretion:** Elimination of the parent toxicant and its metabolites from the body, mainly via urine (kidneys), bile/feces, or exhaled air.

Q86) Water in Relation to Human Health

Water is essential for life, but contaminated or inadequate water supplies serve as a primary vector for human disease. Public health relies heavily on access to clean, safe drinking water.

Categories of Water-Related Diseases

1. **Water-Borne Diseases:** Infections transmitted by drinking water contaminated with pathogenic microorganisms.
 - *Examples:* Cholera (*Vibrio cholerae*), Typhoid fever (*Salmonella typhi*), Amoebic dysentery (*Entamoeba histolytica*).
2. **Water-Washed Diseases:** Infections caused by inadequate water supply for personal hygiene.
 - *Examples:* Trachoma (eye infection), scabies, typhus.
3. **Water-Based Diseases:** Diseases caused by aquatic invertebrate host organisms.
 - *Example:* Schistosomiasis (transmitted via freshwater snails harboring *Schistosoma cercariae*).
4. **Water-Vectored Diseases:** Diseases spread by insects that breed in stagnant water.
 - *Examples:* Malaria (*Anopheles* mosquito), Dengue (*Aedes* mosquito), Filariasis (*Culex* mosquito).

Chemical Contamination Hazards

- **Arsenicosis:** Chronic arsenic exposure leads to skin lesions and hyperkeratosis.
- **Fluorosis:** Excess fluoride intake damages teeth (dental fluorosis) and deforms bones (skeletal fluorosis).

Q87) Differentiate Between Pollutant and Toxicant

Comparison Table

Attribute	Pollutant	Toxicant
Definition	Any biological, chemical, or physical substance introduced into the environment that alters its quality or causes ecological disruption.	Any chemical agent that causes structural or functional harm to a living organism upon exposure.
Scope	Broad environmental concept; includes thermal, physical, and aesthetic degradation.	Specific biological/biochemical toxicity concept.

Attribute	Pollutant	Toxicant
Origin	Can be natural (volcanic ash) or anthropogenic (sewage, industrial smog).	Mostly chemical compounds (man-made synthetic chemicals or biological toxins).
Mechanism of Action	Damages ecosystems by altering physical environments (e.g., reducing light, depleting dissolved oxygen).	Damages organisms by disrupting physiological or biochemical pathways (e.g., enzyme inhibition).
Examples	Plastic waste, thermal hot water discharge, excessive suspended sediments.	Lead (Pb), organophosphate insecticides, Potassium Cyanide (KCN).

Q88) Differentiate Between Point and Nonpoint Pollution Sources

Comparison Table

Attribute	Point Source Pollution	Nonpoint Source Pollution
Definition	Pollutants discharged from a single, identifiable, discrete location.	Pollutants originating from diffuse, widespread areas without a single point of origin.
Traceability	Easy to trace back to a specific source.	Difficult to trace back to a single source.
Discharge Channel	Enters water bodies via pipes, ditches, channels, or smokestacks.	Enters water bodies via overland agricultural runoff, rainfall, or snowmelt.
Regulation & Control	Relatively simple to monitor, manage, and enforce legal discharge limits.	Challenging to monitor and regulate due to widespread geographical distribution.
Examples	Industrial discharge pipes, municipal sewage treatment plant outlets.	Agricultural fertilizer runoff, urban storm runoff, sediment washed from construction sites.

Q89) Parasite That Causes Dysentery and Pyorrhea

1. Parasite Causing Amoebic Dysentery

- **Causative Organism:** *Entamoeba histolytica* (Protozoan parasite, Phylum Sarcomastigophora).
- **Habitat:** Lives in the mucosal and submucosal layers of the human large intestine (colon).
- **Pathogenesis:** Secretes **histolytic** proteolytic enzymes that erode intestinal mucosal tissue, forming characteristic **flask-shaped ulcers**.
- **Symptoms:** Severe abdominal cramps, diarrhea containing blood and mucus, and fever.
- **Transmission:** Ingestion of food or water contaminated with mature tetranucleate cysts.

2. Parasite Associated with Pyorrhea

- **Causative Organism:** *Entamoeba gingivalis* (Protozoan parasite).
- **Habitat:** Found in human oral cavities, specifically within tartar deposits, gum crevices, and periodontal pockets around teeth.
- **Role in Disease:** Feeds on cellular debris, bacteria, and red blood cells. While it acts primarily as a secondary opportunist that aggravates existing **pyorrhea (periodontitis)** rather than initiating it, its presence worsens tissue inflammation, gum bleeding, and tooth socket erosion.
- **Transmission:** Transmitted directly via kiss contact, shared eating utensils, or contaminated drinking cups.

Q90) Behaviour Pattern of Health and Disease

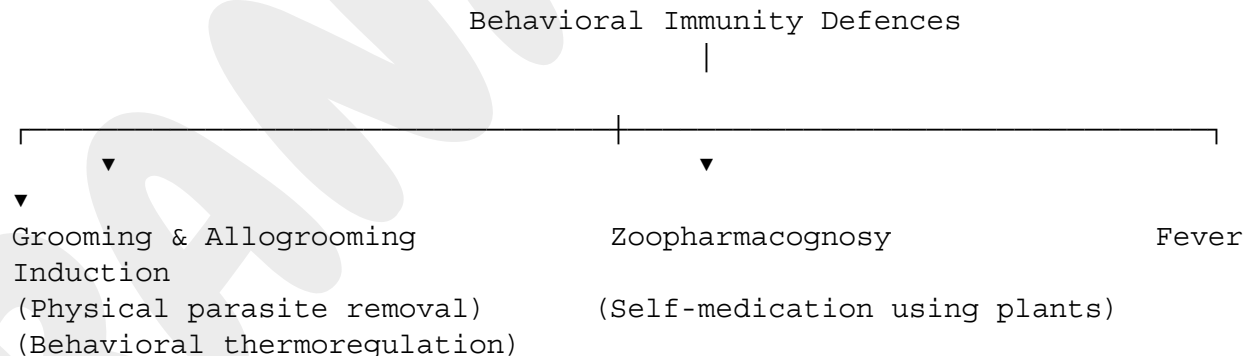
The **behaviour pattern of health and disease** explores how host animal behaviors influence susceptibility to diseases, parasite transmission, and survival outcomes. It includes both risky behaviors that increase infection risk and adaptive defensive behaviors that protect health.

1. Behaviors Increasing Disease Transmission

- **High Aggregation / Sociality:** High population density in social herds or roosts facilitates the transmission of airborne pathogens and ectoparasites.
- **Mating Displays & Contact:** Promiscuous mating behavior increases the transmission of sexually transmitted pathogens.
- **Coprophagy:** Ingestion of feces in certain mammals can introduce intestinal parasites if done indiscriminately.

2. Behavioral Host Defences (Behavioral Immunity)

Animals display specific behavioral adaptations to prevent, reduce, or cure infections:



- **Grooming and Allogrooming:** Primates and birds groom themselves and social partners to physically remove ticks, lice, and fleas.
- **Avoidance Behavior:** Grazing herbivores avoid feeding near fresh dung patches, reducing exposure to infective helminth larvae.
- **Zoopharmacognosy (Self-Medication):** Infected animals consume non-nutritive

medicinal plants. For example, chimpanzees ingest bitter pith (*Vernonia amygdalina*) containing antiparasitic compounds to purge intestinal worms.

- **Behavioral Fever:** Ectothermic animals (like grasshoppers or lizards) deliberately move to warmer microclimates to raise their body temperature, creating an unfavorable environment for invading pathogens.

Q51) Greenhouse Effect

The **Greenhouse Effect** is a natural process that warms the Earth's surface. Atmospheric gases trap solar radiation, keeping the planet warm enough to support life (average global temperature around 15°C instead of -18°C). However, excessive emissions from human activities have enhanced this effect, leading to global warming.

Key Greenhouse Gases (GHGs)

- **Carbon Dioxide (CO₂)**: Primary contributor (~60%), emitted via fossil fuel combustion and deforestation.
- **Methane (CH₄)**: Emitted from paddy fields, enteric fermentation in cattle, and landfills (25x more potent than CO₂).
- **Nitrous Oxide (N₂O)**: Originates from chemical fertilizers and agricultural practices.
- **Chlorofluorocarbons (CFCs)**: Synthetic industrial compounds used in refrigeration.

Mechanism

1. Short-wavelength solar radiation passes through the atmosphere and warms the Earth's surface.
2. The Earth re-radiates this energy as long-wavelength infrared (thermal) radiation.
3. Greenhouse gases absorb and re-emit this infrared radiation back toward Earth, trapping heat within the troposphere.

Consequences

- **Global Warming & Climate Change**: Shifts in weather patterns and extreme climate events.
- **Melting of Ice Caps**: Causes polar ice retreat and thermal expansion of oceans, leading to rising sea levels.
- **Disruption of Ecosystems**: Coral bleaching, loss of biodiversity, and altered migration routes for animals.

Q52) Chipko Movement

The **Chipko Movement** was a non-violent socio-ecological movement that started in 1973 in the Chamoli district of Uttarakhand (then part of Uttar Pradesh). The word "*Chipko*" means "*to hug*", referring to villagers clinging to trees to prevent them from being logged.

Historical Context & Leadership

- Rooted in historical precedents like the 18th-century **Bishnoi movement** led by Amrita Devi in Rajasthan.
- **Prominent Leaders**: Sundarlal Bahuguna, Chandi Prasad Bhatt, and Gaura Devi.

- **Motto:** *"What do the forests bear? Soil, water, and pure air."*

Key Objectives & Mechanism

- To stop large-scale commercial deforestation in fragile Himalayan slopes.
- To secure traditional forest rights for indigenous communities whose livelihoods depended on forest produce.
- Villagers formed human chains and hugged trees when commercial contractors arrived with axes.

Significance & Impact

- Led to a **15-year ban on green felling** in the Himalayan region ordered by Prime Minister Indira Gandhi in 1980.
- Inspired eco-movements across India, such as the **Appiko Movement** in Karnataka.
- Transformed environmental conservation into a grassroots social movement in India.

Q53) Environmental Hazards of Radiation

Radiation hazards stem from the emission of ionizing energy (Alpha, Beta, Gamma rays, X-rays) or non-ionizing energy (UV, Microwaves). Ionizing radiation possesses sufficient energy to remove tightly bound electrons from atoms, causing molecular damage in living organisms.

Sources of Exposure

- **Natural:** Cosmic rays, radon gas from soil and granite rocks.
- **Anthropogenic:** Nuclear power plant accidents (e.g., Chernobyl, Fukushima), nuclear weapons testing, medical X-rays, and improper disposal of radioactive waste.

Biological Effects on Living Organisms

1. **Somatic Damage:** Affects body cells directly, causing tissue necrosis, skin burns, radiation sickness, hair loss, and acute leukemia.
2. **Genetic Damage:** Causes gene mutations and chromosomal aberrations in gametes, leading to congenital defects in future generations.
3. **Ecosystem Biomagnification:** Radioactive isotopes like Strontium-90 (Sr^{90}) substitute for calcium in bones, while Iodine-131 (I^{131}) accumulates in thyroid tissue.

Preventive Measures

- Shielding with dense lead barriers during exposure.
- Safe containment and deep geological disposal of radioactive waste.
- Monitoring radiation worker exposure using personal dosimeters.

Q54) Difference Between a National Park and a Wildlife

Sanctuary

Comparison Table

Attribute	National Park	Wildlife Sanctuary
Primary Focus	Protection of entire ecosystem (flora, fauna, landscape, historical objects).	Protection of specific species or group of species (fauna focus).
Protection Status	Highly protected; stricter legal status under Wildlife Protection Act (1972).	Moderate protection compared to a National Park.
Human Activity	Strictly prohibited (no grazing, forestry, or private ownership allowed).	Limited activity allowed (grazing, timber harvesting with permission).
Boundaries	Fixed and clearly demarcated by legislation.	Boundaries are often not sharply demarcated.
Upgradation	Cannot be downgraded to a sanctuary.	Can be upgraded to a National Park.
Examples	Jim Corbett National Park, Kaziranga National Park.	Periyar Wildlife Sanctuary, Bharatpur Bird Sanctuary.

Q55) Noise Pollution

Noise pollution is defined as any unwanted, unpleasant, or excessive sound that interferes with human comfort, physiological well-being, or ecological communication. Sound intensity is measured in **Decibels (dB)**. Exposure above **85 dB** over long periods can cause permanent hearing impairment.

Major Sources

- **Industrial:** Heavy machinery, textile mills, air compressors.
- **Transport:** Jet aircraft take-offs (130–140 dB), automotive traffic honking, railway engines.
- **Domestic/Community:** Loudspeakers, construction site drills, firecrackers.

Physiological & Psychological Impacts

- **Auditory Damage:** Noise-induced hearing loss (NIHL), acoustic trauma, and persistent tinnitus.
- **Non-Auditory Damage:**
 - Elevated blood pressure, hypertension, and increased cardiovascular stress.
 - Insomnia, fatigue, and psychological irritability.
 - Disruption of animal echolocation and acoustic communication in birds and marine mammals.

Control Measures

- Installation of silencers and acoustic baffles in heavy machinery.
- Creation of **Green Belts** (planting dense trees along highways to absorb noise).
- Setting legal decibel limits and strict zoning laws for Quiet Zones (near hospitals/schools).

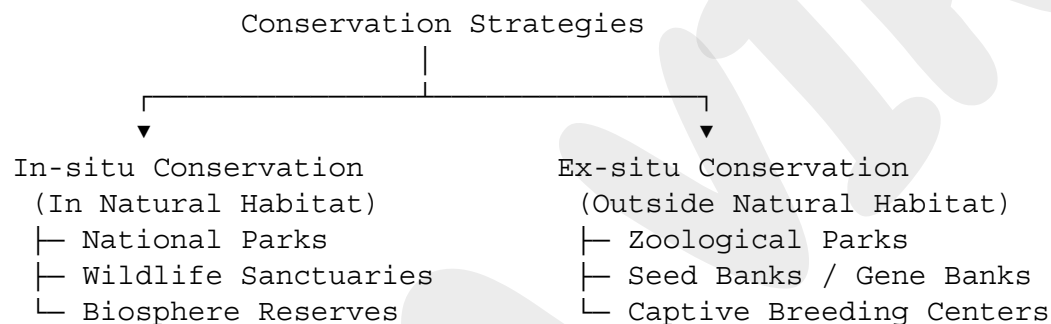
Q56) Conservation of Endangered Species

An **endangered species** is a species categorized by the IUCN (International Union for Conservation of Nature) as facing a very high risk of extinction in the wild. Protecting these species requires multi-tiered conservation strategies.

Causes of Species Decline

- Habitat loss, fragmentation, and degradation.
- Over-exploitation, poaching, and illegal wildlife trade.
- Invasive alien species competition and climate change.

Conservation Strategies



1. In-situ Conservation (On-Site):

- Protecting the species within its natural habitat through National Parks, Sanctuaries, and Biosphere Reserves.
- *Example:* Project Tiger, Project Elephant.

2. Ex-situ Conservation (Off-Site):

- Preserving threatened species outside their natural habitats.
- *Examples:* Zoological gardens, botanical gardens, gene/seed banks, and cryopreservation of gametes.

Legal Measures

- Enforcement of the **Wildlife Protection Act, 1972** in India.
- Implementation of **CITES** (Convention on International Trade in Endangered Species) protocols globally.

Q57) Volcanic Eruptions

A **volcanic eruption** is a geological process where molten rock (magma), ash, steam, and

volatile gases escape to the Earth's surface through a rupture or vent in the crust.

Environmental Effects

- **Atmospheric Impact:** Ash clouds contain Sulfur Dioxide (SO₂), Carbon Dioxide (CO₂), and Hydrogen Fluoride (HF). SO₂ reacts with moisture to form volcanic aerosols that reflect sunlight, leading to **volcanic winter** (short-term global cooling).
- **Acid Rain:** Volcanic gases form atmospheric acids, resulting in localized acid precipitation.
- **Habitat Destruction:** Pyroclastic flows, lahars (mudflows), and basalt lava burn and bury ecosystems instantaneously.

Ecological Restoration & Succession

- **Primary Succession:** Volcanic lava forms bare rock surfaces. Over time, pioneer organisms like lichens and mosses break down rock into soil, initiating ecological succession that eventually restores flora and fauna.

Q58) Urbanization Stress and Health

Rapid **urbanization** leads to population concentration in cities, often outstripping infrastructure. This transformation exerts structural stress on natural systems and creates health hazards for urban populations.

Stressors Associated with Urbanization

1. **Environmental Degradation:** Concentrated air pollution (PM_{2.5}, PM₁₀), industrial effluents, and urban heat island (UHI) effects.
2. **High Population Density:** Leads to slum development, inadequate sanitation, and water contamination.
3. **Loss of Green Spaces:** Deprives urban populations of natural ecosystem services and air filtration.

Impact on Human & Animal Health

- **Respiratory Diseases:** Higher rates of asthma, bronchitis, and lung diseases due to urban smog.
- **Vector-borne Diseases:** Poor drainage creates breeding sites for *Aedes* and *Anopheles* mosquitoes, increasing dengue and malaria transmission.
- **Psychological Stress:** Overcrowding, light pollution, and noise pollution elevate cortisol levels, contributing to anxiety and cardiovascular disorders.
- **Urban Wildlife Alterations:** Species adapt behaviorally or suffer local extinctions; synanthropic animals (e.g., pigeons, rats) proliferate while native species decline.

Q59) Environmental Monitoring

Environmental Monitoring is the systematic collection of physical, chemical, and biological

data using standardized quantitative techniques to assess environmental quality, detect degradation, and manage pollution.

Types of Monitoring

1. **Air Quality Monitoring:** Measures levels of criteria pollutants like SO_x, NO_x, ground-level O₃, carbon monoxide, and particulate matter (PM_{2.5}, PM₁₀).
2. **Water Quality Monitoring:** Evaluates parameters such as dissolved oxygen (DO), biological oxygen demand (BOD), chemical oxygen demand (COD), pH, heavy metal concentration, and coliform counts.
3. **Biological Monitoring (Biomonitoring):** Uses living organisms to assess ecosystem health.
 - *Example:* Lichens act as bioindicators for SO₂ air pollution; benthic macroinvertebrates indicate stream water quality.

Applications & Importance

- Establishes baseline data for Environmental Impact Assessments (EIA).
- Provides early warning systems for ecological disasters and toxic leaks.
- Helps regulatory agencies (e.g., CPCB) enforce environmental standards and policies.
-

Q60) Nesting Behaviour

Nesting behaviour encompasses the sequence of activities performed by animals—particularly birds, insects, and fishes—to construct, maintain, and protect a structure (nest) used for egg laying, incubation, and rearing offspring.

Types & Examples

- **Avian Nests:** Range from simple scrapes on the ground (e.g., plovers) to elaborate woven structures (e.g., Weaverbird *Ploceus philippinus* using intricate knots) and mud nests (e.g., swallows).
- **Fish Nests:** The male Three-spined Stickleback (*Gasterosteus aculeatus*) uses kidney secretions (spiggin) to bind plant materials into a tubular nest.
- **Insect Nests:** Social insects like termites (*Macrotermes*) build complex mud mounds with internal ventilation channels, while honeybees construct hexagonal wax cells.

Hormonal & Evolutionary Control

- Triggered by environmental cues (photoperiod, temperature) that induce hormonal secretion, primarily **prolactin** and **estrogen**.
- Serves crucial evolutionary purposes: protection from predators, microclimate regulation for embryo development, and reproductive success.

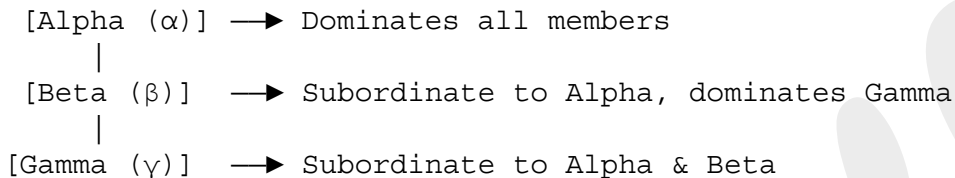
Q61) Social Hierarchies

A **social hierarchy** (or dominance hierarchy) is an organizational structure within an animal

group where individuals are ranked relative to one another. This arrangement reduces energy expenditure and injury from physical combat over resources.

Types of Hierarchies

1. **Linear Hierarchy (Pecking Order):** A clear rank order ($A > B > C > D$) where individual A dominates all, B dominates all except A, and so on. Classic example: chickens established by Schjelderup-Ebbe.
2. **Complex / Despotic Hierarchy:** One dominant individual (Alpha) rules over all other group members, with little clear structure among subordinates (e.g., wolf packs).



Mechanisms & Maintenance

- Established initially through physical contests, threats, or displays.
- Maintained via submissive postures, vocalizations, and behavioral rituals without ongoing actual fights.
- Dominant individuals gain priority access to food, mates, and prime shelter locations.

Q62) Territory Marking

A **territory** is a defended space that an animal or group uses exclusively for feeding, mating, or rearing young. **Territory marking** involves signaling mechanisms used to convey ownership to competitors without engaging in direct combat.

Modes of Territory Marking

1. **Chemical Marking (Scent Marking):**
 - Mammals use specialized glands or bodily fluids (urine, feces).
 - *Examples:* Canids (dogs/wolves) use urine scent posts; felids (tigers) spray urine mixed with glandular secretions; deer use preorbital glands on branches.
2. **Visual Marking:**
 - Visible changes made to the environment.
 - *Examples:* Bears clawing bark off trees; antelopes creating dung heaps; visual display of bright coloration in territorial reef fish.
3. **Acoustic Marking:**
 - Long-distance vocalizations declaring space occupied.
 - *Examples:* Bird song displays at dawn, howler monkey choruses in rainforest canopy, lion roars.

Q63) Fixed Action Patterns (FAP)

A **Fixed Action Pattern (FAP)** is an unlearned, innate, highly stereotyped sequence of behavioral acts defined by ethologists Konrad Lorenz and Niko Tinbergen. Once triggered, it runs to completion even if the triggering stimulus is removed.

Key Characteristics

1. **Innate:** Genetically hardwired, requiring no prior experience or learning.
2. **Stereotyped:** Performed in a uniform manner across all members of a species.
3. **Triggered by Sign Stimulus:** Initiated by a specific external cue known as a **Sign Stimulus** or **Releaser**.
4. **Independent of Feedback:** Once initiated, the action sequence completes regardless of environmental changes.

External Cue (Sign Stimulus) → Innate Releasing Mechanism (IRM) → Fixed Action Pattern (FAP)

Classic Examples

- **Egg-Rolling in Greylag Goose:** If an egg rolls out of the nest, the goose uses its beak in a specific motion to roll it back. If the egg is removed mid-action, the goose completes the rolling motion toward the empty nest anyway.
- **Stickleback Fish Attack:** Male sticklebacks attack any object with a red underside (red acts as the sign stimulus triggering territorial defense).

Q64) Chemical Defence

Chemical defence is an adaptation where organisms produce or accumulate volatile, distasteful, or toxic chemical compounds to deter predators and herbivores.

Mechanisms & Classification

1. **Endogenous Synthesis (Autogenous):**
 - Organisms synthesize toxins internally using metabolic pathways.
 - *Examples:*
 - Bombardier Beetle (*Brachinus*) mixes hydroquinones and hydrogen peroxide inside a reaction chamber, blasting boiling noxious gas at enemies.
 - Poison dart frogs (*Dendrobatidae*) secrete batrachotoxins through cutaneous glands.
2. **Exogenous Acquisition (Sequestration):**
 - Organisms ingest toxins from plant diets and store them safely within their tissues.
 - *Example:* Monarch butterfly caterpillars (*Danaus plexippus*) feed on milkweed (*Asclepias*) to sequester cardiac glycosides, rendering both caterpillar and adult unpalatable to birds.

Warning Mechanisms

- Chemical defence is frequently coupled with **Aposematism** (bright warning coloration) to

signal unpalatability to visual predators.

Q65) Pheromones

Pheromones are ectohormones—chemical substances secreted by an individual into the external environment that elicit a specific behavioral or physiological response in other individuals of the **same species**.

Classification of Pheromones

1. **Releaser Pheromones:** Cause an immediate, direct behavioral response in the recipient.
 - *Sex Pheromones:* **Bombykol**, released by female silk moths (*Bombyx mori*), attracts male moths over long distances.
 - *Alarm Pheromones:* Released by damaged worker honeybees (isopentyl acetate) to alert the hive to defend against threats.
 - *Trail Pheromones:* Used by ants (e.g., formic acid derivatives) to mark routes to food sources.
2. **Primer Pheromones:** Initiate long-term physiological and developmental changes in the recipient via neuroendocrine systems.
 - *Queen Substance (9-ODA):* Secreted by queen honeybees to suppress ovarian development in worker bees.
 - *Bruce Effect in Mice:* Exposure to a novel male's urine causes pregnancy block in recently inseminated female mice.

Q66) Soil Erosion

Soil erosion is the detachment, transport, and deposition of topsoil—the fertile layer rich in organic matter—by physical agents like water and wind. Accelerating soil loss reduces agricultural yields and alters ecological balances.

Agents and Types of Soil Erosion

1. **Water Erosion:**
 - *Splash Erosion:* Raindrops dislodge soil particles.
 - *Sheet Erosion:* Uniform removal of thin topsoil layers over large areas.
 - *Rill & Gully Erosion:* Runoff carves narrow channels (rills) that expand into deep ravines (gullies).
2. **Wind Erosion:** Prevalent in arid zones; lifts fine particles into the air (saltation, suspension).

Causes

- Deforestation, overgrazing by livestock, intensive agricultural practices, and construction activity without soil cover.

Control Measures

- **Terrace farming** and **contour plowing** on hilly terrains.

- Planting **shelterbelts** (windbreaks) and growing ground cover crops to bind soil structure.

Q67) Water Pollution

Water pollution is the addition of organic, inorganic, biological, or thermal substances into water bodies, altering their physical, chemical, or biological properties and making them unsafe for consumption or aquatic life.

Major Types of Pollutants

- **Organic Wastes:** Domestic sewage, agricultural runoff containing animal wastes.
- **Inorganic Chemicals:** Toxic heavy metals (Pb , Hg , Cd , As), nitrates, and phosphates.
- **Pathogens:** Pathogenic bacteria (*Vibrio cholerae*), viruses, and protozoans.
- **Thermal Effluents:** Heated discharge from industrial cooling systems that lowers dissolved oxygen levels.

Excess Nutrients (N & P) → Algal Bloom → High Decomposition → Low Dissolved Oxygen → Fish Kills

Critical Effects

- **Eutrophication:** Nutrient enrichment triggers rapid algal growth (algal blooms), causing oxygen depletion and aquatic dead zones.
- **High BOD (Biological Oxygen Demand):** High organic pollution leads to rapid oxygen consumption by microbial decomposers.

Q68) Deforestation and Desertification

Deforestation and **Desertification** are linked ecological degradation processes that destroy global vegetation and transform productive lands into barren desert environments.

Deforestation

- **Definition:** Large-scale clearance of forest cover for agriculture, timber extraction, urbanization, or mining.
- **Ecological Consequences:**
 - Loss of biodiversity and wild habitats.
 - Disruption of local hydrological cycles and reduced atmospheric transpiration.
 - Increased carbon emissions, accelerating the global greenhouse effect.

Desertification

- **Definition:** Land degradation in arid, semi-arid, and dry sub-humid areas resulting from climatic variations and human activities.
- **The Link:** Deforestation removes vegetation cover, exposing soil to wind and water erosion. Loss of organic topsoil reduces water retention, leading to land degradation and

dry, desert-like conditions.

Q69) Biological Factors Affecting Toxicity

The physiological impact and severity of a toxicant depend on environmental factors and the intrinsic **biological attributes** of the organism exposed to the toxic compound.

Key Biological Factors

1. **Species Differences:** Variations in metabolic pathways affect toxicity. For example, organophosphate insecticides are more toxic to insects than to mammals because mammals possess higher concentrations of detoxifying esterase enzymes.
2. **Age & Stage of Development:** Very young (neonates) and elderly organisms are generally more susceptible due to underdeveloped or declining hepatic enzyme detoxifying systems (e.g., cytochrome P450) and immature blood-brain barriers.
3. **Sex & Hormonal Status:** Differences in body fat distribution, liver metabolism, and sex hormones influence toxicant storage and elimination rates.
4. **Health & Nutritional Status:** Malnutrition or pre-existing liver/kidney disease lowers the body's capacity to detoxify and excrete xenobiotics.
5. **Genetic Factors / Polymorphisms:** Genetic variations in metabolic enzymes cause differing responses to identical chemical exposures across populations.

Q70) Social Interference and Environmental Degradation

Social interference refers to human activities—such as industrial expansion, unstructured urbanization, resource over-exploitation, and unsustainable tourism—that disrupt natural ecosystem balances, accelerating environmental degradation.

Key Aspects of Social Interference

- **Habitat Fragmentation:** Building linear infrastructure (roads, railways, power lines) splits large wildlife ranges into small, isolated patches, causing genetic isolation and higher roadkill mortality.
- **Resource Over-exploitation:** Overfishing, groundwater extraction beyond recharge rates, and rapid logging exceed ecosystem carrying capacities.
- **Pollution Load:** Expanding consumerism generates non-biodegradable waste (plastics, e-waste) that overwhelms natural waste processing capacities.

Societal Impacts

- Leads to loss of ecosystem services (clean air, natural water filtration).
- Triggers human-wildlife conflicts as animals enter urban areas in search of food.
- Causes socio-economic displacement, creating environmental refugees.

Q71) Volcanic Eruptions and Fluid Ion Influence on

Natural Resources

Volcanic eruptions release immense thermal energy along with various gases and **fluid ions** (e.g., F^- , Cl^- , SO_4^{2-} , Na^+ , K^+ , Ca^{2+} , heavy metal cations). These emissions interact directly with surrounding natural resources.

Impact on Natural Water Resources

- **Acidification:** Effluent volcanic gases containing HCl and SO_2 dissolve into surface waters, forming acid solutions that lower pH levels and harm aquatic life.
- **Fluoride Contamination:** High releases of fluoride ions (F^-) contaminate nearby groundwater and surface streams, causing fluorosis in livestock and humans.

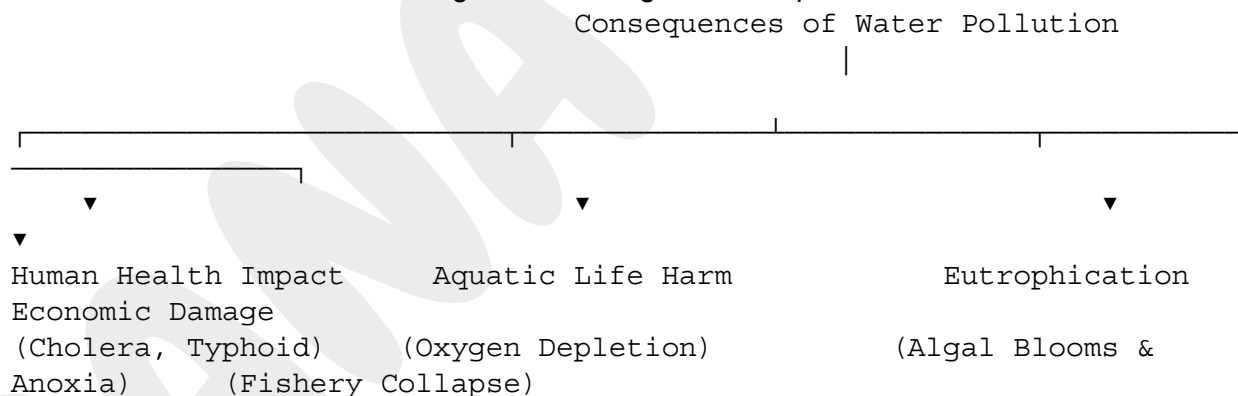
Impact on Soil & Mineral Resources

- **Short-Term Contamination:** Deposition of ash rich in heavy metal ions can temporarily poison soil and make crop yields unsafe.
- **Long-Term Soil Fertility:** Over time, volcanic ash breaks down to enrich soils with mineral ions (K^+ , Ca^{2+} , Mg^{2+}), producing rich, fertile soils ideal for agriculture.

Q72) Consequences of Water Pollution

Introduction

Discharging untreated municipal sewage, industrial effluents, and agricultural runoff into natural water bodies causes severe ecological and biological consequences.



Major Consequences

1. **Human Health Diseases:** Waterborne pathogens cause outbreaks of cholera, dysentery, typhoid, and hepatitis. Heavy metal poisoning leads to targeted diseases:
 - **Minamata Disease:** Caused by Methylmercury (Hg) consumption.
 - **Itai-Itai Disease:** Caused by Cadmium (Cd) toxicity.
2. **Eutrophication & Anoxia:** Excess nitrogen and phosphorus trigger rapid algal blooms. When algae die, microbial decomposition consumes dissolved oxygen, creating hypoxic

"dead zones" where fish cannot survive.

3. **Bioaccumulation & Biomagnification:** Persistent toxic substances (e.g., organochlorine pesticides) accumulate in lower trophic levels and concentrate as they move up the food chain.
4. **Economic & Fishery Losses:** Degradation of aquatic ecosystems destroys local fisheries and increases water treatment costs for domestic use.

Q73) Pesticide

A **pesticide** is a chemical or biological substance used to prevent, destroy, repel, or mitigate pests such as insects, weeds, fungi, or rodents. While useful in agricultural production, excessive pesticide application causes environmental contamination.

Major Classes of Chemical Pesticides

- **Organochlorines (e.g., DDT, Endosulfan):** Highly persistent in the environment, lipid-soluble, and prone to bioaccumulation up the food chain.
- **Organophosphates (e.g., Malathion, Parathion):** Inhibit the **Acetylcholinesterase (AChE)** enzyme in synapses, causing continuous nerve stimulation.
- **Carbamates (e.g., Carbaryl):** Act similarly to organophosphates by inhibiting AChE, but their binding is reversible.
- **Synthetic Pyrethroids (e.g., Cypermethrin):** Synthetic derivatives of natural pyrethrins; lower mammalian toxicity but toxic to aquatic life.

Environmental & Biological Risks

- **Target Resistance:** Pests develop genetic resistance, requiring higher doses or stronger chemicals (the "pesticide treadmill").
- **Non-Target Toxicity:** Kills beneficial pollinators (e.g., honeybees), soil microbes, and natural predators.
- **Biomagnification:** Organochlorine residues weaken bird eggshells by interfering with calcium metabolism.

Q74) Biodiversity

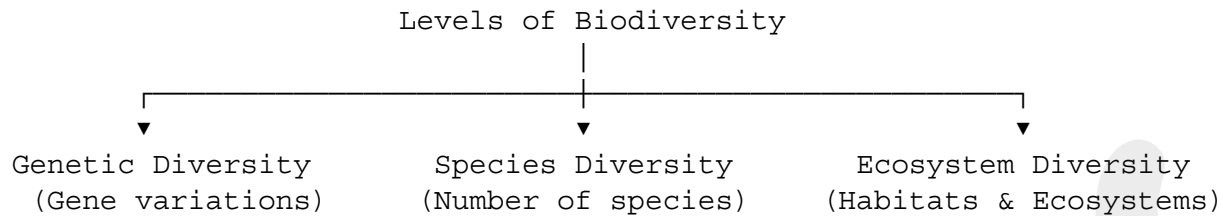
Introduction

Biodiversity (biological diversity) refers to the variety and variability of life on Earth. It encompasses all biological levels, from genetic variations within species to entire functioning ecosystems.

Levels of Biodiversity

1. **Genetic Diversity:** Variation of genes within a single species (e.g., thousands of distinct rice varieties in India).
2. **Species Diversity:** Variety of different species within a region, measured by species richness and species evenness.

3. **Ecosystem Diversity:** Diversity of habitats and ecological processes across biomes (e.g., estuaries, rainforests, coral reefs, deserts).



Importance of Biodiversity

- **Ecosystem Services:** Nutrient cycling, soil formation, crop pollination, climate regulation.
- **Economic Values:** Source of food, pharmaceuticals, timber, and industrial raw materials.
- **Ecological Stability:** High species richness enhances ecosystem resilience against natural environmental disturbances.

Q75) Ozonosphere

The **Ozonosphere** (or Ozone Layer) is a region within the Earth's **stratosphere** located approximately 15 to 30 kilometers above Earth's surface. It contains a higher concentration of ozone gas (O₃) compared to other atmospheric layers.

Mechanism of Protection

The ozonosphere absorbs hazardous **Ultraviolet-B (UV-B)** and **UV-C** radiation emitted by the Sun via the **Chapman Cycle**

This conversion converts solar UV radiation into harmless heat energy.

Ozone Layer Depletion

- **Depleting Agents:** **Chlorofluorocarbons (CFCs)**, Halons, and Nitrogen Oxides (NO_x).
- **Mechanism:** Free chlorine radical released from CFCs by UV light acts as a catalyst, breaking down thousands of O₃ molecules.

Consequences of Depletion

- **Human Health:** Increased rates of skin cancers (melanoma), cataract development, and immune suppression.
- **Ecological Damage:** Inhibits photosynthesis in phytoplankton, disrupting marine food webs.

Q76) National Park and Sanctuary

(Refer to Q54 for full comparative text and details).

Key Highlights

- **National Park:** Created under strict legal protections where **all exploitation and human activities** (grazing, wood collection, private landholding) are strictly banned. Focuses on overall ecosystem conservation.
- **Wildlife Sanctuary:** Individual species-oriented preservation where **limited human activities** (traditional grazing rights, controlled timber harvesting) may be permitted by the Chief Wildlife Warden as long as they do not harm the target species.

Q77) Extinct and Endangered Species

Definitions and Comparison

[Extinct Species] → No living individual exists worldwide (e.g., Dodo)

[Endangered Species] → High risk of extinction in the near future (e.g., Asian Elephant)

1. Extinct Species

- **Definition:** A species is declared extinct when there is no reasonable doubt that its last surviving individual has died worldwide.
- **Examples:** Dodo (*Raphus cucullatus*), Passenger Pigeon (*Ectopistes migratorius*), Tasmanian Tiger (*Thylacinus cynocephalus*).
- **Causes:** Historic over-hunting, extreme habitat destruction, introduction of invasive predators.

2. Endangered Species

- **Definition:** A species categorization by the IUCN indicating a very high risk of extinction in the wild in the near future if habitat degradation or poaching pressures continue.
- **Examples:** Bengal Tiger (*Panthera tigris*), Asian Elephant (*Elephas maximus*), Great Indian Bustard (*Ardeotis nigriceps*).
- **Conservation Needs:** Immediate in-situ and ex-situ habitat protection, captive breeding, and anti-poaching enforcement.

Q78) Renewable and Non-Renewable Resources

Comparison Table

Attribute	Renewable Resources	Non-Renewable Resources
Definition	Resources that regenerate naturally over short time periods.	Resources that exist in fixed amounts and take millions of years to form.
Exhaustibility	Inexhaustible if managed	Depleted over time through

Attribute	Renewable Resources	Non-Renewable Resources
	sustainably.	continuous extraction.
Environmental Impact	Lower carbon footprint; minimal environmental pollution.	High carbon footprint; releases greenhouse gases and pollutants.
Rate of Generation	Rate of replenishment \ge Rate of consumption.	Rate of formation \ll Rate of consumption.
Examples	Solar energy, wind power, biomass, hydro energy.	Fossil fuels (coal, crude oil, natural gas), uranium.

Q79) CNG and SMG

1. CNG (Compressed Natural Gas)

- **Composition:** Composed mainly of **Methane** (CH_4 , ~85–90%) compressed to high pressures (200–250 bar).
- **Environmental Advantages:**
 - Burns much cleaner than petrol or diesel, producing significantly lower amounts of carbon monoxide (CO), hydrocarbons, and particulate matter.
 - Leaves virtually no soot or lead emissions, helping reduce urban smog.
- **Application:** Used extensively as a alternative transport fuel in cities.

2. SMG (Synthetic Methane Gas / Substitute Natural Gas)

- **Composition:** Synthetic natural gas composed primarily of **Methane** (CH_4), produced synthetically from carbon sources.
- **Production Process:** Manufactured via the gasification of solid fossil fuels (like coal) or biomass, followed by a catalytic **methanation reaction**:

$$\text{CO} + 3\text{H}_2 \rightarrow \text{CH}_4 + \text{H}_2\text{O}$$
- **Application:** Serves as a direct pipeline substitute for natural gas in industrial fuel systems and power plants.

Q80) Biomagnification

Biomagnification (also known as bioamplification) is the biological process where persistent, non-biodegradable, lipophilic toxic substances accumulate in increasing concentrations at higher trophic levels of a food chain.

Conditions Required for Biomagnification

1. The toxicant must be persistent and long-lived in the environment.
2. The toxicant must be fat-soluble (lipophilic) and poorly metabolized or excreted by organisms.
3. The chemical must be biologically active.

Water (0.000003 ppm DDT) → Zooplankton (0.04 ppm) → Small Fish (0.5 ppm) → Large Fish (2.0 ppm) → Fish-Eating Bird (25.0 ppm)

Classic Example: DDT Biomagnification

- **DDT in Aquatic Food Chains:** Low concentrations of DDT in water absorb into zooplankton. Small fish consume large quantities of zooplankton, concentrating the DDT.
- **Top Predators:** Fish-eating birds (e.g., Ospreys, Bald Eagles) end up with high concentrations of DDT in their tissues (~25 ppm).
- **Biological Effect:** High levels of DDT inhibit the enzyme **calcium adenosine triphosphatase (Ca-ATPase)**, resulting in thin-shelled eggs that crack prematurely during incubation, causing severe population declines.

Q81) Toxicants Containing Food

Foodborne toxicants are natural or synthetic chemical substances present in food items that can cause adverse physiological reactions or chronic health damage upon ingestion.

Categories of Toxicants in Food

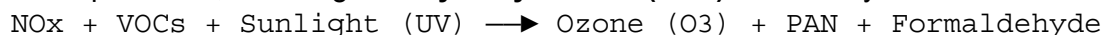
1. **Natural Endogenous Toxins:**
 - **Solanine:** A glycoalkaloid found in green or sprouting potatoes.
 - **Cyanogenic Glycosides:** Present in cassava roots and bitter almonds; releases toxic hydrogen cyanide (HCN) upon hydrolysis.
 - **Tetrodotoxin:** A neurotoxin present in the liver and ovaries of pufferfish (*Fugu*).
2. **Mycotoxins (Fungal Toxins):**
 - **Aflatoxins:** Produced by *Aspergillus flavus* in improperly stored peanuts, maize, and tree nuts; acts as a potent hepatotoxin and liver carcinogen.
3. **Anthropogenic / Contaminant Toxins:**
 - Heavy metals (Pb, Cd, As), pesticide residues on fruits, and adulterants added during processing.

Q82) Photochemical Smog

Photochemical Smog (also known as Los Angeles Smog) is a yellowish-brown atmospheric haze produced by the interaction of solar ultraviolet light with primary air pollutants—mainly **Nitrogen Oxides (NO_x)** and **Volatile Organic Compounds (VOCs)**—released from automobile exhaust.

Chemical Formation Process

1. Internal combustion engines release Nitric Oxide (NO), which oxidizes in air to form Nitrogen Dioxide (NO_2).
2. Sunlight splits NO_2 to release free oxygen atoms, which combine with molecular oxygen to form ground-level **Ozone (O_3)**.
1. Ozone reacts with unburnt hydrocarbon radicals (VOCs) to generate secondary air pollutants, including **Peroxyacetyl Nitrate (PAN)** and aldehydes.



(Photochemical Smog)

Health & Ecological Effects

- **Human Health:** Causes severe eye irritation, coughing, chest tightness, and respiratory damage.
- **Botanical Damage:** PAN and ground-level ozone damage plant leaf tissues (causing chlorosis and necrosis) and depress photosynthetic output.

Q83) Socio-Economic Aspects of Environmental Degradation

Environmental degradation is closely tied to socio-economic conditions. Environmental damage often stems from human economic activities, and its impacts unevenly affect different social classes.

The Vicious Cycle of Poverty and Degradation

- **Poverty as a Cause:** Impoverished communities often rely directly on local natural resources for daily survival, leading to overgrazing, deforestation for fuelwood, and soil depletion.
- **Poverty as an Effect:** Degraded soils, contaminated water, and diminished fish stocks reduce agricultural yields and income, driving affected populations deeper into poverty.

Poverty → Dependence on Natural Resources → Environmental Over-exploitation → Environmental Degradation → Reduced Resource Availability → Deeper Poverty

Socio-Economic Consequences

- **Health Costs:** Low-income communities bear a disproportionate burden of exposure to industrial waste, air pollution, and unsafe drinking water, leading to higher medical expenses and lost productivity.
- **Environmental Refugees:** Land degradation and water scarcity force rural populations to migrate into urban slums, straining city infrastructure.
- **Economic Losses:** Ecosystem degradation reduces agricultural productivity, damages fisheries, and requires costly environmental remediation.

Q84) Effects of Deforestation and Overgrazing

Deforestation Effects

1. **Soil Erosion:** Removing tree roots exposes topsoil to rainfall and wind, washing away fertile layers.
2. **Disruption of Hydrological Cycles:** Decreased tree transpiration lowers local

atmospheric humidity and rainfall, altering regional climate patterns.

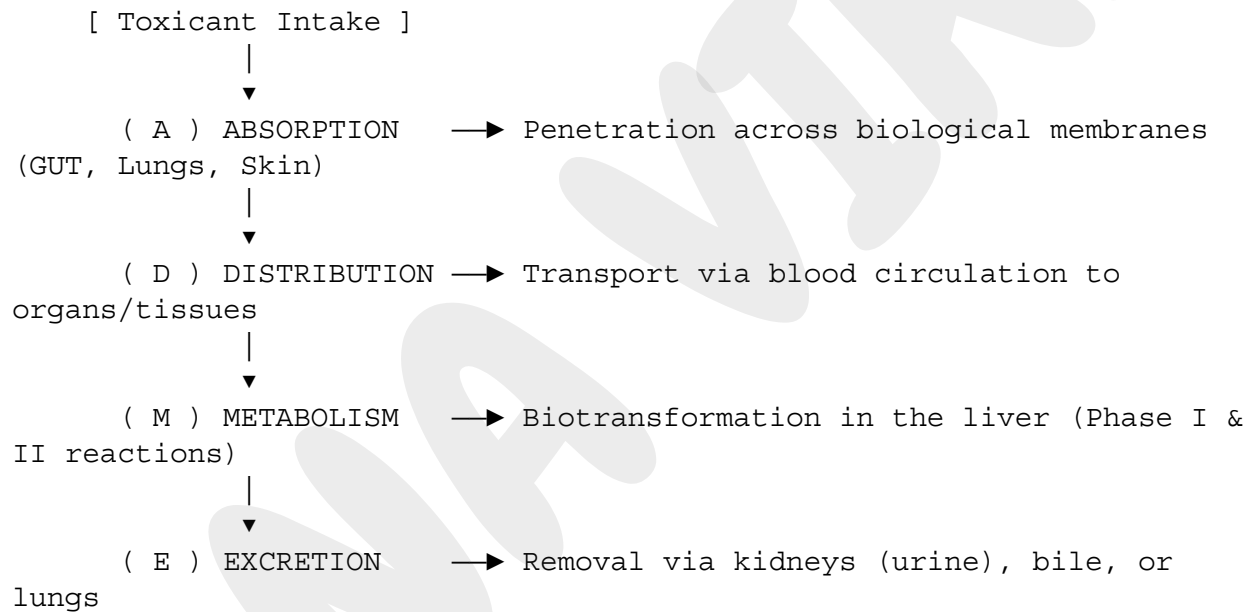
3. **Biodiversity Loss:** Destroys habitat for forest species, accelerating local extinctions.

Overgrazing Effects

1. **Compaction of Soil:** Livestock hooves compact the soil surface, reducing air porosity and water infiltration rates while increasing surface runoff.
2. **Destruction of Vegetation Cover:** Persistent consumption of young shoots prevents natural plant regeneration, allowing unpalatable weeds to take over.
3. **Desertification:** The combined impact of soil exposure and compaction transforms semi-arid pastures into unproductive, desert-like lands.

Q85) Toxicokinetics

Toxicokinetics is the quantitative study of how a toxic substance enters, moves through, and leaves an organism over time. It is summarized by the **ADME** process: **A**bsorption, **D**istribution, **M**etabolism, and **E**xcretion.



The ADME Process

1. **Absorption:** The movement of a toxicant from its site of administration (gastrointestinal tract, lungs, skin) into systemic blood circulation.
2. **Distribution:** The movement of the absorbed toxicant throughout body fluids and tissues, influenced by blood flow rates and tissue binding affinities.
3. **Metabolism (Biotransformation):** Enzymatic alteration of the toxicant, primarily in the liver.
 - *Phase I Reactions:* Oxidation, reduction, or hydrolysis reactions (often mediated by Cytochrome P450 enzymes) that expose polar functional groups.
 - *Phase II Reactions:* Conjugation with endogenous compounds (e.g., glutathione,

glucuronic acid) to render the toxicant water-soluble.

4. **Excretion:** Elimination of the parent toxicant and its metabolites from the body, mainly via urine (kidneys), bile/feces, or exhaled air.

Q86) Water in Relation to Human Health

Water is essential for life, but contaminated or inadequate water supplies serve as a primary vector for human disease. Public health relies heavily on access to clean, safe drinking water.

Categories of Water-Related Diseases

1. **Water-Borne Diseases:** Infections transmitted by drinking water contaminated with pathogenic microorganisms.
 - *Examples:* Cholera (*Vibrio cholerae*), Typhoid fever (*Salmonella typhi*), Amoebic dysentery (*Entamoeba histolytica*).
2. **Water-Washed Diseases:** Infections caused by inadequate water supply for personal hygiene.
 - *Examples:* Trachoma (eye infection), scabies, typhus.
3. **Water-Based Diseases:** Diseases caused by aquatic invertebrate host organisms.
 - *Example:* Schistosomiasis (transmitted via freshwater snails harboring *Schistosoma cercariae*).
4. **Water-Vectored Diseases:** Diseases spread by insects that breed in stagnant water.
 - *Examples:* Malaria (*Anopheles* mosquito), Dengue (*Aedes* mosquito), Filariasis (*Culex* mosquito).

Chemical Contamination Hazards

- **Arsenicosis:** Chronic arsenic exposure leads to skin lesions and hyperkeratosis.
- **Fluorosis:** Excess fluoride intake damages teeth (dental fluorosis) and deforms bones (skeletal fluorosis).

Q87) Differentiate Between Pollutant and Toxicant

Comparison Table

Attribute	Pollutant	Toxicant
Definition	Any biological, chemical, or physical substance introduced into the environment that alters its quality or causes ecological disruption.	Any chemical agent that causes structural or functional harm to a living organism upon exposure.
Scope	Broad environmental concept; includes thermal, physical, and aesthetic degradation.	Specific biological/biochemical toxicity concept.
Origin	Can be natural (volcanic ash) or anthropogenic (sewage,	Mostly chemical compounds (man-made synthetic chemicals

Attribute	Pollutant	Toxicant
	industrial smog).	or biological toxins).
Mechanism of Action	Damages ecosystems by altering physical environments (e.g., reducing light, depleting dissolved oxygen).	Damages organisms by disrupting physiological or biochemical pathways (e.g., enzyme inhibition).
Examples	Plastic waste, thermal hot water discharge, excessive suspended sediments.	Lead (Pb), organophosphate insecticides, Potassium Cyanide (KCN).

Q88) Differentiate Between Point and Nonpoint Pollution Sources

Attribute	Point Source Pollution	Nonpoint Source Pollution
Definition	Pollutants discharged from a single, identifiable, discrete location.	Pollutants originating from diffuse, widespread areas without a single point of origin.
Traceability	Easy to trace back to a specific source.	Difficult to trace back to a single source.
Discharge Channel	Enters water bodies via pipes, ditches, channels, or smokestacks.	Enters water bodies via overland agricultural runoff, rainfall, or snowmelt.
Regulation & Control	Relatively simple to monitor, manage, and enforce legal discharge limits.	Challenging to monitor and regulate due to widespread geographical distribution.
Examples	Industrial discharge pipes, municipal sewage treatment plant outlets.	Agricultural fertilizer runoff, urban storm runoff, sediment washed from construction sites.

Q89) Parasite That Causes Dysentery and Pyorrhea

1. Parasite Causing Amoebic Dysentery

- **Causative Organism:** *Entamoeba histolytica* (Protozoan parasite, Phylum Sarcomastigophora).
- **Habitat:** Lives in the mucosal and submucosal layers of the human large intestine (colon).
- **Pathogenesis:** Secretes **histolytic** proteolytic enzymes that erode intestinal mucosal tissue, forming characteristic **flask-shaped ulcers**.
- **Symptoms:** Severe abdominal cramps, diarrhea containing blood and mucus, and fever.
- **Transmission:** Ingestion of food or water contaminated with mature tetranucleate cysts.

2. Parasite Associated with Pyorrhea

- **Causative Organism:** *Entamoeba gingivalis* (Protozoan parasite).
- **Habitat:** Found in human oral cavities, specifically within tartar deposits, gum crevices, and periodontal pockets around teeth.

- **Role in Disease:** Feeds on cellular debris, bacteria, and red blood cells. While it acts primarily as a secondary opportunist that aggravates existing **pyorrhea (periodontitis)** rather than initiating it, its presence worsens tissue inflammation, gum bleeding, and tooth socket erosion.
- **Transmission:** Transmitted directly via kiss contact, shared eating utensils, or contaminated drinking cups.

Q90) Behaviour Pattern of Health and Disease

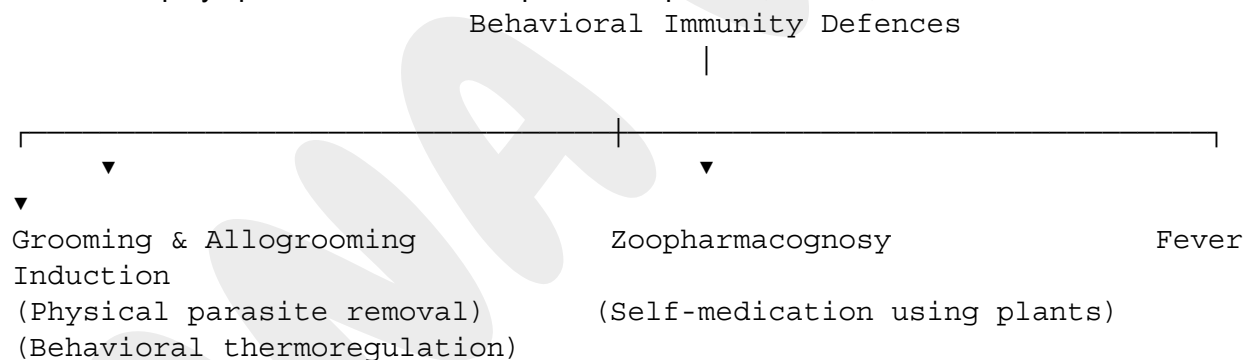
The **behaviour pattern of health and disease** explores how host animal behaviors influence susceptibility to diseases, parasite transmission, and survival outcomes. It includes both risky behaviors that increase infection risk and adaptive defensive behaviors that protect health.

1. Behaviors Increasing Disease Transmission

- **High Aggregation / Sociality:** High population density in social herds or roosts facilitates the transmission of airborne pathogens and ectoparasites.
- **Mating Displays & Contact:** Promiscuous mating behavior increases the transmission of sexually transmitted pathogens.
- **Coprophagy:** Ingestion of feces in certain mammals can introduce intestinal parasites if done indiscriminately.

2. Behavioral Host Defences (Behavioral Immunity)

Animals display specific behavioral adaptations to prevent, reduce, or cure infections:



- **Grooming and Allogrooming:** Primates and birds groom themselves and social partners to physically remove ticks, lice, and fleas.
- **Avoidance Behavior:** Grazing herbivores avoid feeding near fresh dung patches, reducing exposure to infective helminth larvae.
- **Zoopharmacognosy (Self-Medication):** Infected animals consume non-nutritive medicinal plants. For example, chimpanzees ingest bitter pith (*Vernonia amygdalina*) containing antiparasitic compounds to purge intestinal worms.
- **Behavioral Fever:** Ectothermic animals (like grasshoppers or lizards) deliberately move to warmer microclimates to raise their body temperature, creating an unfavorable environment for invading pathogens.

Question 91) Describe the role of various factors responsible for desertification.

Answer:

Desertification refers to the process of land degradation in arid, semi-arid, and dry sub-humid areas resulting from various factors, including climatic variations and human activities. It transforms fertile land into barren, desert-like conditions, severely impairing ecological balance and agricultural productivity. The factors responsible for desertification can be broadly categorized into anthropogenic factors and natural factors.

Anthropogenic (Human-Induced) Factors

1. Deforestation and Vegetation Clearing Forests and native vegetation act as a protective cover for topsoil, binding soil particles together and maintaining soil moisture. Indiscriminate cutting of trees for fuelwood, timber, expansion of agricultural land, and urban development leaves the topsoil directly exposed to intense solar radiation and torrential rainfall. Without root systems to anchor the soil, wind and water erosion rapidly strip away the organic topsoil layer, preventing regeneration and initiating desertification.

2. Overgrazing Uncontrolled and intensive grazing by livestock beyond the carrying capacity of pasture lands is a major contributor to soil degradation. Herbivores strip vegetative cover down to the root level and compact the soil under their hooves. Soil compaction reduces porosity, decreases water infiltration rates, and increases surface runoff. Over time, palatable perennial plant species are destroyed, leaving behind bare ground prone to erosion.

3. Improper Agricultural Practices Modern intensive agricultural practices often undermine soil structural integrity:

- **Over-tilling:** Breaks down natural soil aggregates into fine dust, rendering it highly susceptible to wind transportation.
- **Inappropriate Irrigation:** In arid regions, excessive irrigation combined with poor drainage leads to salinization and alkalinization. As water evaporates, dissolved salts accumulate on the soil surface, forming a toxic crust that inhibits plant growth.
- **Monoculture and Chemical Overuse:** Depletes specific soil nutrients and reduces soil organic matter, destroying microbial biodiversity necessary for healthy soil structure.

4. Over-exploitation of Groundwater Excessive extraction of underground aquifers for irrigation and domestic needs lowers the local water table. Deep-rooted indigenous plants lose access to capillary water and perish, leading to a loss of natural vegetative cover and desiccating the upper soil layers.

Natural and Climatic Factors

1. Prolonged Droughts and Climate Change Recurrent and extended periods of low rainfall reduce soil moisture levels, halting natural plant growth and seed germination. Global climate change alters precipitation patterns, increasing the frequency and intensity of severe droughts in dryland regions.

2. High Temperatures and Evaporation In arid zones, elevated ambient temperatures drive high potential evapotranspiration rates. Water evaporates rapidly from the soil surface, pulling

up subterranean salts and drying out organic matter, making the soil structure light and easily blown away by winds.

3. Wind and Water Erosion High-velocity winds in dry areas carry away nutrient-rich fine soil particles (silt and clay), leaving behind coarse, infertile sand. Rare but intense rainfall events cause flash floods that wash away unanchored soil in sheet and gully erosion.

Consequences and Mitigation

Desertification results in loss of biodiversity, reduced agricultural yield, food insecurity, forced human migration, and worsening atmospheric dust storms. Controlling desertification requires sustainable land management practices, including afforestation, shelterbelt creation, rainwater harvesting, controlled grazing schedules, and the adoption of drip irrigation.

Question 92) Write the names of important endangered species of wildlife in India and describe the steps taken for their conservation.

Answer:

India possesses diverse ecological biomes that support rich biodiversity. However, habitat destruction, poaching, pollution, and climate change have driven many wild species to the brink of extinction.

Important Endangered Species of Wildlife in India

Mammals:

- **Bengal Tiger (*Panthera tigris tigris*)**: Threatened by habitat fragmentation and poaching for bones and skin.
- **Asian Elephant (*Elephas maximus*)**: Threatened by human-wildlife conflict and habitat corridor loss.
- **One-Horned Rhinoceros (*Rhinoceros unicornis*)**: Found mainly in Kaziranga, Assam; threatened by poaching for its horn.
- **Asiatic Lion (*Panthera leo persica*)**: Endemic to the Gir Forest, Gujarat; vulnerable due to a single isolated population.
- **Snow Leopard (*Panthera uncia*)**: Inhabits high-altitude Himalayas; threatened by retaliatory killings and habitat encroachment.
- **Kashmir Stag or Hangul (*Cervus elaphus hanglu*)**: Critically endangered deer species confined to Dachigam National Park, Jammu & Kashmir.
- **Lion-tailed Macaque (*Macaca silenus*)**: Endemic to the Western Ghats rainforests; endangered due to habitat fragmentation.
- **Ganges River Dolphin (*Platanista gangetica*)**: Freshwater mammal threatened by river pollution and dams.

Birds:

- **Great Indian Bustard (*Ardeotis nigriceps*)**: Critically endangered grassland bird threatened by power line collisions and habitat loss.
- **Indian Vulture (*Gyps indicus*)**: Population declined drastically due to veterinary diclofenac poisoning.

Reptiles:

- **Gharial (*Gavialis gangeticus*)**: Critically endangered crocodilian threatened by riverbed

mining and net entanglement.

Steps Taken for Wildlife Conservation in India

Conservation measures in India combine legal protection, habitat restoration, species-specific projects, and community engagement.

1. Legislative Framework

- **The Wildlife (Protection) Act, 1972:** Provides the foundational legal framework for protecting wild animals, birds, and plants. It prohibits hunting, regulates trade in wildlife products, and establishes strict penalties for offenses.
- **Biodiversity Act, 2002:** Focuses on the conservation of biological diversity, sustainable use of its components, and fair sharing of benefits.

2. Protected Area Network (In-Situ Conservation)

India has established an extensive network of protected areas across different biogeographic zones:

- **National Parks and Wildlife Sanctuaries:** Strict preservation zones where habitat destruction and commercial activities are legally banned.
- **Biosphere Reserves:** Larger zones that combine biodiversity conservation with sustainable human development (e.g., Nilgiri, Nanda Devi, Sundarbans).

3. Species-Specific Conservation Projects

- **Project Tiger (1973):** Launched to rescue the Bengal Tiger from extinction. It established designated Tiger Reserves with dedicated core and buffer zones.
- **Project Elephant (1992):** Aims to protect elephant populations, secure biological corridors, and mitigate human-elephant conflict.
- **Crocodile Conservation Project (1975):** Saved the gharial, mugger, and saltwater crocodile through captive breeding and reintroduction programs.
- **Project Snow Leopard (2009):** Focuses on high-altitude ecosystem conservation involving local pastoral communities.
- **Vulture Recovery Plan:** Banned veterinary use of diclofenac and established captive breeding centers across India.

4. Ex-Situ Conservation Programs

Establishing zoological parks, botanical gardens, seed banks, and gene banks provides back-up populations. Captive breeding programs for endangered species—such as the Pygmy Hog, Manipuri Brow-antlered Deer (Sangai), and Gharial—have successfully reintroduced individuals into natural habitats.

5. Community and Eco-Development Initiatives

Integrating local communities through Joint Forest Management (JFM) and eco-tourism reduces reliance on wildlife reserves and creates financial incentives for wildlife preservation.

Question 93) How would you compare the biodegradable and non-biodegradable pollutants? Discuss the causes of global warming due to air pollution.

Answer:

Pollutants are substances introduced into the environment that cause adverse changes to natural ecosystems. Based on their persistence and breakdown mechanisms, pollutants are classified into biodegradable and non-biodegradable pollutants.

Comparison Between Biodegradable and Non-Biodegradable Pollutants

Biodegradable pollutants are substances that can be decomposed and broken down into non-toxic, simpler inorganic compounds by the natural biological action of living organisms, primarily microbial decomposers such as bacteria and fungi. Examples include domestic sewage, livestock manure, agricultural residue, paper, vegetable waste, and wood. These materials have a short persistence time in the environment.

When released in moderate quantities, nature recycles them effectively through natural biogeochemical cycles. However, if discharged in excessive amounts into aquatic systems, their decomposition consumes high levels of dissolved oxygen, leading to biochemical oxygen demand (BOD) elevation and subsequent anaerobic conditions (eutrophication).

Non-biodegradable pollutants are substances that cannot be broken down or decomposed by natural biological processes, or degrade at an extremely slow rate over centuries. Examples include plastics, synthetic polymers, heavy metals (lead, mercury, cadmium), pesticides like DDT, nuclear waste, and glass. These pollutants persist indefinitely in the environment.

Because microbial enzymes cannot cleave their chemical bonds, non-biodegradable pollutants accumulate in soil and aquatic systems. They enter food chains and undergo biomagnification, increasing in concentration at higher trophic levels and causing severe physiological toxicity, reproductive failure, and genetic mutations in higher animals and humans.

Causes of Global Warming Due to Air Pollution

Global warming refers to the long-term rise in Earth's average surface temperature due to the intensified greenhouse effect caused by atmospheric pollutants.

Under normal conditions, Earth absorbs shortwave solar radiation and re-radiates longwave infrared energy back into space. However, certain air pollutants trap this outgoing infrared radiation in the lower atmosphere, acting like a thermal blanket. The key air pollutants causing global warming include:

1. Carbon Dioxide (CO₂) Carbon dioxide is the primary contributor to anthropogenic global warming. Major sources include:

- **Fossil Fuel Combustion:** Burning coal, oil, and natural gas in industrial boilers, power plants, and motor vehicles releases large volumes of CO₂.
- **Deforestation:** Trees act as carbon sinks. Large-scale clearing and burning of forests release stored carbon back into the atmosphere while reducing future carbon assimilation capacity.

2. Methane (CH₄) Methane is a potent greenhouse gas with a global warming potential significantly higher than CO₂ over a 100-year timescale. Key air-polluting sources include:

- Enteric fermentation in ruminant livestock.
- Anaerobic decomposition of organic matter in flooded rice paddies and municipal solid waste landfills.
- Leakage during natural gas extraction, refining, and distribution.

3. Nitrous Oxide (N₂O) Nitrous oxide is released from the heavy application of synthetic nitrogenous fertilizers in agricultural fields, fossil fuel combustion in automobiles, and chemical industries producing nylon and nitric acid.

4. Chlorofluorocarbons (CFCs) and Hydrofluorocarbons (HFCs) These synthetic compounds, used extensively in refrigeration, air conditioning, foam blowing, and aerosol propellants, possess heat-trapping capacities thousands of times greater than carbon dioxide per molecule.

5. Tropospheric Ozone (O₃) and Black Carbon Ground-level ozone, formed by chemical reactions between volatile organic compounds (VOCs) and nitrogen oxides (NO_x) in the presence of sunlight, acts as a short-lived greenhouse gas. In addition, black carbon (soot) emitted from incomplete combustion absorbs solar radiation directly, warming the surrounding air mass and accelerating snowmelt when deposited on ice sheets.

Question 94) Write an essay on water and human diseases.

Answer:

Water is essential for life, serving as the medium for all cellular biochemistry and metabolic processes. However, when water becomes contaminated with biological pathogens or toxic chemicals, it turns into a major vehicle for human disease transmission. The World Health Organization (WHO) estimates that a significant portion of infectious diseases in developing nations is directly linked to unsafe water supplies and poor sanitation.

Classification of Water-Related Human Diseases

Water-related diseases are classified into four main categories based on the role water plays in their transmission cycle:

1. Water-Borne Diseases These diseases are caused by ingesting water contaminated with pathogenic microorganisms originating from human or animal excreta. The pathogen enters the gastrointestinal tract through contaminated drinking water or food preparation.

- **Bacterial Diseases:** Cholera (*Vibrio cholerae*), Typhoid fever (*Salmonella typhi*), and Bacillary dysentery (*Shigella dysenteriae*). They cause severe diarrhea, vomiting, dehydration, and high fever.
- **Viral Diseases:** Gastroenteritis (Rotavirus), Hepatitis A and Hepatitis E viruses, which cause acute liver inflammation and jaundice.
- **Protozoan Diseases:** Amoebic dysentery (*Entamoeba histolytica*) and Giardiasis (*Giardia lamblia*), leading to abdominal cramps, severe diarrhea, and nutrient malabsorption.

2. Water-Washed Diseases These diseases arise from inadequate water quantity for personal hygiene, bathing, and washing clothes, allowing pathogens or parasites to thrive on the body:

- **Trachoma:** Caused by *Chlamydia trachomatis*, leading to chronic conjunctivitis and potential blindness.
- **Skin and Eye Infections:** Scabies (caused by the itch mite *Sarcoptes scabiei*) and fungal ringworm infections spread rapidly under unhygienic conditions.

3. Water-Based Diseases These diseases are caused by aquatic invertebrate organisms or parasitic worms that spend part of their life cycle in an aquatic host (such as freshwater snails) before infecting humans through skin penetration or ingestion:

- **Schistosomiasis (Bilharzia):** Caused by *Schistosoma* blood flukes. Larval cercariae emerge from snails in stagnant water and penetrate human skin during bathing or wading, causing chronic organ damage.
- **Dracunculiasis (Guinea Worm Disease):** Contracted by drinking water containing micro-crustaceans (*Cyclops*) infected with *Dracunculus medinensis* larvae.

4. Water-Related Insect Vector Diseases These diseases are transmitted by insect vectors that breed or feed near water bodies:

- **Malaria:** Transmitted by the female *Anopheles* mosquito, which breeds in clean, stagnant water.

- **Dengue and Chikungunya:** Transmitted by *Aedes aegypti* mosquitoes, which breed in small containers holding clean water.
- **Lymphatic Filariasis:** Transmitted by *Culex* mosquitoes breeding in polluted, organically rich wastewater, leading to elephantiasis.

Chemical Toxicity from Contaminated Water

In addition to infectious agents, chemical contamination of drinking water poses severe long-term health risks:

- **Fluorosis:** Excessive fluoride levels ($>1.5 \text{ mg/L}$) in groundwater cause dental mottling and crippling skeletal fluorosis.
- **Arsenicosis:** Chronic ingestion of inorganic arsenic from deep tube wells leads to skin lesions, hyperkeratosis, peripheral vascular disease (Blackfoot disease), and skin and visceral cancers.
- **Methemoglobinemia (Blue Baby Syndrome):** High nitrate concentrations in drinking water, derived from agricultural runoff, reduce the oxygen-carrying capacity of hemoglobin in infants.
- **Heavy Metal Poisoning:** Industrial effluents containing Lead, Mercury (causing Minamata disease), and Cadmium (causing Itai-Itai disease) result in severe neurological, renal, and skeletal disorders.

Preventive and Control Measures

Mitigating water-related diseases requires an integrated public health strategy:

- Providing safe, treated municipal drinking water via chlorination, filtration, and reverse osmosis.
- Improving basic sanitation infrastructure to separate human waste from water sources.
- Vector control through biological agents (e.g., *Gambusia* fish) and environmental management of standing water.
- Public health education on personal hygiene and safe household water storage.

Question 95) Write an essay on ecotoxicology.

Answer:

Ecotoxicology is an interdisciplinary branch of science that integrates ecology and toxicology to study the adverse effects of toxic chemical agents on natural ecosystems and their biotic components. Coined by René Truhaut in 1969, ecotoxicology extends classical toxicology—which focuses on individual organisms—to higher levels of biological organization, including populations, biological communities, and entire ecosystems.

Core Principles and Scope of Ecotoxicology

While traditional toxicology focuses primarily on evaluating lethal doses (LD_{50}) or toxic mechanisms within individual organisms (typically humans or laboratory mammals), ecotoxicology examines how xenobiotic substances travel through environmental media, transform, enter food webs, and alter ecosystem structure and function.

The scope of ecotoxicology encompasses three main areas:

1. **Release and Transport:** The entry of toxicants into air, water, and soil, and their transport

across spatial scales.

2. **Fate and Transformation:** Chemical transformation, degradation, photolysis, and microbial degradation of toxicants in natural environments.
3. **Biological Impact:** Molecular, physiological, behavioral, population-level, and ecosystem-wide responses to toxic stress.

Key Concepts in Ecotoxicology

1. Xenobiotics Synthetic or foreign chemical substances present within an organism or environment that are not naturally produced or expected to be present (e.g., synthetic pesticides, industrial solvents, pharmaceuticals, microplastics).

2. Bioaccumulation and Bioconcentration

- **Bioconcentration:** Direct uptake of a chemical pollutant by an aquatic organism from the surrounding water medium across respiratory or skin surfaces.
- **Bioaccumulation:** The net accumulation of a toxic substance over time within an organism's tissues from both ambient environmental sources and food intake.

3. Biomagnification (Food Chain Amplification) Biomagnification refers to the progressive increase in toxicant concentration at successively higher trophic levels in a food chain. Lipophilic (fat-soluble) and persistent chemicals—such as dichlorodiphenyltrichloroethane (DDT), polychlorinated biphenyls (PCBs), and methylmercury—resist metabolic breakdown and excretion. As lower-trophic organisms are consumed in large quantities by predators, the toxicant becomes highly concentrated in top predators. A classic example is the thinning of eggshells in raptorial birds (e.g., peregrine falcons and bald eagles) caused by DDT accumulation, leading to reproductive failure.

4. Biomarkers Biomarkers are measurable biological, biochemical, or physiological changes in organisms that indicate exposure to or toxicity from environmental pollutants. Examples include:

- Inhibition of the enzyme Acetylcholinesterase (AChE) by organophosphate pesticides.
- Induction of Cytochrome P450 enzymes (CYP1A) by polycyclic aromatic hydrocarbons (PAHs).
- DNA strand breaks and induction of stress proteins (Heat Shock Proteins - HSPs).

Environmental Testing and Ecotoxicological Assays

To establish safety thresholds, ecotoxicologists perform standardized bioassays using indicator organisms:

- **Acute Toxicity Testing:** Measures short-term mortality to determine the median lethal concentration (LC_{50}) or median effective concentration (EC_{50}). Common test species include the freshwater flea (*Daphnia magna*), green algae (*Chlorella*), and zebrafish (*Danio rerio*).
- **Chronic Toxicity Testing:** Evaluates long-term, subtle endpoints such as growth rate, reproductive output, developmental deformities, and lifespan reduction.

Applications of Ecotoxicology

- **Environmental Risk Assessment (ERA):** Evaluating the potential ecological risk posed by new agrochemicals, industrial compounds, and pharmaceuticals prior to commercial authorization.
- **Setting Environmental Quality Standards:** Establishing maximum permissible

discharge limits for effluents into rivers, lakes, and coastal waters.

- **Bioremediation and Ecological Restoration:** Monitoring ecosystem recovery following chemical spills (e.g., oil spills) using biological indicators.

In conclusion, ecotoxicology provides essential scientific tools for diagnosing environmental degradation, establishing regulatory framework parameters, and protecting biodiversity from anthropogenic chemical stress.

Question 96) Discuss the environmental impact of air pollution.

Answer:

Air pollution refers to the addition of harmful solids, liquids, or gases into the atmosphere in concentrations that harm living organisms and disrupt ecosystem processes. While human health impacts are severe, the environmental impacts of air pollution extend across terrestrial, aquatic, and atmospheric systems.

1. Acid Deposition (Acid Rain)

Acid deposition occurs when sulfur dioxide (SO₂) and nitrogen oxides (NO_x), emitted by fossil fuel combustion, industrial smelters, and power plants, react with atmospheric moisture, oxygen, and oxidants to form sulfuric acid (H₂SO₄) and nitric acid (HNO₃). These fall to the ground as wet deposition (rain, snow, fog) or dry deposition (acidic particles and gas).

- **Terrestrial Impact:** Acid rain leaches essential plant nutrients such as calcium, magnesium, and potassium from the soil, depriving trees of vital elements. Concurrently, it solubilizes toxic aluminum ions naturally present in soil minerals. Free aluminum destroys plant root systems, leading to forest dieback, particularly in high-altitude coniferous forests.
- **Aquatic Impact:** Acidic runoff lowers the pH of lakes and streams. Below a pH of 5.0, most fish eggs fail to hatch, and adult fish suffer branchial mucus clogging and osmoregulatory failure due to dissolved aluminum toxicity. Entire aquatic food webs collapse as sensitive macroinvertebrates perish.

2. Atmospheric Warming and Global Climate Change

Emissions of greenhouse gases—such as Carbon Dioxide (CO₂), Methane (CH₄), Nitrous Oxide (N₂O), and Chlorofluorocarbons (CFCs)—trap outgoing longwave thermal radiation within the troposphere. The environmental consequences include:

- Accelerated melting of polar ice caps and glaciers, causing sea-level rise and coastal inundation.
- Altered global atmospheric circulation and rainfall patterns, increasing the frequency of severe droughts, heatwaves, and extreme tropical storms.
- Shifts in species distribution ranges, disrupting phenological synchrony between flowering plants and their pollinators.

3. Stratospheric Ozone Depletion

Anthropogenic emissions of halogenated compounds, particularly Chlorofluorocarbons (CFCs), Hydrochlorofluorocarbons (HCFCs), and Halons, migrate into the stratosphere. Under intense ultraviolet light, these molecules release free chlorine and bromine radicals that catalytically

destroy the protective stratospheric ozone layer.

- **Impact on Marine Life:** Increased UV-B radiation penetrates surface ocean waters, killing surface-dwelling phytoplankton—the primary producers of marine ecosystems. This impairs carbon sequestration and reduces fish yields.
- **Impact on Terrestrial Plants:** High UV-B exposure damages plant DNA, impairs photosynthetic machinery, reduces leaf surface area, and lowers crop productivity.

4. Ground-Level (Tropospheric) Ozone Damage

Unlike stratospheric ozone, ground-level ozone is a secondary air pollutant formed by photochemical reactions between Nitrogen Oxides (NO_x) and Volatile Organic Compounds (VOCs) in the presence of intense sunlight.

- Ground-level ozone enters plant leaves through stomata during normal gas exchange.
- It produces reactive oxygen species (ROS) that destroy chloroplast membranes, causing foliar stippling, chlorosis, and premature leaf senescence.
- It significantly suppresses agricultural crop yields in wheat, rice, soybean, and cotton.

5. Photochemical Smog and Reduced Solar Radiation

In urban and industrial areas, photochemical smog forms a persistent brown haze that attenuates incoming sunlight, reducing photosynthetically active radiation (PAR) reaching plant canopies. Particulate matter (PM_{2.5}) and PM₁₀) and soot settle on plant leaves, physically clogging stomatal pores, blocking gas exchange, and inhibiting photosynthesis.

6. Eutrophication and Nitrogen Saturation

Excessive atmospheric emissions of nitrogen oxides from motor vehicles and ammonia (NH₃) from intensive agriculture are deposited back onto land and water bodies through atmospheric deposition. This input acts as an unintended nutrient overload, causing terrestrial plant community shifts where fast-growing nitrophilic weeds outcompete indigenous wild species, reducing plant biodiversity. In aquatic systems, excess nitrogen fuels toxic algal blooms and subsequent hypoxia (dead zones).

Question 97) Discuss the causes and consequences of soil erosion.

Answer:

Soil erosion is the physical detachment, displacement, and transport of the fertile topsoil layer from one location to another by natural agents, primarily water and wind. Topsoil contains the highest concentration of organic matter, microbial activity, and plant nutrients. Its loss degrades land and threatens food security.

Causes of Soil Erosion

Soil erosion results from the interplay of natural environmental driving forces and destructive human activities.

Natural Causes:

- **Rainfall Intensity and Runoff:** Torrential rainfall drops large raindrops with high kinetic energy that dislodge soil particles upon impact (splash erosion). When rainfall exceeds

soil infiltration capacity, surface runoff carries dislodged soil downhill.

- **Wind Velocity:** In arid and semi-arid regions devoid of ground cover, high-velocity winds pick up loose, dry, fine soil particles and transport them over long distances via suspension, saltation, and surface creep.
- **Topography:** Steep slopes accelerate surface water velocity, increasing its energy to erode and transport topsoil.
- **Soil Erodibility:** Soils with low organic matter content, coarse texture, or poor aggregation possess weak structural stability and erode easily.

Anthropogenic (Human-Induced) Causes:

- **Deforestation:** Trees and forest roots anchor soil structure and break the physical impact of falling rain. Large-scale tree removal leaves the soil exposed to rain and wind.
- **Overgrazing:** Excessive livestock stocking rates remove protective grass cover and compact topsoil through hoof action, suppressing plant regrowth and accelerating runoff.
- **Inappropriate Agricultural Practices:** Continuous plowing along vertical slope lines creates channels that turn into gullies during rain. Leaving fields bare between crop cycles (fallowing) exposes the soil during storm seasons.
- **Construction and Mining Activities:** Large-scale land clearance, excavation, road building, and open-cast mining destabilize soil structure, generating huge volumes of easily transportable sediment.

Types of Water and Wind Erosion

- **Splash Erosion:** Initial detachment of soil particles caused by falling raindrops.
- **Sheet Erosion:** The uniform removal of thin layers of topsoil over a wide area by overland water flow, often subtle and unnoticed initially.
- **Rill Erosion:** Small, well-defined streams of water carve tiny channels (rills) into the soil.
- **Gully Erosion:** Advanced stage where rills merge into deep, wide channels (gullies) that cannot be cleared by normal tillage equipment.
- **Streambank Erosion:** Washing away of soil along the banks of rivers and streams during high flows.

Consequences of Soil Erosion

1. Loss of Agricultural Productivity and Soil Fertility Erosion selectively removes fine silt, clay, and organic humus—the components responsible for nutrient binding and moisture retention. The remaining subsoil is dense, nutrient-poor, and difficult to till, leading to declining crop yields and increased dependence on synthetic chemical fertilizers.

2. Siltation of Rivers and Water Reservoirs Eroded sediments end up in downstream aquatic ecosystems, rivers, and hydroelectric reservoirs. Reservoir sedimentation reduces water storage capacity, lowers hydroelectric power generation efficiency, and shortens the operational lifespan of dams.

3. Increased Risk of Flash Floods Siltation raises riverbed levels, reducing channel carrying capacity. Concurrently, degraded, compacted soils upstream lose their water absorption capacity, drastically increasing peak surface runoff during heavy rainfall and causing severe downstream flash flooding.

4. Desertification and Land Degradation Continuous loss of topsoil in dryland biomes leads to irreversible land degradation, transforming arable land into barren, rocky deserts incapable of supporting vegetation or agriculture.

5. Impact on Aquatic Ecosystems and Water Quality Suspended sediment increases water turbidity, blocking sunlight penetration and inhibiting aquatic plant photosynthesis. Settling fine silt smothers benthic invertebrates and covers fish spawning gravels. Furthermore, eroded soil carries attached agricultural pesticides and phosphorus fertilizers into lakes, triggering toxic algal blooms and eutrophication.

Preventive and Conservation Measures

Mitigating soil erosion involves soil conservation practices:

- **Contour Plowing and Terracing:** Plowing across slope contours and creating step-like terraces on hillsides to slow water runoff.
- **Afforestation and Shelterbelts:** Planting rows of trees across wind directions to break wind speed and binding soil roots.
- **Mulching and Cover Cropping:** Covering bare soil with organic mulch or planting leguminous cover crops between primary harvests.
- **Check Dams:** Constructing small barrier dams across gullies to trap sediment and reduce water flow velocity.

Question 98) Describe the concept and procedure of wildlife management.

Answer:

Wildlife management is the science and art of manipulating wildlife populations, their habitats, and human interactions to achieve specific ecological, aesthetic, and socio-economic objectives. It aims to maintain balanced ecosystem processes while ensuring the long-term survival and sustainable utilization of wild animal species.

Core Concept of Wildlife Management

The fundamental concept of wildlife management rests on the understanding that wild species cannot be managed in isolation; they must be managed as dynamic components of their natural habitats.

Management strategies typically focus on three core objectives depending on the species' conservation status:

1. **Preservation/Protection:** Applied to rare, endangered, or threatened species to prevent population decline and habitat loss.
2. **Sustained Yield Management:** Applied to abundant or game species to allow controlled harvesting (e.g., regulated hunting or fishing) without depleting the breeding stock.
3. **Control:** Applied to overabundant, invasive, or pest species that threaten agricultural crops, native biodiversity, or public health.

The key ecological foundation of wildlife management is the **Carrying Capacity (K)**—the maximum population size of a species that a given habitat can sustainably support over time without environmental degradation, determined by limiting factors such as food, water, cover, and spatial territory.

Procedure of Wildlife Management

Wildlife management follows a structured, scientific procedure consisting of sequential steps:

1. Habitat and Population Inventory (Data Collection) Before implementing management interventions, field biologists collect baseline ecological data:

- **Census and Population Estimation:** Determining species density, age structure, sex ratios, birth rates, and mortality rates. Methods include direct line transect counts, camera trapping, pugmark identification, distance sampling, aerial surveys, and DNA-based scat analysis.
- **Habitat Evaluation:** Mapping vegetation types, assessing food availability, identifying water sources, and evaluating cover quality across seasons.

2. Identifying Management Objectives Setting clear goals based on gathered field data, legislative mandates, and ecological priorities. Examples include increasing the population of an endangered predator by 20% over ten years or securing a biological corridor between two fragmented national parks.

3. Habitat Management Interventions Habitat quality determines population health. Primary habitat management techniques include:

- **Habitat Restoration:** Re-establishing native vegetation, removing invasive alien plant species (such as *Lantana camara* or *Parthenium*), and restoring degraded wetlands.
- **Water Development:** Constructing artificial water holes, earthen check dams, and solar-pumped water troughs to ensure year-round water availability during summer months.
- **Creation of Edge and Openings:** Controlled burning or selective clearing to create open glades and early successional habitats that supply fresh forage for wild herbivores.
- **Corridor Establishment:** Protecting and establishing green wildlife corridors linking isolated reserves, allowing safe animal movement, seasonal migration, and gene flow.

4. Direct Population Management

- **In-situ Conservation:** Establishing core zones in national parks free from all human interference.
- **Anti-Poaching Operations:** Deploying armed forest guards, sniffer dog units, drone surveillance, and wireless communication networks to suppress illegal poaching and wildlife trade.
- **Captive Breeding and Reintroduction:** Breeding critically endangered species in specialized facilities and reintroducing them into secured historical habitats (e.g., the reintroduction of the Indian Rhinoceros into Dudhwa National Park).
- **Translocation:** Relocating excess or problem animals from high-density or human-conflict areas to suitable underpopulated reserves.
- **Disease Monitoring and Vaccination:** Screening wild populations for infectious livestock diseases (e.g., Foot-and-Mouth disease, Anthrax, Rinderpest) and vaccinating domestic cattle grazing near buffer zones.

5. Human-Wildlife Conflict Mitigation

- Installing solar-powered electric fencing, trench barriers (Elephant Proof Trenches), and acoustic deterrents around forest boundary villages.
- Establishing rapid compensation schemes for livestock loss, crop damage, or human injuries caused by wild animals to prevent retaliatory killings by local communities.

6. Continuous Monitoring and Adaptive Management Wildlife management is an ongoing, adaptive process. Regular monitoring of population trends and habitat changes allows managers to evaluate the success of interventions and adjust policies based on emerging ecological data.

Question 99) Define migration. Write a brief essay on fish migration.

Answer:

Definition of Migration

Migration in animals is defined as the periodic, synchronized, directional movement of a population or group of individuals from one region or habitat to another, followed by a return journey to the original habitat. It is usually tied to specific seasons or life history stages and is primarily driven by the search for optimal feeding grounds, favorable climatic conditions, or suitable breeding and spawning sites.

Essay on Fish Migration

Fish migration represents one of the most remarkable physiological and behavioral adaptations in the animal kingdom. Many species of teleost (bony) and elasmobranch (cartilaginous) fishes travel vast distances across marine and freshwater systems to complete their life cycles.

Types of Fish Migration

Based on the environment and salinity gradients traversed during migration, fish migration is broadly categorized as follows:

1. Diadromous Migration Diadromous fishes travel between marine (saltwater) and freshwater environments during their life cycles. Diadromous migration is further divided into three distinct sub-types:

- **Anadromous Migration:** Fishes spend the majority of their adult life feeding and growing in the ocean, but migrate upstream into freshwater rivers and streams to spawn.
 - **Classic Examples:** Atlantic Salmon (*Salmo salar*), Pacific Salmon (*Oncorhynchus* species), and the Indian Shad or Hilsa (*Tenualosa ilisha*).
 - **Process:** Adult salmon travel thousands of miles through open oceans back to their natal freshwater streams, relying on olfactory memory (olfactory imprinting). During this journey, they undergo major physiological adaptations, including osmoregulatory changes, stop feeding, and change body coloration. After depositing and fertilizing eggs in gravel nests (redds), most adult Pacific salmon die of exhaustion. The hatched larvae (alevins and parr) grow into smolts, adapt to saltwater, and migrate back to the ocean.
- **Catadromous Migration:** Fishes spend most of their adult life feeding and maturing in freshwater rivers and lakes, but migrate down to deep marine oceanic waters to spawn.
 - **Classic Examples:** Freshwater Eels (*Anguilla rostrata* - American eel, and *Anguilla anguilla* - European eel).
 - **Process:** Adult eels live in European and North American rivers for 10–20 years. Upon reaching sexual maturity, they undergo metamorphosis (turning into silver eels), leave the rivers, and cross the Atlantic Ocean to reach their spawning ground in the Sargasso Sea. Spawning occurs at great depths, after which the adult eels die. The leaf-like transparent larvae, called **leptocephali**, drift back toward coastal waters via ocean currents, metamorphose into glass eels and elvers, and ascend freshwater rivers to repeat the cycle.
- **Amphidromous Migration:** Fishes move between freshwater and marine environments

at specific life stages, but this movement is not for spawning purposes (e.g., migrating for feeding or shelter, such as certain species of Gobies).

2. Potamodromous Migration Fishes execute migratory journeys entirely within freshwater systems, moving between rivers, streams, and lakes (e.g., *Tor tor* - Mahseer migrating upstream to clear, cold headwaters for spawning).

3. Oceanodromous Migration Fishes migrate entirely within marine environments across open oceans, traveling vast distances along major ocean currents between temperate feeding areas and warm tropical spawning grounds (e.g., Tunas, Mackerels, Herrings, and Cod).

Factors Regulating Fish Migration

Fish migration is governed by a complex interplay of internal and external factors:

- **Environmental (Extrinsic) Factors:** Photoperiod (day length changes), water temperature variations, river discharge volume and current velocity, water turbidity, lunar phases, and salinity gradients.
- **Endocrine and Physiological (Intrinsic) Factors:** Hormonal secretions play a pivotal role. The thyroid hormone controls osmoregulatory acclimatization and swimming activity; sex steroids (estrogen and testosterone) drive sexual maturation and migratory instincts; and pituitary hormones regulate metabolic fuel utilization.

Adaptive Significance of Fish Migration

- **Reproductive Success:** Ensures eggs and delicate larvae are deposited in sheltered, oxygen-rich environments with fewer predators.
- **Optimal Foraging:** Allows adult populations to exploit seasonal, food-rich marine or estuarine environments, maximizing growth rates.
- **Survival in Extreme Conditions:** Helps fishes avoid extreme winter freezing or low-oxygen conditions in seasonal river channels.

Threats to Migratory Fishes

Human activities have severely impacted migratory fish populations worldwide. Construction of hydroelectric dams and barrages creates physical barriers that block upstream migration routes (e.g., impacting Hilsa and Salmon runs). In response, modern fish conservation employs structural measures such as **fish ladders, fish passes, and fish lifts** alongside regulated seasonal fishing bans during spawning runs.

Question 100) Describe briefly the methods used in ecological studies.

Answer:

Ecology is the scientific study of interactions between organisms and their physical and biological environment. To investigate ecosystem dynamics, population trends, community structures, and energy flows, ecologists use a variety of field, laboratory, analytical, and technological methods.

1. Population Estimation and Sampling Methods

Because counting every individual in a wild population is rarely feasible, ecologists rely on quantitative sampling techniques to estimate population size, density, and distribution.

A. Quadrats and Transects (For Sessile and Slow-Moving Organisms)

- **Quadrat Method:** Standard square frames (e.g., $1\text{ m} \times 1\text{ m}$) are placed randomly or systematically along a grid within a study area. Organisms (such as plants, ground insects, or sessile intertidal invertebrates) inside each quadrat are counted. Population density is calculated as: $\text{Density} = \frac{\text{Total number of individuals counted in all quadrats}}{\text{Total area of sampled quadrats}}$
- **Line and Belt Transects:** A calibrated line or narrow tape strip is laid across an environmental gradient (e.g., from an ocean shore inland). In line transects, organisms touching the line are recorded; in belt transects, organisms within a specified width along the line are recorded. This method evaluates species zonation and community changes.

B. Mark-Release-Recapture Method (For Mobile Animals) Used to estimate population size (N) of mobile animals like mammals, birds, fish, and insects.

- **Procedure:** A sample of individuals (M) is captured from the wild, marked with non-harmful tags, paint, or rings, and released back into the population. After allowing time for the marked individuals to mix randomly, a second sample (n) is captured, and the number of marked individuals (R) in this second sample is counted.
- **Lincoln-Petersen Index Formula:** $N = \frac{M \times n}{R}$ Where:
- $N = \text{Estimated total population size}$
- $M = \text{Number of individuals marked and released initially}$
- $n = \text{Total number of individuals captured in the second sample}$
- $R = \text{Number of marked individuals recaptured in the second sample}$

2. Methods for Assessing Abiotic Factors

Understanding ecosystem function requires measuring key physical and chemical variables:

- **Microclimate & Soil Parameters:** Measurement of ambient air and soil temperature, relative humidity using psychrometers, light intensity using lux meters, soil moisture content, and soil pH using electronic probes.
- **Aquatic Parameters:** Water temperature, dissolved oxygen (DO) via Winkler's titration method or electronic DO meters, turbidity using a Secchi disc, salinity using refractometers, and chemical measurement of nitrates and phosphates.

3. Community and Biodiversity Analysis Methods

- **Species Richness and Diversity Indices:** Mathematical metrics used to quantify community complexity:
 - **Shannon-Wiener Diversity Index (H'):** Measures species diversity taking into account both richness and relative abundance. $H' = -\sum (p_i \ln p_i)$ Where p_i is the proportion of individuals belonging to the i -th species.
 - **Simpson's Diversity Index:** Measures the probability that two randomly selected individuals belong to the same species.
- **Trophic Structure and Biomass Determination:** Harvest techniques where sample biomass is dried in hot-air ovens at $60\text{--}70^\circ\text{C}$ to constant weight to measure dry organic matter per unit area (g/m^2).

4. Modern Technological Methods in Ecological Studies

A. Remote Sensing and Geographic Information Systems (GIS) Satellite imaging (e.g., Landsat, Sentinel) and aerial photography are used to map large-scale land-use changes, track deforestation rates, monitor desertification, measure marine chlorophyll levels, and evaluate forest canopy health over time. GIS software integrates spatial data to model species distribution ranges and identify potential conservation corridors.

B. Radio-Telemetry and Satellite Tracking Miniaturized electronic transmitters or GPS collars are fitted to wild animals (e.g., tigers, migratory birds, sea turtles). Receiver units or satellites track real-time spatial movements, home range sizes, habitat utilization patterns, and migratory routes.

C. Camera Trapping Infrared motion-sensing camera traps deployed along forest paths photograph nocturnal, secretive, or low-density wildlife without human disturbance. Individual identification using unique stripe or spot patterns (e.g., in tigers and leopards) allows accurate population estimation using spatially explicit capture-recapture models.

D. Environmental DNA (eDNA) Analysis A modern molecular technique where water, soil, or air samples are collected to extract trace DNA shed by organisms through skin cells, feces, or mucus. Amplifying this DNA via Polymerase Chain Reaction (PCR) enables detection of rare, elusive, or invasive species without physically capturing them.

Question 101) Give an account of biological factors affecting the toxicity of xenobiotics.

Answer:

A xenobiotic is any synthetic or foreign chemical compound found within an organism that is not naturally produced or expected to be present (e.g., synthetic pesticides, pharmaceuticals, industrial solvents, heavy metals). The toxic response elicited by a xenobiotic depends not only on the chemical's physical properties and dosage, but also on the physiological and biochemical status of the biological organism exposed to it. The primary biological factors modulating xenobiotic toxicity include the following:

1. Species Differences

Different animal species often show marked variations in their sensitivity to the same xenobiotic. These differences stem from variations in metabolic pathways, enzymatic profiles, and target-site receptor affinities.

- **Enzymatic Profiles:** Phase I biotransformation enzymes, particularly the Cytochrome P450 (CYP) monooxygenase system, vary across species. For instance, cats have a low capacity for glucuronide conjugation due to a deficiency in UDP-glucuronosyltransferase, making them exceptionally sensitive to paracetamol and phenol toxicity.
- **Anatomical and Physiological Variations:** Ruminants digest chemicals differently than monogastric animals due to microflora present in their rumen, which can either detoxify or bioactivate incoming toxicants.

2. Genetic Factors (Pharmacogenetics/Toxicogenetics)

Genetic polymorphism within a single species leads to variable individual responses to xenobiotics.

- **Enzyme Polymorphism:** Variations in genes encoding metabolic enzymes produce "slow metabolizers" and "fast metabolizers." Individuals with a genetic deficiency in glucose-6-phosphate dehydrogenase (G6PD) experience severe hemolysis when exposed to oxidative drugs or environmental oxidants.
- **Target Site Modifications:** Genetic mutations altering the structural conformation of cellular receptors or ion channels can render an individual resistant or hypersensitive to specific toxicants.

3. Age and Developmental Stage

Sensitivity to xenobiotics varies across an organism's lifespan, with neonates/juveniles and elderly individuals being the most vulnerable.

- **Neonates and Young Animals:** Immaturity of detoxifying organ systems increases toxicity. The hepatic Cytochrome P450 system and Phase II conjugation pathways are not fully developed in newborns. Furthermore, the blood-brain barrier is incomplete, and renal filtration rates are low, allowing toxicants to enter the central nervous system and persist in the bloodstream.

- **Aged Animals:** Aging leads to a progressive decline in hepatic blood flow, reduced cytochrome levels, impaired renal clearance, and decreased tissue regeneration capacity, resulting in prolonged toxicant retention and higher tissue toxicity.

4. Sex and Hormonal Status

Sex-dependent differences in xenobiotic toxicity stem from hormonal influences on metabolic enzyme expression and body composition.

- **Hormonal Regulation of Enzymes:** Sex steroids (androgens and estrogens) regulate the expression of specific Cytochrome P450 isoenzymes. In laboratory rats, males metabolize many compounds faster than females due to higher androgen-driven CYP expression.
- **Body Composition:** Females generally possess a higher ratio of adipose tissue to lean muscle mass compared to males. Lipophilic toxicants (such as DDT, PCBs, and organochlorines) accumulate in greater quantities within fat stores in females, altering their distribution dynamics.

5. Nutritional Status and Diet

An organism's nutritional condition affects its metabolic handling of xenobiotics.

- **Protein Deficiency:** Low dietary protein intake reduces the synthesis of biotransformation enzymes and depletes intracellular glutathione stores, severely compromising Phase II conjugation pathways and rendering cells vulnerable to reactive intermediates.
- **Micronutrient Availability:** Deficiencies in essential trace elements (such as zinc, copper, and selenium) or vitamins (A, C, E) impair antioxidant defense mechanisms, exacerbating oxidative stress induced by toxicants.

6. Health and Disease State

Pre-existing pathological conditions modify xenobiotic toxicity.

- **Hepatic Impairment:** Liver diseases (such as cirrhosis, hepatitis, or fatty liver) reduce functional hepatocyte mass and enzyme activity, leading to impaired biotransformation and systemic accumulation of toxicants.
- **Renal Failure:** Impaired kidney function reduces the excretion of polar metabolic products, causing toxic metabolites to accumulate to dangerous levels.

Question 102) Describe the various aspects of urbanization stress and health.

Answer:

Urbanization is the mass migration of human populations from rural regions to urban centers, leading to the rapid growth of towns and cities. While urbanization drives economic growth, rapid and unplanned urban expansion creates a set of physical, biological, psychological, and social stressors collectively referred to as **urbanization stress**. This stress severely impacts human physical and mental health.

Physical and Environmental Aspects of Urbanization Stress

1. Ambient Air and Noise Pollution Urban areas host concentrated motor vehicle traffic, industrial manufacturing, and power generation units.

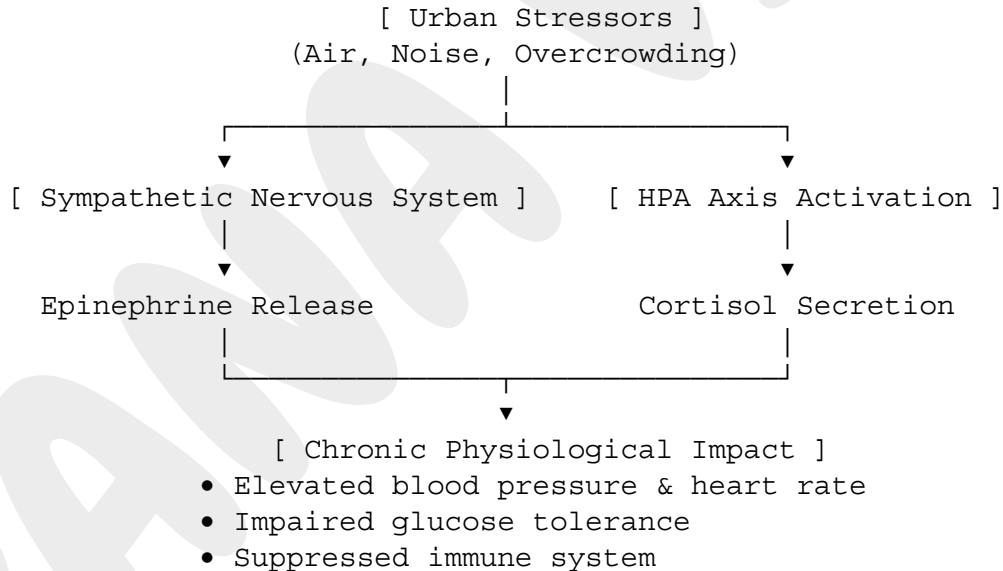
- **Air Pollutants:** Fine particulate matter (PM_{2.5} and PM₁₀), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), and ground-level ozone (O₃) penetrate deep into human pulmonary alveoli, causing chronic obstructive pulmonary disease (COPD), asthma, cardiovascular inflammation, and lung cancer.
- **Noise Pollution:** Continuous exposure to high decibel levels from traffic, construction, and sirens triggers chronic activation of the sympathetic nervous system, leading to elevated cortisol levels, hypertension, sleep disruption, and cardiovascular disease.

2. Urban Heat Island (UHI) Effect Replacing natural vegetation with heat-absorbing asphalt, concrete roads, and tall buildings, combined with anthropogenic heat from vehicles and air conditioners, creates the Urban Heat Island effect. Elevated urban temperatures exacerbate heat exhaustion, heat stroke, and respiratory distress, particularly among elderly and vulnerable populations.

3. Inadequate Water, Sanitation, and Waste Accumulation Uncontrolled urban growth overwhelms municipal infrastructure, leading to overcrowding in informal settlements and slums. Inadequate clean water access and accumulated solid waste create breeding grounds for disease vectors, causing frequent outbreaks of water-borne (cholera, typhoid) and vector-borne (dengue, chikungunya, malaria) infectious diseases.

Physiological Mechanisms of Urbanization Stress

The continuous exposure to environmental stressors activates two neuroendocrine pathways in the human body:



1. **Sympathetic-Adrenal-Medullary (SAM) Axis:** Triggers the release of epinephrine and norepinephrine, raising heart rate, blood pressure, and metabolic rate.
2. **Hypothalamic-Pituitary-Adrenal (HPA) Axis:** Stimulates sustained cortisol secretion. Chronic elevation of cortisol leads to immune suppression, insulin resistance, visceral fat accumulation, and elevated risks of type-2 diabetes and atherosclerosis.

Psychological and Behavioral Aspects

- **Psychological Strain and Overcrowding:** High population density, high living costs, long daily commutes, and reduced exposure to natural green spaces increase risks for anxiety disorders, depression, chronic insomnia, and neuroses.
- **Sedentary Lifestyle and Dietary Shifts:** Urban living often reduces physical activity while increasing consumption of processed, high-calorie, ultra-refined foods. This transition fuels an epidemic of non-communicable lifestyle diseases, including central obesity, metabolic syndrome, and coronary artery disease.
- **Social Isolation and Violence:** Breakdown of traditional community support networks in dense urban environments fosters social isolation, substance abuse, and increased interpersonal violence.

Question 103) Short questions:

a) Radiation Biology

Radiation biology (radiobiology) is the branch of science that studies the action of ionizing and non-ionizing radiation on biological systems and living organisms. Ionizing radiation includes alpha (α) particles, beta (β) particles, gamma (γ) rays, X-rays, and neutrons.

- **Primary Mechanisms of Damage:**
 - **Direct Action:** Radiation directly collides with target biological macromolecules (such as DNA, RNA, or proteins), ionizing them and causing single-strand or double-strand DNA breaks.
 - **Indirect Action:** Radiation collides with intracellular water molecules (H_2O), inducing **radiolysis of water**. This process produces highly reactive free radicals, primarily the hydroxyl radical ($\cdot OH$) and hydrogen peroxide (H_2O_2). These free radicals diffuse through the cytoplasm and oxidize lipid membranes, structural proteins, and nucleic acids.

$Radiation + H_2O \longrightarrow H_2O^+ + e^- \longrightarrow \text{Free Radicals } (\cdot OH, H\cdot) \longrightarrow \text{Oxidative DNA \& Tissue Damage}$

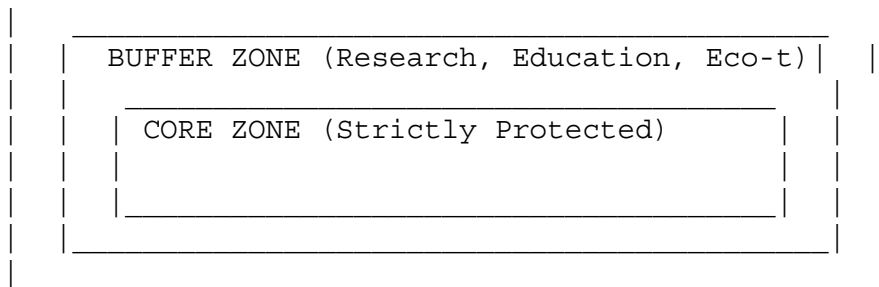
- **Biological Effects:** Radiosensitivity varies according to the **Law of Bergonie and Tribondeau**, which states that rapidly dividing, undifferentiated cells (e.g., bone marrow stem cells, germinal epithelium, gastrointestinal crypt cells) are the most sensitive to radiation damage. Radiations induce chromosomal aberrations, somatic mutations, carcinogenesis (such as leukemia and thyroid carcinoma), and hereditary genetic defects.

b) Biosphere Reserves

A Biosphere Reserve is an internationally designated protected area under UNESCO's **Man and the Biosphere (MAB) Program**, created to promote a balanced relationship between human activity and biodiversity conservation. Biosphere reserves combine ecosystem preservation with sustainable socio-economic development.

Each Biosphere Reserve is divided into three interconnected functional zones:

| TRANSITION ZONE (Sustainable Human Settlements) |



1. **Core Zone:** A strictly protected ecosystem devoted exclusively to conserving wild species, habitats, and genetic resources. Human activities—except non-destructive monitoring—are legally prohibited.
2. **Buffer Zone:** Surrounds or adjoins the core zone. Controlled activities compatible with conservation goals—such as environmental education, scientific research, eco-tourism, and traditional resource gathering—are permitted here.
3. **Transition Zone (Zone of Cooperation):** The outermost area where local communities, management agencies, and scientists work together to practice sustainable agricultural techniques, forestry, and human settlements.

Examples in India include the Nilgiri, Sundarbans, Nanda Devi, and Gulf of Mannar Biosphere Reserves.

c) Environmental Monitoring

Environmental monitoring is the systematic, continuous, or periodic measurement and assessment of physical, chemical, and biological indicators within an environment. Its purpose is to quantify baseline environmental quality, identify pollution sources, track degradation trends over time, and evaluate the effectiveness of regulatory policies.

- **Key Objectives:**
 - To collect reliable scientific data on air, water, and soil quality.
 - To detect toxic chemical discharges early before ecological damage occurs.
 - To verify industrial compliance with statutory environmental standards.
- **Core Components:**
 1. **Physical and Chemical Monitoring:** Continuous measurement of parameters such as PM_{2.5}, SO₂, dissolved oxygen, pH, heavy metal concentrations, and pesticide residues using analytical instruments.
 2. **Biological Monitoring (Biomonitoring):** Using sensitive indicator species (such as lichens for air quality or benthic macroinvertebrates for aquatic health) to detect biological impacts of pollution over time.

Question 104) Zoological names of three vectors of disease that are potentially dangerous to human health. Illustrate the role of animals in relation to human health giving suitable examples.

Answer:

Zoological Names of Three Important Vectors of Human Disease

1. ***Anopheles stephensi*** (or ***Anopheles culicifacies***)
 - **Disease Transmitted:** Malaria (caused by the protozoan parasite *Plasmodium falciparum* and *Plasmodium vivax*).
2. ***Aedes aegypti***
 - **Disease Transmitted:** Dengue fever, Chikungunya, Yellow fever, and Zika virus.
3. ***Xenopsylla cheopis*** (Oriental Rat Flea)
 - **Disease Transmitted:** Bubonic Plague (caused by the bacterium *Yersinia pestis*) and Murine Typhus (*Rickettsia typhi*).

The Role of Animals in Relation to Human Health

Animals play a complex role in human health. They can act as disease vectors, reservoirs, or venomous threats, while also providing critical benefits as models for medical research, sources of pharmaceuticals, and key components of balanced ecosystems.

1. Negative Aspects: Disease Transmission and Health Hazards

A. Animals as Vectors and Hosts of Zoonotic Diseases Zoonoses are infectious diseases naturally transmitted between vertebrate animals and humans.

- **Direct Vectoring:** Blood-sucking arthropods transmit pathogens between animal reservoirs and human hosts. For example, *Culex vishnui* mosquitoes transmit the Japanese Encephalitis virus from wild wading birds and domestic pigs to humans.
- **Mammalian Reservoirs:** Stray and wild carnivores like dogs, bats, and foxes serve as reservoirs for the **Rabies virus**, which is transmitted to humans via infected saliva from bites, causing fatal encephalomyelitis.
- **Intermediate Hosts:** Freshwater snails such as *Bulinus* and *Lymnaea* serve as mandatory intermediate hosts for human blood flukes (*Schistosoma* species) and liver flukes (*Fasciola hepatica*), facilitating parasitic life cycles.

B. Envenomation and Animal Injuries Bites and stings from venomous animals pose immediate threats to human life. Neurotoxic venoms from elapid snakes (e.g., *Naja naja* - Indian Cobra; *Bungarus caeruleus* - Common Krait) cause respiratory paralysis, while hemotoxic venoms from vipers (e.g., *Daboia russelii* - Russell's Viper) induce systemic hemorrhaging and acute renal failure.

2. Positive Aspects: Biomedical Advancement and Therapeutic Contributions

A. Animal Models in Biomedical Research Laboratory animals serve as essential biological models for understanding human disease pathways, evaluating drug safety, and developing vaccines:

- **Mice (*Mus musculus*) and Rats (*Rattus norvegicus*):** Widely used in genetics, oncology, immunology, and toxicology testing due to their physiological similarities to humans and short generation times.
- **Non-Human Primates (e.g., Rhesus Macaques - *Macaca mulatta*):** Crucial for neurobiology research and testing vaccines (such as polio, hepatitis, and COVID-19 vaccines) prior to human clinical trials.

B. Source of Life-Saving Pharmaceuticals and Therapeutics

- **Antivenoms:** Produced by immunizing horses or sheep with small doses of snake venom and harvesting hyperimmune serum containing specific neutralizing antibodies.

- **Hormones and Enzymes:** Insulin was historically extracted from bovine and porcine pancreases for diabetes therapy. Heparin, an anticoagulant, is derived from bovine lungs and intestinal mucosa.
- **Venom-Derived Drugs:** Captopril, a drug used to treat hypertension, was developed from peptides found in the venom of the pit viper *Bothrops jararaca*.

C. Psychological Well-being and Ecological Stability Animals contribute to human mental health through companion therapy, which lowers blood pressure and anxiety. Ecologically, wild animals maintain ecosystem services—such as pollination by bats and bees, and seed dispersal by birds and primates—which support food production and agricultural biosecurity.

Question 105) Justify the statement—"Man-made alterations have enhanced the threat and ferociousness of natural calamities."

Answer:

Natural calamities such as floods, droughts, landslides, cyclonic storms, and heatwaves are driven by earth system dynamics. However, human activities—including environmental degradation, unplanned urbanization, and industrial pollution—have exacerbated these natural phenomena. Unregulated human interventions have increased both the frequency and severity of natural disasters, transforming manageable hazards into catastrophic events.

1. Deforestation and Landside Risk

Trees and natural vegetation stabilize slopes and manage hydrological flow. Their removal amplifies natural hazards:

- **Mechanics of Damage:** In mountainous biomes (such as the Himalayas), root systems anchor the topsoil to underlying bedrock. Uncontrolled logging and highway expansion remove this vegetative anchor. During heavy monsoon rains, water-saturated soils lose cohesion and collapse, causing frequent, high-energy landslides and mudflows that destroy downstream communities.
- **Loss of Canopy:** Deforestation eliminates the forest canopy, which normally intercepts falling rain. Raindrops strike the ground directly, increasing surface runoff and soil erosion.

2. Wetland Destruction and Urban Flooding

Natural wetlands, floodplains, and mangroves act as shock absorbers during extreme weather events:

- **Encroachment on Wetlands:** Lakes, marshes, and urban floodplains absorb excess storm runoff. Converting these natural sponges into concrete housing complexes prevents water infiltration. During heavy rain, runoff rapidly accumulates on impermeable surfaces, causing urban floods (as seen in the Chennai and Mumbai floods).
- **Destruction of Coastal Mangroves:** Mangrove forests along coastal belts absorb cyclonic wave energy and storm surges with their complex aerial root systems. Converting mangroves into commercial aquaculture ponds or coastal developments leaves shorelines exposed to intense cyclonic storms, increasing loss of life and saltwater intrusion into arable farmland.

3. Climate Change and Extreme Weather Events

Anthropogenic emissions of greenhouse gases (CO₂, CH₄, N₂O) from fossil fuel combustion and industrial activity drive global warming, disrupting global climate systems:

- **Intensified Tropical Cyclones:** Rising ocean surface temperatures provide excess thermal energy to atmospheric low-pressure systems, increasing the power and destructive potential of tropical cyclones.
- **Altered Rainfall Patterns:** Climate change alters jet stream dynamics, producing prolonged atmospheric blocking patterns. This leads to intense rainfall over short periods—triggering sudden flash floods and cloudbursts—alternating with extended severe droughts.

4. Riverbed Mining, Damming, and Altered Hydrology

Structural alterations to river systems heighten flood severity:

- **Indiscriminate Sand Mining:** Excessive extraction of riverbed sand destabilizes river banks, lowers water tables, and changes natural flow channels, causing riverbanks to collapse during high flows.
- **Improper Reservoir Management:** Dam construction modifies natural sediment transport, causing sediment to settle in reservoirs and reduce their storage capacity. During extreme monsoon events, sudden large-scale water releases from overfilled reservoirs worsen downstream flooding.

5. Soil Degradation and Desertification

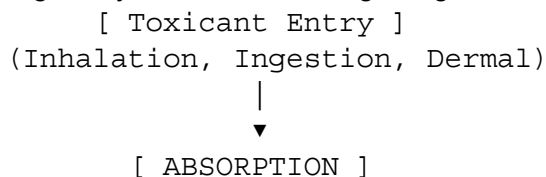
Improper agricultural practices—such as overgrazing, intensive tilling, and monoculture—destroy natural soil structure and lower organic matter content. Depleted soil loses its moisture-retention capacity, turning seasonal rain shortfalls into severe, prolonged agricultural droughts.

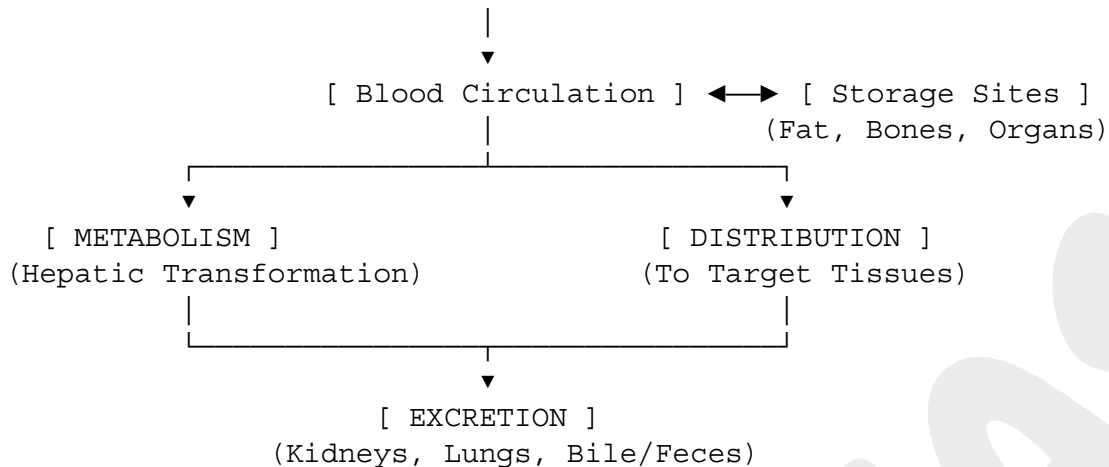
In conclusion, while natural hazards stem from natural processes, human interventions have compromised the earth's natural buffers. Unplanned development, ecosystem destruction, and atmospheric warming have amplified the scale and destruction of natural calamities.

Question 106) Explain the generalized mechanism of absorption, distribution, and excretion of toxicants.

Answer:

Toxicokinetics describes how a toxic compound moves through an organism over time. It encompasses four major processes: **Absorption, Distribution, Metabolism (Biotransformation), and Excretion**. Understanding these mechanisms reveals how xenobiotics enter biological systems, access target organs, and are cleared by the body.





1. Mechanism of Absorption

Absorption is the process by which a toxicant crosses biological cell membranes from the site of exposure (gastrointestinal tract, respiratory tract, or skin) and enters systemic blood circulation.

- **Mechanisms of Trans-Membrane Transport:**

- **Passive Diffusion:** The primary entry mechanism for most toxicants. Non-ionized, lipophilic compounds pass through the phospholipid bilayer along a concentration gradient without expending cellular energy.
- **Facilitated Diffusion & Active Transport:** Polar or charged toxicants structurally resembling essential nutrients utilize carrier protein channels. Active transport moves substances against concentration gradients using ATP.
- **Filtration (Aqueous Pores):** Small, water-soluble, low-molecular-weight molecules pass through aqueous membrane pores along hydrostatic or osmotic gradients.
- **Endocytosis (Phagocytosis/Pinocytosis):** Ingestion of particulate matter (e.g., insoluble dust particles by alveolar macrophages).

- **Major Routes of Entry:**

- **Gastrointestinal Tract:** Ingested toxicants pass through the mucosal epithelium of the stomach or intestine. Non-ionized lipophilic substances are absorbed rapidly in the small intestine due to its large surface area.
- **Respiratory System:** Inhaled gases, vapors, and fine particulate matter (PM_{2.5}) cross the thin, highly vascularized alveolar membrane directly into pulmonary capillaries.
- **Dermal Route:** Toxicants pass through the outer epidermal layer (*stratum corneum*). Lipophilic chemicals (e.g., organophosphate pesticides, organic solvents) readily penetrate this barrier.

2. Mechanism of Distribution

Distribution is the process by which an absorbed toxicant travels via blood and lymph circulation throughout the body to reach various organs, tissues, and toxicological target sites.

- **Factors Controlling Distribution:**

- **Blood Flow Rate:** Highly perfused organs (liver, kidneys, brain, heart) receive systemic toxicants rapidly, whereas poorly perfused tissues (fat, bone, cartilage)

- take up toxicants more slowly.
- **Plasma Protein Binding:** Toxicants in the bloodstream bind reversibly to plasma proteins, primarily **albumin**. The protein-bound fraction cannot cross capillary walls or enter cells; only the unbound ("free") toxicant fraction diffuses into tissues and produces toxic effects.
- **Tissue Affinity and Storage Repositories:** Certain toxicants concentrate selectively in specific tissues:
 - *Lipophilic Chemicals* (DDT, PCBs, Dioxins) accumulate in adipose tissue.
 - *Lead (Pb) and Fluoride (F⁻)* substitute for calcium in bone matrices, accumulating in bone tissue.
 - *Cadmium (Cd)* binds to metallothionein proteins and concentrates in the renal cortex.
- **Physiological Barriers:**
 - *Blood-Brain Barrier (BBB):* Brain capillary endothelial cells feature tight junctions and active efflux transporters (P-glycoprotein) that restrict the entry of polar toxicants into central nervous tissue.
 - *Placental Barrier:* Separates maternal and fetal circulatory systems, though many lipophilic toxicants cross it and cause developmental defects.

3. Mechanism of Excretion

Excretion is the definitive physiological clearance route by which a toxicant or its metabolic products are removed from systemic circulation to the exterior environment.

- **Renal Excretion (Kidneys):** The primary elimination route for water-soluble (hydrophilic) toxicants and Phase II metabolic conjugates.
 - *Glomerular Filtration:* Unbound, low-molecular-weight toxicants in plasma pass through the glomerular membrane into the nephron filtrate.
 - *Tubular Secretion:* Active transport mechanisms in proximal tubule cells transfer organic acids and bases into tubular urine.
 - *Tubular Reabsorption:* Non-ionized, lipophilic compounds in the filtrate diffuse back across tubular epithelial cells into systemic blood, delaying clearance.
- **Biliary and Fecal Excretion:** The liver excretes high-molecular-weight, lipophilic toxicants and Phase II conjugates into bile, which flows into the intestinal lumen for elimination via feces. Certain conjugated toxicants are hydrolyzed by gut bacteria back into parent compounds, which are reabsorbed into the bloodstream via **enterohepatic circulation**.
- **Pulmonary Excretion:** Volatile gases, liquids, and organic solvents (e.g., ethanol, benzene, acetone) diffuse out of pulmonary blood across alveolar membranes and are exhaled.
- **Minor Routes:** Toxicants and their metabolites are also excreted through sweat, saliva, milk (posing risks to nursing neonates), hair, and nails (e.g., arsenic deposition).

Question 107) Mention various sources of water pollution, its consequences, and methods of control.

Answer:

Water pollution is the addition of physical, chemical, or biological substances to water bodies

that alter their natural quality, making them unsafe for human use and harmful to aquatic ecosystems.

Sources of Water Pollution

Water pollution sources fall into two main spatial categories: **Point Sources** (identifiable, localized discharge points like industrial pipe outfalls) and **Non-Point Sources** (diffuse, unconfined runoff from agricultural fields and urban areas).

1. Domestic Sewage and Municipal Wastewater Untreated human excreta, household greywater, synthetic detergents, and food wastes discharged into rivers and lakes. Sewage contains high concentrations of organic matter, disease-causing pathogens, nitrates, and phosphates.

2. Industrial Effluents Factories discharge toxic chemical waste directly into water systems:

- *Textile and Leather Tanneries*: Chrome salts, heavy metals, acid dyes, and sulfides.
- *Chemical and Pharmaceutical Plants*: Synthetic organic compounds, solvents, and antibiotics.
- *Electroplating Units*: Cyanides, lead, cadmium, nickel, and hexavalent chromium.

3. Agricultural Runoff Rainwater washes synthetic chemical fertilizers (nitrates and phosphates), persistent pesticides (organochlorines, organophosphates, carbamates), and animal farm wastes off agricultural fields into surface waters and subterranean aquifers.

4. Thermal and Radioactive Sources Power plants and industrial boilers draw water for cooling and return heated water to natural water bodies (thermal pollution). Nuclear facilities and research laboratories can release radioactive isotopes like Iodine-131, Strontium-90, and Cesium-137.

Consequences of Water Pollution

1. Oxygen Depletion and Eutrophication Discharging organic waste or agricultural fertilizer runoff into water bodies causes nutrient enrichment (high N and P levels), triggering rapid algal growth (**algal blooms**). When these algae die, aerobic microbial decomposers consume large amounts of dissolved oxygen (DO) to break down the organic matter. This increases the **Biochemical Oxygen Demand (BOD)**, depleting DO levels and causing hypoxic "dead zones" where fish and macroinvertebrates suffocate.

Nutrient Runoff (N & P) → Rapid Algal Growth → Microbial Decomposition → DO Depletion → Hypoxic Dead Zones

2. Biomagnification of Toxic Compounds Non-biodegradable lipophilic toxicants (e.g., methylmercury, DDT) enter aquatic food webs through primary producers and accumulate at higher concentrations in top predators, causing reproductive failure, mutations, and organ damage.

3. Spread of Water-Borne Diseases Pathogens in domestic sewage spread water-borne infectious diseases—such as cholera, typhoid, amoebic dysentery, and hepatitis A—among populations relying on untreated water sources.

4. Heavy Metal Toxicity in Humans

- *Minamata Disease*: Caused by eating fish contaminated with methylmercury, leading to severe neurological damage and muscular degeneration.
- *Itai-Itai Disease*: Caused by drinking water contaminated with cadmium, causing severe bone softening and renal failure.

- *Arsenic Poisoning:* Chronic exposure to arsenic from contaminated groundwater leads to skin hyperkeratosis, Blackfoot disease, and internal cancers.

Methods of Controlling Water Pollution

1. Sewage Treatment Plants (STP) Domestic sewage should undergo three treatment stages before environmental discharge:

- **Primary Treatment:** Physical screening and sedimentation to remove large solids and suspended debris.
- **Secondary Treatment:** Biological degradation of organic matter using aerobic microbes in activated sludge units or trickling filters, significantly reducing BOD levels.
- **Tertiary Treatment:** Advanced chemical treatment involving filtration, nitrogen and phosphorus removal, and disinfection using chlorine, ozone, or UV light to kill remaining pathogens.

2. Industrial Effluent Treatment Plants (ETP) Industries must treat wastewater onsite using neutralization, chemical precipitation (to settle heavy metals), activated carbon absorption, and reverse osmosis (RO) before discharging effluent. Advanced plants adopt **Zero Liquid Discharge (ZLD)** systems to recycle all processed water.

3. Sustainable Agriculture Adopting Integrated Pest Management (IPM), applying slow-release organic fertilizers, and establishing vegetated riparian buffer strips along riverbanks to trap agricultural runoff before it reaches water bodies.

4. Legislative Enforcement and Environmental Standards Strictly enforcing statutory regulations—such as the **Water (Prevention and Control of Pollution) Act, 1974**—and imposing heavy fines on industries violating legal discharge limits.

Question 108) "Water is the elixir of life"—but in relation to human disease presents an exception to this statement. Justify.

Answer:

Water is universally recognized as the "elixir of life." It serves as the primary fluid medium for cell metabolism, cellular transport, temperature regulation, and physiological homeostasis in living organisms. No life can exist without water. However, when water becomes contaminated with pathogenic microorganisms, biological parasites, or toxic chemical pollutants, this vital resource becomes a primary carrier of human disease, suffering, and death—presenting a clear paradox to its role as life's elixir.

The Dual Nature of Water: Life-Giver vs. Disease Carrier

While clean water sustains biological processes, contaminated water disrupts human health across global populations.

1. Biological Contamination: Water as a Vector for Pathogens

Water acts as an ideal transport medium for microbial pathogens excreted in human and animal waste. In areas lacking basic sanitation infrastructure, contaminated water supplies can initiate widespread disease outbreaks.

A. Water-Borne Infectious Outbreaks Ingesting drinking water contaminated with human fecal

waste introduces dangerous pathogens into the human gastrointestinal tract:

- **Bacterial Pathogens:** *Vibrio cholerae* releases enterotoxins in the small intestine, causing rapid loss of fluids and electrolytes through acute watery diarrhea. Without swift rehydration, severe dehydration and circulatory collapse can cause death within hours. *Salmonella typhi* causes systemic typhoid fever.
- **Viral Pathogens:** Enteric viruses like Hepatitis A and E cause liver inflammation and acute jaundice, while Rotaviruses cause severe infant gastroenteritis.
- **Protozoan Parasites:** *Entamoeba histolytica* and *Giardia lamblia* invade intestinal mucosa, causing severe dysentery, chronic abdominal pain, and nutrient malabsorption.

B. Water-Based Parasitic Cycles Water provides the required habitat for intermediate hosts in parasitic life cycles that infect humans:

- **Schistosomiasis:** *Schistosoma* blood flukes depend on freshwater snails to produce free-swimming larval cercariae. These larvae penetrate human skin during bathing or wading, migrating through blood vessels to damage liver, intestinal, and bladder tissues.
- **Dracunculiasis (Guinea Worm Disease):** Contracted by drinking water containing micro-crustaceans (*Cyclops*) infected with *Dracunculus medinensis* larvae, which mature and emerge through painful skin blisters.

C. Breeding Environment for Vector-Borne Diseases Stagnant water bodies serve as breeding grounds for disease vectors:

- *Anopheles* mosquitoes breed in stagnant water, transmitting the malaria parasite (*Plasmodium*).
- *Aedes* mosquitoes breed in clean water containers, transmitting Dengue, Chikungunya, and Zika viruses.

2. Chemical Contamination: Toxic Hazards in Water

Industrial effluents, agricultural runoff, and geogenic processes introduce toxic chemicals into drinking water supplies, turning water into a agent of chronic poisoning:

- **Arsenic Toxicity:** Natural leaching of inorganic arsenic into deep groundwater causes chronic arsenicosis, leading to painful skin hyperkeratosis, Blackfoot disease, and visceral cancers.
- **Fluorosis:** Drinking water containing elevated fluoride concentrations ($>1.5 \text{ mg/L}$) leads to permanent dental mottling and crippling skeletal fluorosis.
- **Heavy Metal Toxicity:** Industrial discharge of Lead, Mercury, and Cadmium causes irreversible neurological impairment, renal failure, and skeletal deformities.
- **Agricultural Nitrates:** High nitrate levels in drinking water induce methemoglobinemia ("Blue Baby Syndrome") in infants by converting hemoglobin into methemoglobin, blocking blood oxygen transport.

Conclusion

Water is fundamental to life, but its physical and solvent properties also make it vulnerable to contamination. When water carries microscopic pathogens and toxic pollutants, the very fluid required to sustain human life causes severe illness and mortality. Preserving water's role as the "elixir of life" requires consistent water purification, effective sewage treatment, and protective environmental sanitation.

Question 109) Describe the primary and secondary air pollutants and

give your suggestions to control air pollution.

Answer:

Air pollution refers to the release of foreign substances—gaseous, liquid, or solid particulates—into the atmosphere at concentrations that harm human health, living organisms, and ecosystems. Atmospheric pollutants are classified into two main categories based on how they are formed: **Primary Air Pollutants** and **Secondary Air Pollutants**.

1. Primary Air Pollutants

Primary air pollutants are substances emitted directly into the atmosphere from identifiable sources, such as industrial stacks, motor vehicle exhausts, agricultural activities, or volcanic eruptions. They retain their basic chemical structure after emission.

- **Carbon Monoxide (CO):** A colorless, odorless gas produced by the incomplete combustion of carbon-containing fuels in automobiles and industrial furnaces. It binds reversibly to hemoglobin with an affinity roughly 200 times higher than oxygen, forming **carboxyhemoglobin (COHb)**, which inhibits blood oxygen transport and causes hypoxia, dizziness, and asphyxiation.
- **Sulfur Dioxide (SO₂):** Generated by burning sulfur-rich fossil fuels (coal and oil) in thermal power plants and metal smelting operations. SO₂ irritates upper respiratory tracts, triggers severe asthma, and reacts in the air to produce acid rain.
- **Nitrogen Oxides (NO_x - NO and NO₂):** Emitted from high-temperature combustion in motor vehicle engines and power plants. NO₂ causes pulmonary irritation, lowers host resistance to respiratory infections, and acts as a necessary precursor for smog formation.
- **Particulate Matter (PM_{2.5} and PM₁₀):** Fine solid particles or liquid droplets suspended in the air (soot, fly ash, silica dust). Particulate matter below 2.5 μm (PM_{2.5}) penetrates deep into lung alveoli, entering systemic circulation to cause chronic cardiovascular and pulmonary disease.
- **Volatile Organic Compounds (VOCs):** Hydrocarbons (such as benzene, toluene, and formaldehyde) emitted from unburnt fuel, industrial solvents, and chemical manufacturing. Many VOCs are direct carcinogens.

2. Secondary Air Pollutants

Secondary air pollutants are not emitted directly into the atmosphere. Instead, they form in the air through physical and chemical reactions between primary air pollutants, atmospheric gases, moisture, and solar ultraviolet radiation.

- **Ground-Level (Tropospheric) Ozone (O₃):** Formed by complex photochemical reactions between primary NO_x emissions and Volatile Organic Compounds (VOCs) in the presence of intense sunlight: $\text{NO}_2 + h\nu (\text{solar UV}) \rightarrow \text{NO} + \text{O}$
 $\text{O} + \text{O}_2 \rightarrow \text{O}_3$ Ground-level ozone damages pulmonary tissue, reduces plant photosynthetic capacity, and lowers agricultural crop yields.
- **Photochemical Smog:** A brown atmospheric haze formed in sunny urban environments with high vehicular traffic. It consists of ground-level ozone, Peroxyacetyl Nitrate (PAN), aldehydes, and nitrogen oxides.
- **Peroxyacetyl Nitrate (PAN):** A potent chemical component of photochemical smog produced when VOC oxidation products react with NO₂. PAN causes intense eye

irritation, lacrimation, and severe damage to plant leaf chloroplasts.

- **Acid Deposition (Acid Rain):** Formed when primary emissions of SO₂ and NO_x react with atmospheric moisture and oxygen to yield sulfuric acid (H₂SO₄) and nitric acid (HNO₃).
- **Sulfur Trioxide (SO₃):** Formed when primary SO₂ oxidizes in the atmosphere before reacting with water vapor to create sulfuric acid mist.

Suggestions and Control Strategies for Air Pollution

Addressing air pollution requires a combination of technological innovations, regulatory measures, and alternative energy adoption:

1. Technological and Engineering Control Devices Industrial facilities and power plants should install flue-gas cleaning technologies:

- **Electrostatic Precipitators (ESP):** Use high-voltage electric fields to charge and collect fine fly ash and particulate matter from industrial smoke before discharge, reaching removal efficiencies above 99%.
- **Scrubbers (Wet and Dry):** Use alkaline sprays (such as lime slurry) to neutralize and scrub acidic gaseous pollutants like SO₂ from industrial exhaust streams.
- **Fabric Filters (Baghouses):** Pass industrial exhaust through woven fabric bags to trap fine dust particles.
- **Catalytic Converters:** Install three-way catalytic converters (containing platinum, palladium, and rhodium catalysts) on motor vehicles to convert toxic CO, unburnt hydrocarbons, and NO_x into harmless CO₂, H₂O, and N₂.

Industrial Exhaust → ESP / Scrubbers → Chemical & Particulate Removal → Clean Stack Gas

2. Clean Energy Transition Progressively replace fossil-fuel power plants with renewable energy infrastructure, including solar, wind, hydroelectric, and geothermal installations.

3. Fuel Quality and Vehicular Emission Standards Mandate cleaner fuel standards with minimal sulfur content (such as Bharat Stage VI / BS-VI standards) and accelerate the adoption of electric vehicles (EVs) supported by public charging networks.

4. Urban Planning and Green Infrastructure Expand municipal mass public transit networks (metros, bus rapid transit systems) to reduce private vehicle usage. Plant urban forest shelterbelts along roadways; broad-leaf trees act as natural bio-filters that trap suspended dust particles and absorb gaseous air pollutants.

5. Regulatory Enforcement Enforce statutory standards under the **Air (Prevention and Control of Pollution) Act, 1981**, accompanied by real-time continuous air quality monitoring and strict penalties for open crop residue burning and industrial non-compliance.

Question 110) What role do animals play to influence human health? Explain.

Answer:

Animals exert a profound influence on human health. This relationship is dual-faceted: animals can pose health risks as disease reservoirs, vectors, and venomous hazards, but they also offer significant health benefits by providing biomedical research models, essential pharmaceuticals,

therapeutic companion support, and foundational ecosystem services.

1. Negative Influences on Human Health

A. Transmission of Zoonotic Pathogens Zoonoses are infectious diseases naturally transmitted between vertebrate animals and humans. Animals act as biological reservoirs for major pathogens:

- **Viral Zoonoses:** Rabies is transmitted via infected mammalian saliva (dogs, bats), causing fatal central nervous system inflammation. Wildlife species (e.g., bats, non-human primates, birds) act as natural reservoirs for emerging viruses, including Ebola, Avian Influenza (H5N1), Nipah virus, and Coronaviruses.
- **Bacterial Zoonoses:** *Brucella* species from infected livestock cause brucellosis, while *Bacillus anthracis* from cattle causes anthrax.
- **Parasitic Zoonoses:** Toxoplasmosis (*Toxoplasma gondii*) is transmitted to humans via feline feces, posing severe risks to fetal development during pregnancy.

B. Arthropod Vectors of Pathogens Animals act as biological or mechanical vectors that carry pathogens from animal reservoirs to human hosts:

- **Mosquitoes:** *Anopheles* transmits malaria; *Aedes* transmits dengue, chikungunya, and yellow fever.
- **Fleas:** *Xenopsylla cheopis* (rat flea) transmits *Yersinia pestis*, the causative agent of bubonic plague, from rodent hosts to humans.
- **Ticks:** *Ixodes* ticks transmit Lyme disease (*Borrelia burgdorferi*) from wild deer and rodent reservoirs to humans.

[Animal Reservoir] —> [Arthropod Vector] —> [Human Host] —> [Disease State]
(e.g., Rodent / Bird) (e.g., Flea / Mosquito)

C. Envenomation, Allergies, and Physical Injury Venomous bites and stings from snakes (cobras, vipers), scorpions, and hymenopterans (bees, wasps) induce systemic neurotoxicity, hemotoxicity, or fatal anaphylactic shock. In addition, animal dander, microscopic dust mites (*Dermatophagoides*), and cockroach exuviae are major environmental allergens that trigger allergic asthma and rhinitis.

2. Positive Influences on Human Health

A. Biomedical Models for Research Laboratory animals serve as essential biological models for investigating human physiological processes, disease mechanisms, and drug toxicities:

- **Genetically Modified Rodents:** Transgenic mice and knock-out rat models allow researchers to study human cancer genetics, cardiovascular diseases, and metabolic disorders.
- **Non-Human Primates:** Used in neurobiology research and preclinical safety testing of complex vaccines and gene therapies due to their close physiological and genetic similarity to humans.

B. Source of Life-Saving Pharmaceuticals and Medical Tools

- **Biological Sera and Vaccines:** Polivalent antivenoms are produced by immunizing horses with snake venoms and extracting neutralizing antibodies from their blood plasma. Vaccines against diseases like rabies and smallpox were developed using animal tissues.
- **Therapeutic Biomarkers and Anticoagulants:** Heparin (derived from porcine intestine)

is used to prevent blood clots during surgery. The diagnostic limulus amebocyte lysate (LAL) assay, extracted from the blue blood of horseshoe crabs (*Limulus polyphemus*), is used globally to test injectable drugs for bacterial endotoxin contamination.

C. Psychological Well-being and Animal-Assisted Therapy Interacting with companion animals (dogs, cats) reduces human physiological stress parameters—lowering circulating cortisol levels, resting heart rates, and blood pressure. Animal-Assisted Therapy (AAT) is widely used to assist in the rehabilitation of post-traumatic stress disorder (PTSD), autism spectrum disorders, and clinical depression.

D. Ecosystem Services and Nutrition Animals provide essential nutritional intake (high-quality protein, essential amino acids, and micronutrients) via dairy, poultry, and fisheries. In addition, wild fauna maintains essential ecosystem services—such as natural pest control by insectivorous bats and birds, and crop pollination by wild bees—which sustain agricultural productivity and global food biosecurity.

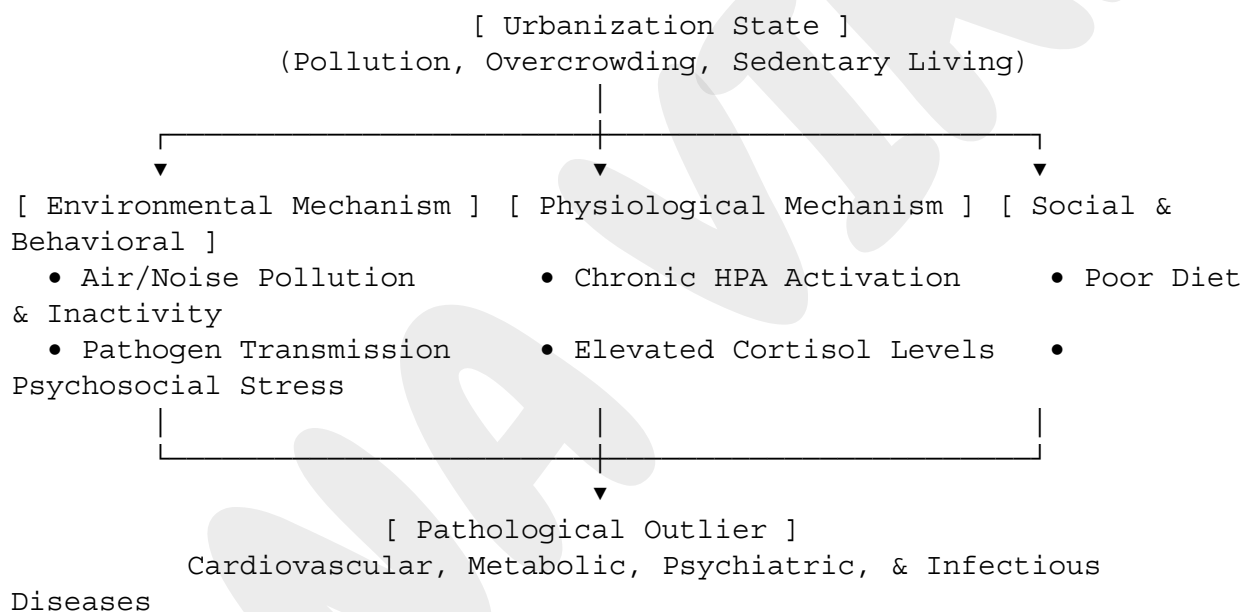
Question 111) With the spread of urbanization in human society the concern of human health have increased due to organization state. Explain its mechanism.

Answer:

The transition from traditional rural environments to dense urban centers represents one of the most rapid demographic shifts in human history. While urbanization brings modern healthcare facilities and economic development, the "urban state"—characterized by overcrowding, industrial activity, altered microclimates, and sedentary lifestyle patterns—exerts complex environmental and physiological stresses on human populations.

Mechanistic Pathways of Urbanization Stress on Human Health

The detrimental health impacts associated with urban expansion operate through three primary interconnected mechanisms:



1. Environmental Mechanistic Pathways

- **Air Quality Degradation and Respiratory Dynamics:** Urban centers concentrate motor vehicles, industrial operations, and energy production facilities. High atmospheric densities of PM_{2.5}, nitrogen dioxide (NO₂), and ground-level ozone (O₃) enter human respiratory tracts. Fine particulate matter (PM_{2.5}) crosses the alveolar-capillary barrier into systemic circulation, inducing localized pulmonary inflammation, endothelial dysfunction, systemic oxidative stress, and heightened risks of chronic obstructive pulmonary disease (COPD), asthma, and myocardial infarction.
- **Vector and Pathogen Transmission Dynamics:** Unplanned urban density overwhelms municipal sanitation, leading to open sewage drainage, uncollected solid waste, and

artificial water storage. This creates breeding grounds for disease vectors (such as *Aedes aegypti* and *Culex* mosquitoes), accelerating the transmission mechanics of vector-borne illnesses (dengue, chikungunya, lymphatic filariasis) and water-borne pathogens (cholera, typhoid).

2. Neuroendocrine and Physiological Mechanistic Pathways

Continuous exposure to urban environmental stressors—specifically chronic ambient noise pollution (traffic, construction) and severe physical crowding—triggers continuous neuroendocrine stress responses:

- **Sympathetic-Adrenal-Medullary (SAM) Axis Overdrive:** Chronic exposure to high ambient decibel levels keeps the sympathetic nervous system continuously activated, raising circulating epinephrine and norepinephrine levels. This leads to persistent peripheral vasoconstriction, elevated resting heart rates, and sustained arterial hypertension.
- **Hypothalamic-Pituitary-Adrenal (HPA) Axis Dysregulation:** Chronic urban psychological stress causes the hypothalamus to secrete Corticotropin-Releasing Hormone (CRH), driving the anterior pituitary to release Adrenocorticotrophic Hormone (ACTH), which stimulates sustained cortisol secretion from the adrenal cortex.
- **Pathological Consequences of Hypercortisolemia:**
 - **Immune Suppression:** Cortisol inhibits lymphocyte proliferation and suppresses inflammatory cytokine signaling, rendering urban populations more susceptible to opportunistic infections.
 - **Metabolic Dysfunction:** Sustained cortisol promotes gluconeogenesis, induces peripheral insulin resistance, and accelerates visceral fat deposition, driving high rates of type-2 diabetes mellitus and metabolic syndrome.

3. Behavioral and Socio-Psychological Mechanistic Pathways

- **Sedentary Transition and Nutritional Shift:** Urban built environments favor motorized transit over walking and replace natural open spaces with high-density concrete infrastructure. Concurrently, traditional diets rich in fiber are replaced by ultra-processed foods dense in refined sugars, sodium, and saturated fats. This shift drives an epidemic of obesity, coronary artery disease, and gastrointestinal disorders.
- **Psychosocial Disconnection and Mental Health Disruption:** The breakdown of extended family networks, high financial pressures, long commutes, and reduced contact with natural ecosystems disrupt neurochemical regulation. This increases the incidence of major depressive disorders, chronic anxiety, sleep fragmentation, and substance abuse in urban populations.

Question 112) What is environmental monitoring? Analyse its tools and methodology in modern day's world.

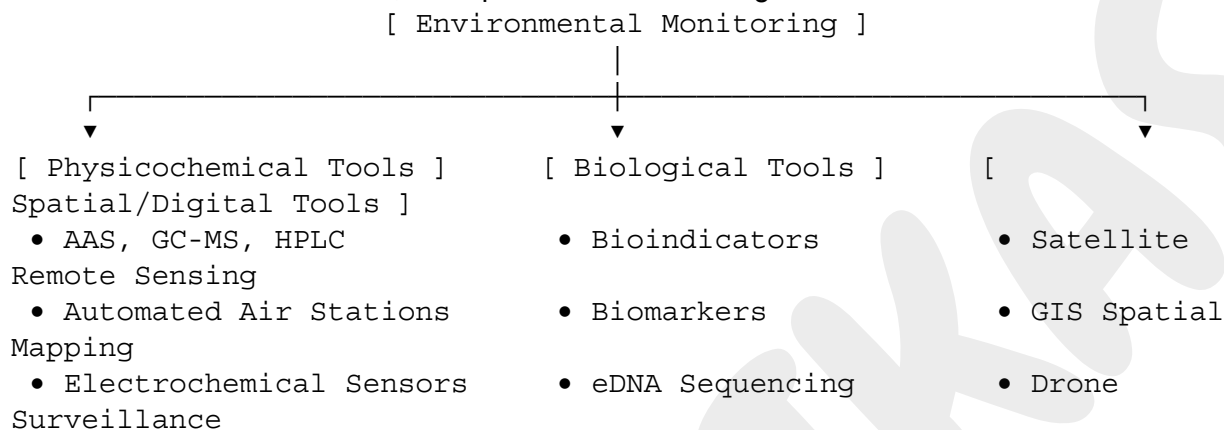
Answer:

Environmental monitoring is the systematic, continuous, or periodic measurement and assessment of physical, chemical, and biological parameters in air, water, soil, and biotic communities. Its primary objectives are to establish baseline environmental conditions, detect

pollution events, assess ecosystem health, evaluate compliance with statutory standards, and inform environmental management policies.

Modern Tools and Methodology of Environmental Monitoring

Modern environmental monitoring integrates analytical chemistry, biological indicators, automated electronic sensors, and space-based technologies.



1. Chemical and Physical Analytical Methodology

Chemical monitoring measures pollutant concentrations across environmental media using high-precision instrumentation:

- **Atmospheric Air Quality Monitoring:**
 - **Continuous Ambient Air Monitoring Stations (CAAQMS):** Use non-dispersive infrared (NDIR) spectroscopy for measuring CO, chemiluminescence for nitrogen oxides (NO_x), ultraviolet fluorescence for sulfur dioxide (SO₂), and beta-attenuation monitors (BAM) for real-time mass concentration measurement of fine particulate matter (PM_{2.5} and PM₁₀).
- **Aquatic and Soil Chemical Analysis:**
 - **Atomic Absorption Spectroscopy (AAS) & Inductively Coupled Plasma Mass Spectrometry (ICP-MS):** Analytical tools used to detect trace heavy metal contamination (Lead, Mercury, Cadmium, Arsenic) down to parts-per-billion (ppb) levels in water and soil samples.
 - **Gas Chromatography-Mass Spectrometry (GC-MS) & High-Performance Liquid Chromatography (HPLC):** Used to isolate, identify, and quantify organic pollutants, including pesticide residues, polychlorinated biphenyls (PCBs), and pharmaceutical compounds in environmental samples.

2. Biological Monitoring (Biomonitoring) Tools

Biological monitoring evaluates how pollutants affect living organisms, providing a direct assessment of ecological impact.

- **Bioindicators:** Sensitive wild organisms used to monitor environmental quality:
 - **Lichens (*Usnea*, *Parmelia*):** Epiphytic lichens are sensitive to atmospheric SO₂ and heavy metals; their decline indicates air degradation.

- **Benthic Macroinvertebrates:** The presence of sensitive larvae (Mayflies - *Ephemeroptera*, Stoneflies - *Plecoptera*) indicates clean, oxygen-rich stream water, whereas a dominance of bloodworms (*Chironomus*) signals severe organic pollution and hypoxia.
- **Biomarkers:** Molecular, physiological, or biochemical changes in indicator organisms:
 - Measurement of **Acetylcholinesterase (AChE) inhibition** in fish brains to evaluate exposure to organophosphate pesticides.
 - Measurement of **Metallothionein protein induction** in aquatic organisms to track heavy metal exposure.
- **Environmental DNA (eDNA) Metabarcoding Analysis:** A modern molecular tool where total genomic DNA shed by organisms (via skin, mucus, or feces) is extracted directly from water or soil samples and amplified using Polymerase Chain Reaction (PCR). This technique enables non-invasive monitoring of biodiversity, endangered species, and invasive pathogens.

3. Spatial, Remote Sensing, and Digital Tools

- **Satellite Remote Sensing:** Multi-spectral and hyper-spectral sensors mounted on Earth-observation satellites (such as Landsat, Sentinel, and MODIS) capture surface reflectance data across different wavelengths. This allows large-scale monitoring of:
 - Deforestation and forest canopy health via the Normalized Difference Vegetation Index (NDVI).
 - Marine algal blooms via chlorophyll-a spectral signatures.
 - Land surface temperature variations and desertification trends.
- **Geographic Information Systems (GIS):** GIS software integrates spatial location data with environmental attribute data, allowing ecologists to build spatial pollution heatmaps, model contaminant migration pathways, and map habitat corridors.
- **Wireless Sensor Networks (WSN) and IoT Devices:** Internet-of-Things (IoT) enabled water quality probes and low-cost air sensors deployed across urban areas stream continuous, real-time environmental data to centralized cloud servers for instant analysis and early warning alerts.

Question 113) Give an account of general principles of radiation biology and its environmental hazards.

Answer:

Radiation biology (radiobiology) is the study of the action of ionizing and non-ionizing radiation on biological systems. Ionizing radiation—including high-energy electromagnetic waves (gamma-rays, X-rays) and particulate radiation (alpha-particles, beta-particles, neutrons)—possesses sufficient energy to displace orbital electrons from atoms, creating reactive ions and free radicals that damage biological macromolecules.

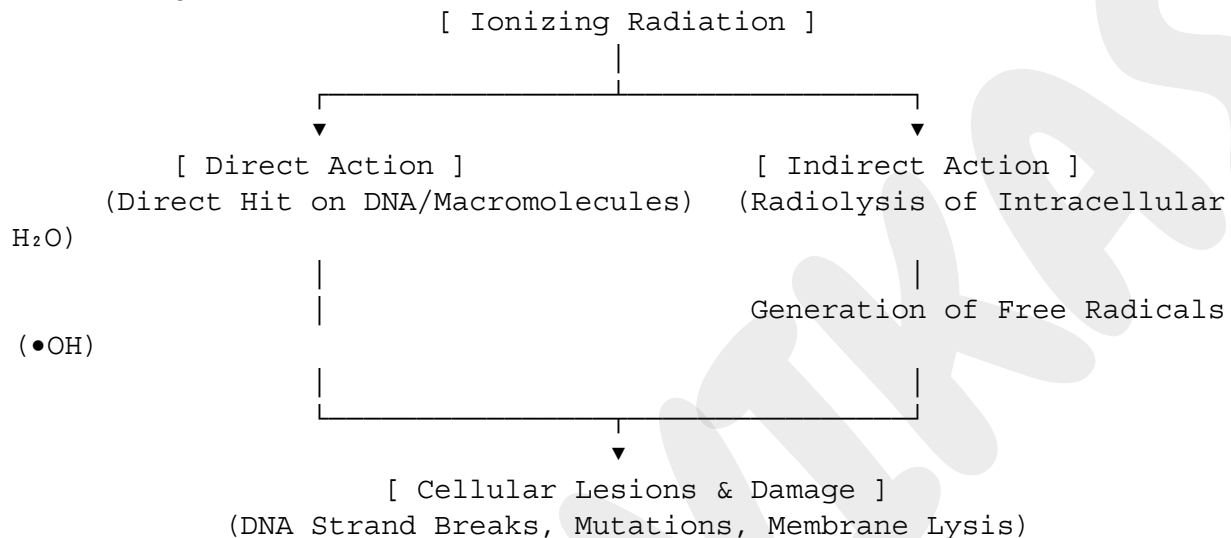
General Principles of Radiation Biology

1. Primary Mechanisms of Action

Radiation damages cellular components through two main pathways:

- **Direct Action:** The radiation particle or photon directly strikes a target cellular macromolecule—most critically the DNA double helix—causing single-strand breaks (SSBs), double-strand breaks (DSBs), or base modifications.
- **Indirect Action (Radiolysis of Water):** Because water constitutes roughly 70 to 80% of cellular volume, radiation photons interact primarily with intracellular water molecules (H_2O), inducing **radiolysis**.

The resulting free radicals, particularly the hydroxyl radical ($\cdot OH$), are highly reactive. They diffuse through the cell and oxidize membrane lipids, structural proteins, and nuclear DNA.



2. The Law of Bergonié and Tribondeau (Radiosensitivity)

This principle states that the radiosensitivity of a biological tissue is directly proportional to its metabolic activity and reproductive capacity, and inversely proportional to its degree of differentiation.

- **Highly Radiosensitive Tissues:** Rapidly dividing, undifferentiated cells—such as hematopoietic bone marrow stem cells, intestinal crypt stem cells, spermatogonia, and embryonic tissues.
- **Radioresistant Tissues:** Differentiated, non-dividing cells—such as mature neurons and skeletal muscle cells.

3. Deterministic vs. Stochastic Effects

- **Deterministic Effects (Tissue Reactions):** Show a clear dose threshold below which the effect does not occur. Severity increases with radiation dose (e.g., radiation burns, cataracts, acute radiation syndrome, and sterility).
- **Stochastic Effects:** Radiation-induced effects that occur by chance, with no dose threshold. The *probability* of occurrence—rather than severity—increases linearly with radiation dose (e.g., somatic mutations, radiation-induced cancers like leukemia, and hereditary genetic mutations).

Environmental Hazards of Radiation

Radiation hazards enter natural ecosystems through nuclear weapon testing, nuclear power plant accidents (e.g., Chernobyl, Fukushima), uranium mining tailings, and industrial nuclear waste dumping.

1. Bioaccumulation and Radionuclide Dynamics in Food Chains Radioactive isotopes resemble essential nutrient elements chemically and are absorbed by living organisms, cycling through ecosystems:

- **Strontium-90** : Structurally mimics calcium (Ca). It accumulates in bone tissue, where continuous emission of β -particles damages bone marrow stem cells, causing leukemia and bone sarcoma.
- **Cesium-137** : Chemically resembles potassium (K). It distributes throughout muscular tissues in animals and bioaccumulates across terrestrial and aquatic food webs.
- **Iodine-131** : Concentrates rapidly in the thyroid gland, increasing the risk of thyroid carcinoma, particularly in children.

2. Damage to Natural Ecosystems and Wild Fauna High exposure to ionizing radiation reduces reproductive output, induces congenital deformities, and increases mortality in wild populations:

- **Plant Communities**: High radiation fields cause forest dieback (e.g., the "Red Forest" near Chernobyl), killing sensitive coniferous trees and altering plant community composition toward resistant herbaceous species.
- **Animal Populations**: Wild mammals and birds exposed to radioactive fallout show high rates of partial sterility, cataracts, embryonic mortality, and elevated germline DNA mutation rates.

3. Long-Term Genetic and Evolutionary Consequences Sub-lethal, persistent radiation exposure causes germline mutations in animal populations. Accumulating deleterious recessive mutations in gene pools degrades population fitness, increases hereditary malformations, and threatens long-term species survival.

Question 114) Giving an inventory of wildlife in India, describe the wildlife management in detail.

Answer:

India is one of the world's 17 megadiverse nations. Occupying roughly 2.4% of the global land area, India supports approximately 7-8% of all recorded biological species, including a rich diversity of wild fauna distributed across its 10 biogeographic zones.

Inventory of Wildlife in India

India's wild fauna spans diverse taxonomic groups:

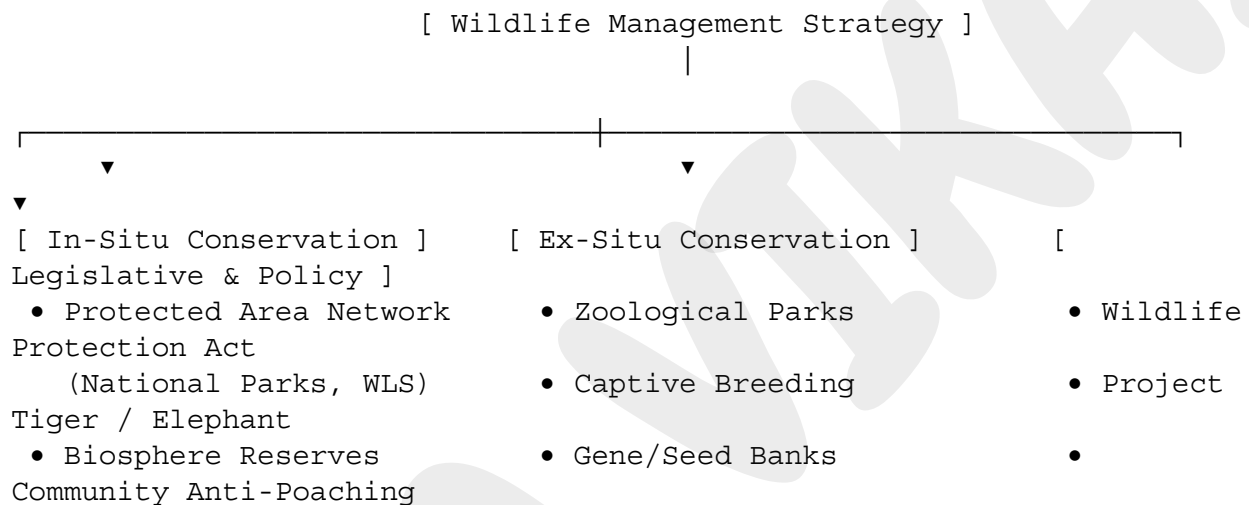
- **Mammalian Fauna**: Includes iconic apex predators and large herbivores:
 - **Carnivores**: Bengal Tiger (*Panthera tigris tigris*), Asiatic Lion (*Panthera leo persica* - endemic to Gir, Gujarat), Snow Leopard (*Panthera uncia*), Indian Leopard (*Panthera pardus fusca*), and Dhole (*Cuon alpinus*).
 - **Herbivores**: Asian Elephant (*Elephas maximus*), Great Indian One-Horned Rhinoceros (*Rhinoceros unicornis*), Gaur or Indian Bison (*Bos gaurus*), Wild Water Buffalo (*Bubalus arnee*), and endemic ungulates like the Hangul (*Cervus elaphus hanglu*) and Sangai Deer (*Rucervus eldii eldii*).
- **Avian Fauna**: Over 1,300 recorded bird species, including the critically endangered Great

Indian Bustard (*Ardeotis nigriceps*), Sarus Crane (*Antigone antigone*), Indian Vultures (*Gyps indicus*), and the Malabar Pied Hornbill (*Anthracoceros coronatus*).

- **Reptilian Fauna:** King Cobra (*Ophiophagus hannah*), Gharial (*Gavialis gangeticus*), Marsh Crocodile (*Crocodylus palustris*), and the Saltwater Crocodile (*Crocodylus porosus*).
- **Aquatic Mammal Fauna:** Ganges River Dolphin (*Platanista gangetica*) and the Dugong or Sea Cow (*Dugong dugon*).

Comprehensive Account of Wildlife Management in India

Wildlife management involves legal protection, habitat restoration, population monitoring, and community engagement to ensure the sustainable persistence of wild populations and their habitats.



1. In-Situ Conservation Strategies (Protected Area Network)

In-situ conservation focuses on protecting wild species within their natural habitats:

- **National Parks and Wildlife Sanctuaries:** India has established an extensive network of over 100 National Parks and 500 Wildlife Sanctuaries under the Wildlife (Protection) Act, 1972.
 - *National Parks* enjoy the highest level of legal protection; all commercial activities, human settlements, and livestock grazing are prohibited.
 - *Wildlife Sanctuaries* afford protection to wild fauna while permitting regulated traditional activities like firewood collection or grazing in buffer areas.
- **Biosphere Reserves:** Large ecosystem reserves under UNESCO's MAB program (e.g., Nilgiri, Sundarbans, Nanda Devi) that integrate core conservation zones with surrounding buffer and transition zones supporting sustainable human development.
- **Conservation and Community Reserves:** Designated corridors and privately or community-owned lands protected to maintain ecological connectivity between isolated national parks.

2. Species-Specific Conservation Projects

Targeted national initiatives designed to recover endangered species facing immediate extinction risks:

- **Project Tiger (1973):** Established dedicated Tiger Reserves featuring core areas managed exclusively for tiger breeding, linked by buffer zones supporting eco-development.
- **Project Elephant (1992):** Focuses on protecting Asian elephant populations, securing historical migratory corridors, and mitigating human-elephant conflict.
- **Gharial Conservation Project:** Rescued the gharial from near-extinction using "rear-and-release" captive hatching and reintroduction into protected river systems (e.g., National Chambal Sanctuary).
- **Project Snow Leopard:** Combines high-altitude ecosystem conservation with community-based pastoral management across Himalayan states.

3. Habitat Management and Restoration Techniques

Maintaining habitat quality is central to successful wildlife management:

- **Water and Forage Development:** Constructing artificial water holes and earthen check dams to ensure water availability during dry summer months. Managed burning and clearing of invasive plant species (e.g., *Lantana camara*, *Eupatorium*) are used to maintain grassy glades for wild herbivores.
- **Biological Corridors:** Securing vegetated landscape corridors that link fragmented forest patches, allowing animal dispersal, seasonal migration, and gene flow.

4. Legislative Framework and Enforcement

- **The Wildlife (Protection) Act, 1972:** The central legal framework in India. It categorizes wild species into protective Schedules (Schedule I granting absolute legal protection with maximum penalties for offenses) and restricts commercial trade in wildlife products.
- **Anti-Poaching Operations:** Deploying armed forest protection units, dog squads, and real-time electronic surveillance tools (such as M-STrIPES for digital tiger monitoring) to prevent illegal poaching and wildlife trafficking.

5. Ex-Situ Conservation and Community Involvement

- **Ex-Situ Conservation:** Zoological gardens, botanical parks, and specialized breeding facilities serve as genetic reservoirs. Captive breeding programs (e.g., for the Pygmy Hog and Vultures) preserve species faced with extreme wild population declines until threats are mitigated.
- **Community-Based Conservation:** Integrating local communities into conservation management through eco-development committees, Joint Forest Management (JFM), eco-tourism profit sharing, and rapid crop/livestock compensation schemes to convert local populations into active conservation partners.

Question 115) Describe critical issues related to organizational stress and discuss mechanism of its influence on human health.

Answer:

Organizational stress (occupational or workplace stress) refers to the adverse physiological and psychological responses that occur when job demands, environmental pressures, and workplace expectations exceed an employee's perceived resources, capabilities, or coping mechanisms. In modern industrial societies, organizational stress represents a major occupational health hazard.

Critical Issues Related to Organizational Stress

Organizational stress stems from several key workplace factors:

1. High Job Demands vs. Low Decision Control According to Karasek's Demand-Control model, severe workplace stress occurs in high-strain jobs that combine high workload demands (tight deadlines, long working hours) with low decision-making authority or autonomy over task execution.

2. Effort-Reward Imbalance When an employee expends high mental or physical effort but receives low extrinsic rewards (compensation, job security, promotion opportunities) or low intrinsic recognition, it generates chronic psychological strain.

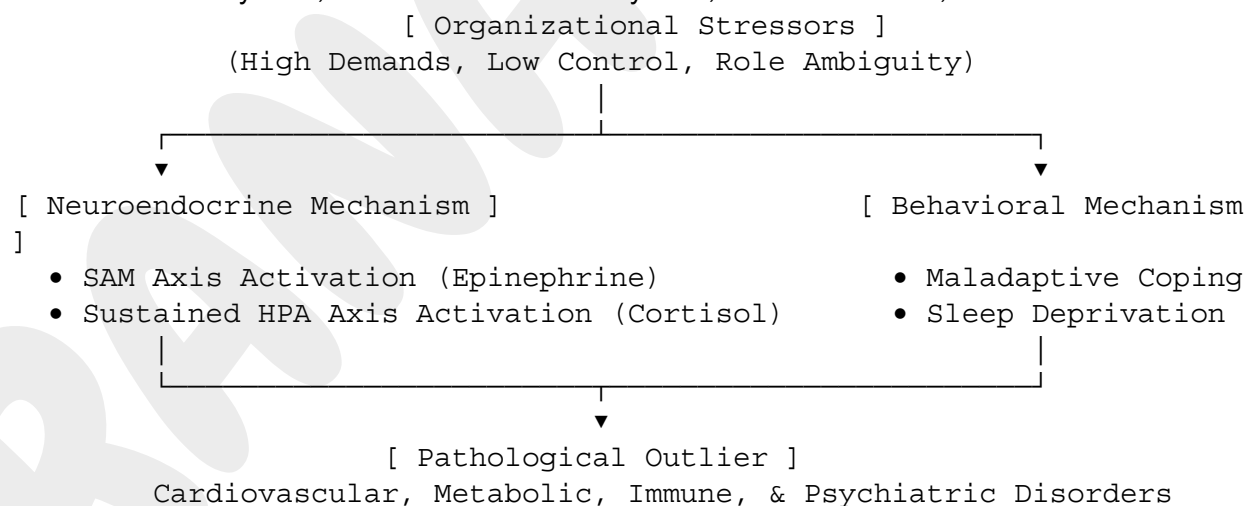
3. Organizational Role Ambiguity and Conflict

- *Role Ambiguity*: Unclear job objectives, responsibilities, and performance evaluation metrics create continuous anxiety.
- *Role Conflict*: Conflicting demands from different supervisors force employees into incompatible operational pressures.

4. Workplace Insecurity and Dysfunctional Interpersonal Dynamics Fears of restructuring, lay-offs, or contract termination, combined with workplace harassment, bullying, or isolation from peer support networks, drive chronic emotional strain.

Mechanism of Organizational Stress Influence on Human Health

Organizational stress affects human health through physiological pathways involving the neuroendocrine system, autonomous nervous system, immune function, and health behaviors.



1. The Neuroendocrine Mechanism

Chronic workplace stress continuously triggers the body's physiological stress systems:

- **Sympathetic-Adrenal-Medullary (SAM) Axis Overactivity:** Acute workplace stressors trigger the release of epinephrine and norepinephrine from the adrenal medulla. This raises resting heart rates, increases cardiac output, and causes peripheral vasoconstriction. Over time, persistent SAM activation injures vascular endothelial linings, accelerating atherosclerosis and essential arterial hypertension.
- **Hypothalamic-Pituitary-Adrenal (HPA) Axis Dysregulation:** Chronic organizational strain leads to sustained secretion of Corticotropin-Releasing Hormone (CRH) from the hypothalamus and Adrenocorticotrophic Hormone (ACTH) from the pituitary, driving prolonged cortisol elevation from the adrenal cortex.
 - *Metabolic Impacts:* Sustained high cortisol levels promote hepatic gluconeogenesis and induce peripheral insulin resistance, driving visceral fat accumulation and accelerating the onset of metabolic syndrome and type-2 diabetes.
 - *Immune System Suppression:* Chronic cortisol exposure inhibits nuclear factor kappa B (NF- κ B), suppressing T-lymphocyte proliferation and natural killer (NK) cell activity. This weakens host immune responses, increasing susceptibility to viral infections and slowing tissue repair.

2. Pathological Health Outcomes

- **Cardiovascular Disease:** Chronic organizational stress is an independent risk factor for coronary artery disease, angina pectoris, and stroke due to sustained hypertension, systemic inflammation (elevated C-reactive protein), and heightened platelet aggregation.
- **Gastrointestinal Disorders:** Chronic stress alters brain-gut axis signaling, causing mucosal ischemia, altered gut motility, and increased intestinal permeability. This exacerbates gastroesophageal reflux disease (GERD), irritable bowel syndrome (IBS), and peptic ulceration.
- **Psychiatric and Neurodevelopmental Disturbances:** Persistent workplace stress depletes brain-derived neurotrophic factor (BDNF) and alters serotonin and dopamine receptor sensitivity, leading to clinical burnout syndrome, major depressive disorder, severe anxiety, and chronic sleep fragmentation.

3. Behavioral Mechanisms

To cope with chronic workplace pressure, individuals frequently adopt unhealthy behaviors—such as increased alcohol consumption, heavy tobacco smoking, poor nutritional choices, and physical inactivity. These habits further compound physiological damage, accelerating health deterioration.

Question 116) What role remote sensing plays in environmental monitoring? Describe its various constituents.

Answer:

Remote sensing is the science and technology of acquiring quantitative information about Earth's surface, atmosphere, and aquatic systems from a distance, without coming into direct physical contact with the object or area under investigation. It relies on detecting and recording reflected or emitted electromagnetic radiation (EMR) using sensors mounted on airborne

(drones, aircraft) or spaceborne (satellite) platforms.

Role of Remote Sensing in Environmental Monitoring

Remote sensing provides broad spatial coverage, high temporal frequency, and multi-spectral imaging capability, making it an essential tool for monitoring large-scale environmental processes:

1. Deforestation and Forest Canopy Monitoring: Satellite sensors measure light reflection in the red and near-infrared (NIR) spectrums to calculate the **Normalized Difference Vegetation Index (NDVI)**

Healthy, green vegetation absorbs red light for photosynthesis and strongly reflects near-infrared light. Tracking changes in NDVI over time allows researchers to quantify global deforestation, monitor forest degradation, assess canopy health, and track post-fire vegetation recovery.

2. Water Resource Management and Aquatic Quality Monitoring: Remote sensing maps surface water dynamics, measures lake and reservoir storage variations, tracks flood inundation extents, and evaluates aquatic pollution. Specific spectral signatures detect chlorophyll-a concentrations (identifying algal blooms), suspended sediment loads, and thermal discharge plumes from power plants.

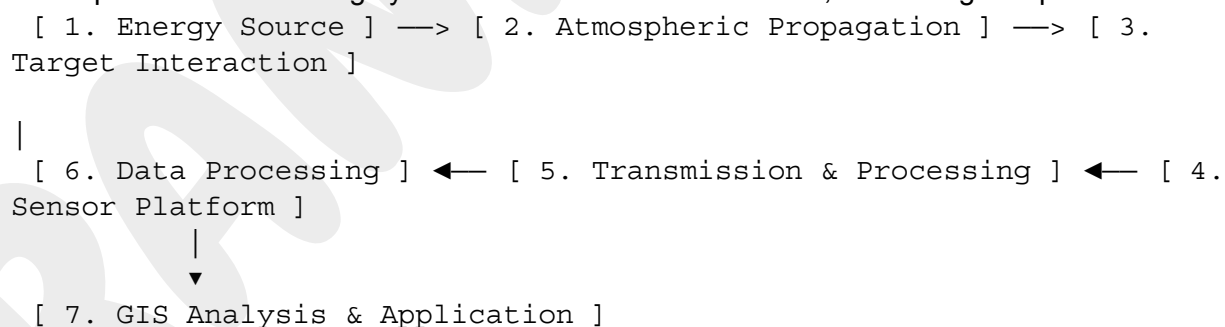
3. Urban Sprawl and Land-Use/Land-Cover (LULC) Change Analysis: Multi-temporal satellite imagery maps the conversion of agricultural land, wetlands, and natural forests into urban concrete infrastructure, supporting sustainable land-use planning and urban heat island analysis.

4. Desertification and Soil Degradation Tracking: Hyper-spectral sensors detect topsoil moisture depletion, surface salinization, and wind erosion trends across arid regions, enabling early desertification interventions.

5. Disaster Management and Climate Monitoring: Remote sensing provides real-time data during natural disasters—tracking tropical cyclone trajectories, mapping flood extents, detecting forest fire hotspots using thermal infrared bands, and measuring polar ice cap melting rates.

Various Constituents of Remote Sensing Systems

A complete remote sensing system consists of seven essential, interacting components:



1. The Energy or Radiation Source (Constituent A)

Every remote sensing system requires an energy source that illuminates or emits electromagnetic energy toward the target.

- **Passive Remote Sensing:** Relies on natural energy sources, primarily solar radiation reflected by Earth's surface or thermal heat naturally emitted by objects.
- **Active Remote Sensing:** The sensor provides its own energy source, emitting pulses of electromagnetic radiation toward the target and measuring the reflected backscatter (e.g., Radar and LiDAR systems).

2. Radiation and the Atmosphere (Constituent B)

As electromagnetic energy travels from the source to Earth's surface and back up to the sensor, it interacts with atmospheric gases and suspended particles. This interaction causes **scattering** (Rayleigh, Mie, and Non-selective) and **absorption** (by water vapor, ozone, and CO₂). Remote sensing operates within specific spectral regions where the atmosphere is transparent, known as **Atmospheric Windows**.

3. Interaction with the Target (Constituent C)

When electromagnetic energy hits Earth's surface, it interacts with different features depending on their physical and chemical properties. The energy is absorbed, transmitted, or reflected. The unique pattern of reflected energy across different wavelengths forms an object's **Spectral Signature**, allowing different features (such as water, soil, and vegetation) to be distinguished.

4. The Sensor and Platform (Constituent D)

- **Platforms:** The vehicle or structure that carries the sensing device. Platforms are categorized by altitude:
 - *Ground-borne:* Handheld radiometers, towers.
 - *Airborne:* Helicopters, airplanes, unmanned aerial vehicles (drones).
 - *Spaceborne:* Satellites orbiting Earth (e.g., Landsat, Sentinel, IRS, SPOT).
- **Sensors:** The optical or electronic instruments that collect and record reflected or emitted electromagnetic radiation. Sensors convert optical signals into digital values across specified spatial, spectral, radiometric, and temporal resolutions.

5. Transmission, Reception, and Processing (Constituent E)

The electromagnetic energy recorded by the sensor is converted into electronic digital data. These digital signals are transmitted via radio telemetry to ground receiving stations, where they are converted into raw digital satellite images.

6. Image Processing and Data Interpretation (Constituent F)

Raw digital image data undergoes pre-processing (radiometric, atmospheric, and geometric corrections) followed by digital image processing (contrast enhancement, spectral classification, ratioing). Remote sensing specialists interpret the processed imagery to identify spatial patterns and extract environmental metrics.

7. Integration and Application (Constituent G)

The final constituent involves integrating interpreted remote sensing data into **Geographic**

Information Systems (GIS) software. This allows spatial data to be combined with field survey observations to inform environmental decision-making, natural resource management, and policy development.